ETSI TS 129 519 V17.17.0 (2026-10)



TECHNICAL SPECIFICATION

5G;
5G System;
Usage of the Unified Data Repository Service for Policy Data, Application Data and Structured Data for Exposure;
Stage 3
(3GPP TS 29.519 version 17.17.0 Release 17)

Reference
RTS/TSGC-0329519vhh0

Keywords
5G

ETSI

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Contents

Intellectual Property Rights 2
Legal Notice 2
Modal verbs terminology 2
Foreword 9
1 Scope 10
2 References 10
3 Definitions and abbreviations 11
3.1 Definitions 11
3.2 Abbreviations 11
4 Overview 11
5 Usage of Nudr_DataRepository Service API for Policy Data 12
5.1 Introduction 12
5.2 Resources 12
5.2.1 Overview 12
5.2.2 Resource Structure 12
5.2.3 Resource: AccessAndMobilityPolicyData 18
5.2.3.1 Description 18
5.2.3.2 Resource definition 18
5.2.3.3 Resource Standard Methods 18
5.2.3.3.1 GET 18
5.2.4 Resource: UEPolicySet 19
5.2.4.1 Description 19
5.2.4.2 Resource definition 19
5.2.4.3 Resource Standard Methods 19
5.2.4.3.1 PUT 19
5.2.4.3.2 PATCH 20
5.2.4.3.3 GET 20
5.2.5 Resource: SessionManagementPolicyData 21
5.2.5.1 Description 21
5.2.5.2 Resource definition 21
5.2.5.3 Resource Standard Methods 21
5.2.5.3.1 GET 21
5.2.5.3.2 PATCH 22
5.2.6 Resource: UsageMonitoringInformation 23
5.2.6.1 Description 23
5.2.6.2 Resource definition 23
5.2.6.3 Resource Standard Methods 23
5.2.6.3.1 PUT 23
5.2.6.3.2 DELETE 24
5.2.6.3.3 GET 24
5.2.7 Resource: SponsorConnectivityData 25
5.2.7.1 Description 25
5.2.7.2 Resource definition 25
5.2.7.3 Resource Standard Methods 25
5.2.7.3.1 GET 25
5.2.8 Resource: BdtData 26
5.2.8.1 Description 26
5.2.8.2 Resource definition 26
5.2.8.3 Resource Standard Methods 26
5.2.8.3.1 GET 26
5.2.9 Resource: IndividualBdtData 27
5.2.9.1 Description 27
5.2.9.2 Resource definition 27

5.2.9.3 Resource Standard Methods 27
5.2.9.3.1 GET 27
5.2.9.3.2 PUT 28
5.2.9.3.3 DELETE 28
5.2.9.3.4 PATCH 29
5.2.10 Resource: PolicyDataSubscriptions 29
5.2.10.1 Description 29
5.2.10.2 Resource definition 29
5.2.10.3 Resource Standard Methods 30
5.2.10.3.1 POST 30
5.2.11 Resource: IndividualPolicyDataSubscription 30
5.2.11.1 Description 30
5.2.11.2 Resource definition 30
5.2.11.3 Resource Standard Methods 31
5.2.11.3.1 PUT 31
5.2.11.3.2 DELETE 31
5.2.12 Resource: OperatorSpecificData 32
5.2.12.1 Description 32
5.2.12.2 Resource Definition 32
5.2.12.3 Resource Standard Methods 32
5.2.12.3.1 GET 32
5.2.12.3.2 PATCH 33
5.2.12.3.3 PUT 33
5.2.12.3.4 DELETE 34
5.2.13 Resource: PlmnUePolicySet 35
5.2.13.1 Description 35
5.2.13.2 Resource definition 35
5.2.13.3 Resource Standard Methods 35
5.2.13.3.1 GET 35
5.2.14 Resource: PolicyDataForIndividualUe 35
5.2.14.1 Description 35
5.2.14.2 Resource definition 35
5.2.14.3 Resource Standard Methods 36
5.2.14.3.1 GET 36
5.2.15 Resource: SlicePolicyControlData 36
5.2.15.1 Description 36
5.2.15.2 Resource definition 36
5.2.15.3 Resource Standard Methods 37
5.2.15.3.1 GET 37
5.2.15.3.2 PATCH 37
5.2.16 Resource: MBSSessionPolicyControlData 38
5.2.16.1 Description 38
5.2.16.2 Resource definition 38
5.2.16.3 Resource Standard Methods 38
5.2.16.3.1 GET 38
5.3 Notifications 39
5.3.1 General 39
5.3.2 Policy Data Change Notification 39
5.4 Data Model 39
5.4.1 General 39
5.4.2 Structured data types 43
5.4.2.1 Introduction 43
5.4.2.2 Type AmPolicyData 44
5.4.2.3 Type UePolicySection 44
5.4.2.4 Type UePolicySet 45
5.4.2.5 Type SmPolicyData 46
5.4.2.6 Type UsageMonDataLimit 47
5.4.2.7 Type UsageMonData 48
5.4.2.8 Type SponsorConnectivityData 48
5.4.2.9 Type BdtData 49
5.4.2.10 Type PolicyDataSubscription 50
5.4.2.11 Type PolicyDataChangeNotification 53

5.4.2.12 Type TimePeriod 55
5.4.2.13 Type UsageMonDataScope 55
5.4.2.14 Type SmPolicySnssaiData 56
5.4.2.15 Type SmPolicyDnnData 57
5.4.2.16 Type LimitIdToMonitoringKey 59
5.4.2.17 Type UePolicySetPatch 60
5.4.2.18 Type PlmnRouteSelectionDescriptor 60
5.4.2.19 Type SnssaiRouteSelectionDescriptor 60
5.4.2.20 Type DnnRouteSelectionDescriptor 60
5.4.2.21 Type SmPolicyDataPatch 61
5.4.2.22 Type SmPolicySnssaiDataPatch 61
5.4.2.23 Type SmPolicyDnnDataPatch 61
5.4.2.24 Type ResourceItem 61
5.4.2.25 Type NotificationItem 62
5.4.2.26 Type UpdatedItem 62
5.4.2.27 Type BdtDataPatch 62
5.4.2.28 Type PolicyDataForIndividualUe 62
5.4.2.29 Type SlicePolicyData 63
5.4.2.31 Type MbsSessPolCtrlData 64
5.4.2.32 Type MbsSessPolDataId 64
5.4.3 Simple data types and enumerations 64
5.4.3.1 Introduction 64
5.4.3.2 Simple data types 64
5.4.3.3 Enumeration: UsageMonLevel 65
5.4.3.4 Enumeration: Periodicity 65
5.4.3.5 Enumeration: BdtPolicyStatus 65
5.5 Error handling 66
5.5.1 General 66
5.5.2 Protocol Errors 66
5.5.3 Application Errors 66
5.6 Feature negotiation 66

6 Usage of Nudr_DataRepository Service API for Application Data 67
6.1 Introduction 67
6.2 Resources 67
6.2.1 Overview 67
6.2.2 Resource Structure 67
6.2.3 Resource: PFD Data 73
6.2.3.1 Description 73
6.2.3.2 Resource definition 73
6.2.3.3 Resource Standard Methods 73
6.2.3.3.1 GET 73
6.2.4 Resource: Individual PFD Data 74
6.2.4.1 Description 74
6.2.4.2 Resource definition 74
6.2.4.3 Resource Standard Methods 74
6.2.4.3.1 GET 74
6.2.4.3.2 DELETE 74
6.2.4.3.3 PUT 75
6.2.5 Resource: Influence Data 76
6.2.5.1 Description 76
6.2.5.2 Resource definition 76
6.2.5.3 Resource Standard Methods 76
6.2.5.3.1 GET 76
6.2.6 Resource: Individual Influence Data 77
6.2.6.1 Description 77
6.2.6.2 Resource definition 77
6.2.6.3 Resource Standard Methods 77
6.2.6.3.1 PUT 77
6.2.6.3.2 PATCH 78
6.2.6.3.3 DELETE 78
6.2.7 Resource: Influence Data Subscription 79

6.2.7.1 Description 79
6.2.7.2 Resource definition 79
6.2.7.3 Resource Standard Methods 79
6.2.7.3.1 POST 79
6.2.7.3.2 GET 80
6.2.8 Resource: Individual Influence Data Subscription 80
6.2.8.1 Description 80
6.2.8.2 Resource definition 81
6.2.8.3 Resource Standard Methods 81
6.2.8.3.1 GET 81
6.2.8.3.2 PUT 81
6.2.8.3.3 DELETE 82
6.2.9 Resource: Applied BDT Policy Data 82
6.2.9.1 Description 82
6.2.9.2 Resource definition 82
6.2.9.3 Resource Standard Methods 83
6.2.9.3.1 GET 83
6.2.10 Resource: Individual Applied BDT Policy Data 83
6.2.10.1 Description 83
6.2.10.2 Resource definition 84
6.2.10.3 Resource Standard Methods 84
6.2.10.3.1 PUT 84
6.2.10.3.2 PATCH 84
6.2.10.3.3 DELETE 85
6.2.11 Resource: IPTV Configurations 85
6.2.11.1 Description 85
6.2.11.2 Resource definition 86
6.2.11.3 Resource Standard Methods 86
6.2.11.3.1 GET 86
6.2.12 Resource: Individual IPTV Configuration 87
6.2.12.1 Description 87
6.2.12.2 Resource definition 87
6.2.12.3 Resource Standard Methods 87
6.2.12.3.1 PUT 87
6.2.12.3.2 DELETE 88
6.2.12.3.3 PATCH 88
6.2.13 Resource: ApplicationDataSubscriptions 89
6.2.13.1 Description 89
6.2.13.2 Resource definition 89
6.2.13.3 Resource Standard Methods 89
6.2.13.3.1 POST 89
6.2.13.3.2 GET 90
6.2.14 Resource: IndividualApplicationDataSubscription 91
6.2.14.1 Description 91
6.2.14.2 Resource definition 91
6.2.14.3 Resource Standard Methods 91
6.2.14.3.1 PUT 91
6.2.14.3.2 DELETE 91
6.2.14.3.3 GET 92
6.2.15 Resource: Service Parameter Data 92
6.2.15.1 Description 92
6.2.15.2 Resource definition 92
6.2.15.3 Resource Standard Methods 93
6.2.15.3.1 GET 93
6.2.16 Resource: Individual Service Parameter Data 94
6.2.16.1 Description 94
6.2.16.2 Resource definition 94
6.2.16.3 Resource Standard Methods 95
6.2.16.3.1 PUT 95
6.2.16.3.2 PATCH 95
6.2.16.3.3 DELETE 96
6.2.17 Resource: AM Influence Data 96

6.2.17.1 Description 96
6.2.17.2 Resource definition 96
6.2.17.3 Resource Standard Methods 97
6.2.17.3.1 GET 97
6.2.18 Resource: Individual AM Influence Data 98
6.2.18.1 Description 98
6.2.18.2 Resource definition 98
6.2.18.3 Resource Standard Methods 98
6.2.18.3.1 PUT 98
6.2.18.3.2 PATCH 99
6.2.18.3.3 DELETE 99
6.2.19 Resource: EAS Deployment Information Data 100
6.2.19.1 Description 100
6.2.19.2 Resource definition 100
6.2.19.3 Resource Standard Methods 100
6.2.19.3.1 GET 100
6.2.20 Resource: Individual EAS Deployment Information Data 101
6.2.20.1 Description 101
6.2.20.2 Resource definition 101
6.2.20.3 Resource Standard Methods 101
6.2.20.3.1 GET 101
6.2.20.3.2 PUT 102
6.2.20.3.3 DELETE 102
6.3 Notifications 103
6.3.1 General 103
6.3.2 Void 103
6.3.3 Influence Data Update Notification 103
6.3.3.1 Description 103
6.3.3.2 Operation Definition 103
6.3.4 Application Data Change Notification 104
6.4 Data Model 104
6.4.1 General 104
6.4.2 Structured data types 108
6.4.2.1 Introduction 108
6.4.2.2 Type TrafficInfluData 109
6.4.2.5 Void 116
6.4.2.6 Type: PfdDataForAppExt 116
6.4.2.7 Type BdtPolicyData 116
6.4.2.9 Type IptvConfigData 117
6.4.2.10 Type ApplicationDataSubs 118
6.4.2.11 Type ApplicationDataChangeNotif 119
6.4.2.12 Type DataFilter 120
6.4.2.13 Void 123
6.4.2.14 Type TrafficInfluDataNotif 123
6.4.2.15 Type ServiceParameterData 124
6.4.2.15A Type ServiceParameterDataPatch 127
6.4.2.16 Type AmInfluData 128
6.4.2.17 Type AmInfluDataPatch 129
6.4.3 Simple data types and enumerations 129
6.4.3.1 Introduction 129
6.4.3.2 Simple data types 129
6.4.3.3 Enumeration: DataInd 129
6.5 Error handling 130
6.5.1 General 130
6.5.2 Protocol Errors 130
6.5.3 Application Errors 130
6.6 Feature negotiation 130

7 Usage of Nudr_DataRepository Service API for Structured Data for Exposure 130
7.1 Introduction 130
7.2 Resources 130
7.2.1 Overview 130

7.2.2 Resource Structure 131
7.2.3 Resource: AccessAndMobilityData 132
7.2.3.1 Description 132
7.2.3.2 Resource definition 132
7.2.3.3 Resource Standard Methods 132
7.2.3.3.1 PUT 132
7.2.3.3.2 GET 133
7.2.3.3.3 DELETE 133
7.2.3.3.4 PATCH 134
7.2.4 Resource: PduSessionManagementData 134
7.2.4.1 Description 134
7.2.4.2 Resource definition 134
7.2.4.3 Resource Standard Methods 135
7.2.4.3.1 PUT 135
7.2.4.3.2 GET 135
7.2.4.3.3 DELETE 136
7.2.5 Resource: ExposureDataSubscriptions 137
7.2.5.1 Description 137
7.2.5.2 Resource Definition 137
7.2.5.3 Resource Standard Methods 137
7.2.5.3.1 POST 137
7.2.6 Resource: IndividualExposureDataSubscription 138
7.2.6.1 Description 138
7.2.6.2 Resource Definition 138
7.2.6.3 Resource Standard Methods 138
7.2.6.3.1 PUT 138
7.2.6.3.2 DELETE 138
7.3 Data Model 139
7.3.1 General 139
7.3.2 Structured data types 140
7.3.2.1 Introduction 140
7.3.2.2 Type AccessAndMobilityData 141
7.3.2.3 Type PduSessionManagementData 142
7.3.2.4 Type ExposureDataSubscription 143
7.3.2.5 Type ExposureDataChangeNotification 143
7.3.3 Simple data types and enumerations 143
7.3.3.1 Introduction 143
7.3.3.2 Simple data types 144
7.3.3.3 Enumeration: PduSessionStatus 144
7.4 Error handling 144
7.4.1 General 144
7.4.2 Protocol Errors 144
7.4.3 Application Errors 144
7.5 Feature negotiation 144
7.6 Notifications 145
7.6.1 General 145
7.6.2 Exposure Data Change Notification 145

Annex A (normative): OpenAPI specification 146

A.1 General 146

A.2 Nudr_DataRepository API for Policy Data 146

A.3 Nudr_DataRepository API for Application Data 177

A.4 Nudr_DataRepository API for Exposure Data 220

Annex B (informative): Change history 232

History 238

Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

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1 Scope

The present specification provides the stage 3 definition for the usage of the Unified Data Repository service (Nudr_DataRepository) of 5G System for Policy Data, Application Data and Structured Data for Exposure.

The Unified Data Repository (UDR) provides the Unified Data Repository service. This service provides a repository for policy data, application data and exposure data.

The stage 3 protocol definition for Nudr Service Based Interface which is common for subscription data, policy data, application data and exposure data is specified in 3GPP TS 29.504 [6].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

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[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] Void.

[3] OpenAPI: "OpenAPI Specification Version 3.0.0",https://spec.openapis.org/oas/v3.0.0.

[4] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[5] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".

[6] 3GPP TS 29.504:"5G System; Unified Data Repository Services; Stage 3".

[7] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces Stage 3".

[8] 3GPP TS 29.551: "5G System; Packet Flow Description Management Service; Stage 3".

[9] 3GPP TS 29.122: "T8 reference point for Northbound APIs".

[10] 3GPP TS 29.518: "5G System; Access and Mobility Management Services; Stage 3".

[11] 3GPP TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3".

[12] 3GPP TS 29.512: "5G System; Session Management Policy Control Service; Stage 3".

[13] 3GPP TS 29.554: "5G System; Background Data Transfer Policy Control Service; Stage 3".

[14] Void.

[15] 3GPP TS 29.505: "5G System; Usage of the Unified Data Repository services for Subscription Data; Stage 3".

[16] 3GPP TS 29.514: "5G System; Policy Authorization Service; Stage 3".

[17] IETF RFC 4122: "A Universally Unique IDentifier (UUID) URN Namespace".

[18] 3GPP TR 21.900: "Technical Specification Group working methods".

[19] 3GPP TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3".

[20] IETF RFC 6901: "JavaScript Object Notation (JSON) Pointer".

[21] IETF RFC 7230: "Hypertext Transfer Protocol (HTTP/1.1): Message Syntax and Routing".

[22] 3GPP TS 29.534: "5G System; Access and Mobility Policy Authorization Service; Stage 3".

[23] 3GPP TS 29.591: "5G System; Network Exposure Function Southbound Services; Stage 3".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

AM	Access and Mobility Management
ATSSS	Access Traffic Steering, Switching, Splitting
BDT	Background Data Transfer
DNAI	DN Access Identifier
DNN	Data Network Name
GPSI	Generic Public Subscription Identifier
JSON	JavaScript Object Notation
MA	Multi Access
NEF	Network Exposure Function
NR	New Radio
NRF	Network Repository Function
PCF	Policy Control Function
PFD	Packet Flow Description
PSA	PDU Session Anchor
SM	Session Management
S-NSSAI	Single Network Slice Selection Assistance Information
SUPI	Subscription Permanent Identifier
UDR	Unified Data Repository
UPSI	UE Policy Section Identifier
URSP	UE Route Selection Policy

4 Overview

The resources, data model and usage defined in the present document are used by the 5GC NFs (e.g. PCF, NEF) to access the policy data, application data or exposure data via the Unified Data Repository services as specified in 3GPP TS 29.504 [6].

5 Usage of Nudr_DataRepository Service API for Policy Data

5.1 Introduction

The following clauses specify the usage of Nudr_DataRepository service for policy data. The principles specified in 3GPP TS 29.504 [6] are followed unless explicitly specified otherwise in the following clauses.

5.2 Resources

5.2.1 Overview

This clause specifies the resources for the usage of the Unified Data Repository service for Policy Data.

5.2.2 Resource Structure

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 5.2.2-1 depicts the resource URIs structure for the Nudr_DataRepository API for policy data.

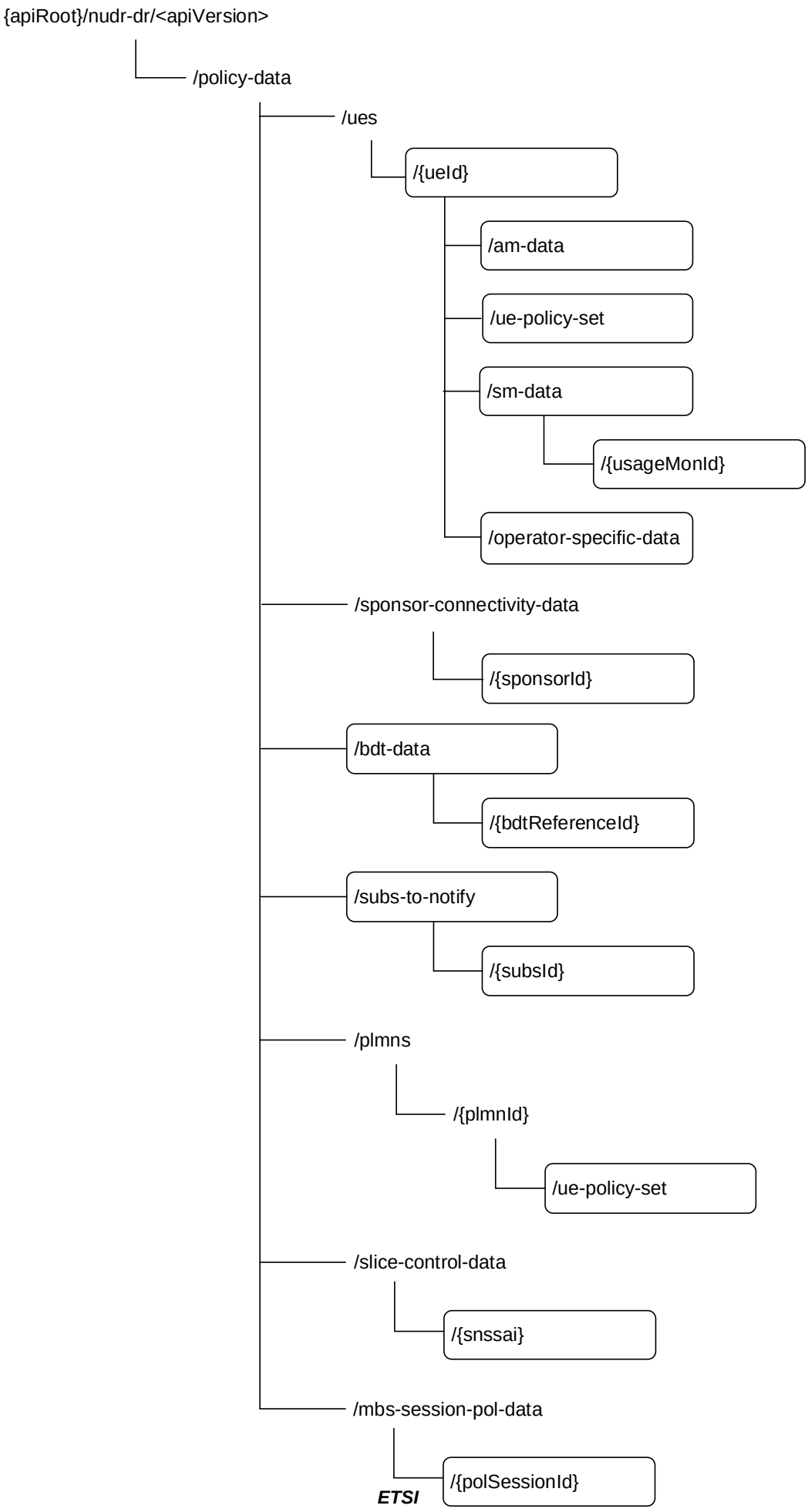

{apiRoot}/nudr-dr/<apiVersion>
/policy-data
/ues
/{ueId}
/am-data
/ue-policy-set
/sm-data
/{usageMonId}
/operator-specific-data
/sponsor-connectivity-data
/{sponsorId}
/bdt-data
/{bdtReferenceId}
/subs-to-notify
/{subsId}
/plmns
/{plmnId}
/ue-policy-set
/slice-control-data
/{snssai}
/mbs-session-pol-data
/{polSessionId}

Figure 5.2.2-1: Resource URI structure of the Nudr_DataRepository API for policy data

Table 5.2.2-1 provides an overview of the resources and applicable HTTP methods.

Table 5.2.2-1: Resources and methods overview

Resource name	Resource URI	HTTP method or custom operation	Description
PolicyDataForIndividualUe	/policy-data/ues/{ueId}	GET	Retrieve multiple policy data sets (access and mobility policy data, UE policy set data, session management policy data, usage monitoring data, operator specific policy data) for a subscriber.
AccessAndMobilityPolicyData	/policy-data/ues/{ueId}/ am-data	GET	Retrieve the access and mobility policy data for a subscriber.
UEPolicySet	/policy-data/ues/{ueId}/ ue-policy-set	GET	Retrieve the UE policy set data for a subscriber.
		PUT	Create or modify the UE policy set data for a subscriber.
		PATCH	Modify the UE policy set data for a subscriber.
SessionManagementPolicyData	/policy-data/ues/{ueId}/ sm-data	GET	Retrieve the session management policy data for a subscriber.
		PATCH	When the feature "SessionManagementPolicyDataPatch" is supported, modify the session management policy data for a subscriber.
UsageMonitoringInformation	/policy-data/ues/{ueId}/ sm-data/{usageMonId}	GET	Retrieve a usage monitoring resource.
		PUT	Create a usage monitoring resource.
		DELETE	Delete a usage monitoring resource.
OperatorSpecificData	/policy-data/ues/{ueId}/ operator-specific-data	GET	Retrieve the operator specific policy data of a UE
		PATCH	Modify the operator specific policy data of a UE.
		PUT	Create or modify the operator specific policy data of a UE.
		DELETE	Delete the operator specific policy data for a UE.
SponsorConnectivityData	/policy-data/sponsor-connectivity-data/{sponsorId}	GET	Retrieve the sponsored connectivity information for a given sponsorId.
BdtData	/policy-data/bdt-data	GET	Retrieve the BDT data collection.
IndividualBdtData	/policy-data/bdt-data/ {bdtReferenceId}	GET	Retrieve the BDT data information associated with a BDT reference Id.
		PUT	Create a BDT data resource associated with a BDT reference Id.
		PATCH	Modifies an BDT data resource associated with an BDT reference Id.
		DELETE	Delete a BDT data resource associated with a BDT reference Id.
PolicyDataSubscriptions	/policy-data/subs-to-notify	POST	Create a subscription to receive notifications of policy data changes.
IndividualPolicyDataSubscription	/policy-data/subs-to-notify/ {subsId}	PUT	Modify a subscription to receive notifications of policy data changes.
		DELETE	Delete a subscription identified by {subsId}.

PlmnUePolicySet	/policy-data/plmns/{plmnId}/ ue-policy-set	GET	Retrieve the UE policy set data for an H-PLMN.
SlicePolicyControlData	/policy-data/slice-control-data/{snssai}	GET	Retrieve network slice specific policy control data for a given S-NSSAI.
		PATCH	Modify network slice specific policy control data for a given S-NSSAI.
MBSSessionPolicyControlData	/policy-data/mbs-session-pol-data/{polSessionId}	GET	Retrieve MBS Session Policy Control Data for an MBS Session based on the MBS Session Identifier or an AF Application Identifier.

5.2.3 Resource: AccessAndMobilityPolicyData

5.2.3.1 Description

The resource represents all UE related mobility and access policies attributes in the UDR for a given "ueId".

5.2.3.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/ues/{ueId}/am-data**

This resource shall support the resource URI variables defined in table 5.2.3.2-1.

Table 5.2.3.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1
ueId	VarUeId	Represents the Subscription Identifier SUPI or GPSI. Data type VarUeId is defined in 3GPP TS 29.571 [7].

5.2.3.3 Resource Standard Methods

5.2.3.3.1 GET

This method shall support the URI query parameters specified in table 5.2.3.3.1-1.

Table 5.2.3.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.3.3.1-2 and the response data structures and response codes specified in table 5.2.3.3.1-3.

Table 5.2.3.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.3.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AmPolicyData	M	1	200 OK	Upon success, a response body containing access and mobility policies shall be returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.4 Resource: UEPolicySet

5.2.4.1 Description

The resource represents UE policy set for a given "ueId" stored in the UDR of the H-PLMN.

5.2.4.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/ues/{ueId}/ue-policy-set**

This resource shall support the resource URI variables defined in table 5.2.4.2-1.

Table 5.2.4.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1
ueId	VarUeId	Represents the Subscription Identifier SUPI or GPSI. Data type VarUeId is defined in 3GPP TS 29.571 [7].

5.2.4.3 Resource Standard Methods

5.2.4.3.1 PUT

This method shall support the URI query parameters specified in table 5.2.4.3.1-1.

Table 5.2.4.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.4.3.1-2 and the response data structures and response codes specified in table 5.2.4.3.1-3.

Table 5.2.4.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
UePolicySet	M	1	Create or modify the UE policy set resource for a given subscriber.

Table 5.2.4.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
UePolicySet	M	1	201 Created	Successful case. The resource has been successfully created and a response body containing a representation of the created UEPolicySet resource shall be returned.
UePolicySet	M	1	200 OK	Successful case. The resource has been successfully updated and a response body containing UE policies shall be returned.
n/a			204 No Content	Successful case. The resource has been successfully updated and no additional content is to be sent in the response message.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 5.2.4.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/policy-data/ues/{ueId}/ue-policy-set

5.2.4.3.2 PATCH

This method shall support the URI query parameters specified in table 5.2.4.3.2-1.

Table 5.2.4.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.4.3.2-2 and the response data structures and response codes specified in table 5.2.4.3.2-3.

Table 5.2.4.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
UePolicySetPatch	M	1	Modify the UE policy set resource for a given subscriber.

Table 5.2.4.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful case. The resource has been successfully updated and no additional content is to be sent in the response message.
NOTE: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.4.3.3 GET

This method shall support the URI query parameters specified in table 5.2.4.3.3-1.

Table 5.2.4.3.3-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
supp-feat	SupportedFeatures	O	0..1	The features supported by the NF service consumer.

This method shall support the request data structures specified in table 5.2.4.3.3-2 and the response data structures and response codes specified in table 5.2.4.3.3-3.

Table 5.2.4.3.3-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.4.3.3-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
UePolicySet	M	1	200 OK	Upon success, a response body containing the UE policies shall be returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.5 Resource: SessionManagementPolicyData

5.2.5.1 Description

The resource represents all PDU session related subscription attributes in the UDR for a given ueId.

5.2.5.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/ues/{ueId}/sm-data**

This resource shall support the resource URI variables defined in table 5.2.5.2-1.

Table 5.2.5.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
ueId	VarUeId	Represents the Subscription Identifier SUPI or GPSI. Data type VarUeId is defined in 3GPP TS 29.571 [7].

5.2.5.3 Resource Standard Methods

5.2.5.3.1 GET

This method shall support the URI query parameters specified in table 5.2.5.3.1-1.

Table 5.2.5.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
snssai	Snssai	O	0..1	Identifies a Single Network Slice Selection Assistance Information.
dnn	Dnn	O	0..1	Identifies a Data Network Name.
fields	array(string)	C	1..N	When the NF consumer only retrieves a subset of the resource, the "fields" query parameter shall be included. The "fields" query parameter contains the pointers of the attribute(s) to be retrieved.
supp-feat	SupportedFeatures	O	0..1	Identifies the features supported by the NF service consumer.

When an optional query parameter is not included, the UDR shall return session management policy data per DNN matching the value of the included query parameter(s) for all the possible values of the omitted query parameter. E.g. if "snssai" is not included, and "dnn" is included, UDR shall return the session management policy data per DNN for the DNN identified by "dnn" for all network slices where such DNN is available.

This method shall support the request data structures specified in table 5.2.5.3.1-2 and the response data structures and response codes specified in table 5.2.5.3.1-3.

Table 5.2.5.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.5.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
SmPolicyData	M	1	200 OK	Upon success, a response body containing SmPolicyData shall be returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

NOTE: This release supports that only one PCF can access to the UDR to retrieve the SM Policy Data under one S-NSSAI and DNN association.

5.2.5.3.2 PATCH

This method shall support the URI query parameters specified in table 5.2.5.3.2-1.

Table 5.2.5.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description

This method shall support the request data structures specified in table 5.2.5.3.2-2 and the response data structures and response codes specified in table 5.2.5.3.2-3.

Table 5.2.5.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
SmPolicyDataPatch	M	1	When the "SessionManagementPolicyDataPatch" feature is supported, it requests the modification of the session management policy data for a subscriber.

Table 5.2.5.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful case. The resource has been successfully updated and no additional content is to be sent in the response message. (NOTE 2)(NOTE 3)
SmPolicyData	M	1	200 OK	Upon success, the execution report is returned. (NOTE 2)(NOTE 3)
NOTE 1: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply. NOTE 2: If all the modification instructions in the PATCH request have been implemented, the UDR shall respond with 204 No Content response; if some of the modification instructions in the PATCH request have been discarded, and the NF service consumer supported the "PatchReport" feature, the UDR shall respond with SmPolicyData. NOTE 3: When the resource is successfully updated, if the SmPolicyDataPatch in the request requires the creation, removal, or update of an entry of the "umData" attribute, the UDR shall create, remove or update the corresponding UsageMonitoringInformation resource for the indicated limit identifier.				

5.2.6 Resource: UsageMonitoringInformation

5.2.6.1 Description

The resource represents an individual usage monitoring resource created in the UDR and associated with a ueId and a usageMonId.

5.2.6.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/ues/{ueId}/sm-data/{usageMonId}**

This resource shall support the resource URI variables defined in table 5.2.6.2-1.

Table 5.2.6.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
ueId	VarUeId	Represents the Subscription Identifier SUPI or GPSI. Data type VarUeId is defined in 3GPP TS 29.571 [7].
usageMonId	string	Unique identifier of the individual SM Policy usage monitoring resource. It contains the identity of the corresponding limit as defined in the "limitId" attribute of the UsageMonDataLimit data type (see clause 5.4.2.6).

5.2.6.3 Resource Standard Methods

5.2.6.3.1 PUT

This method shall support the URI query parameters specified in table 5.2.6.3.1-1.

Table 5.2.6.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.6.3.1-2 and the response data structures and response codes specified in table 5.2.6.3.1-3.

Table 5.2.6.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
UsageMonData	M	1	Create the usage monitoring resource.

Table 5.2.6.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
UsageMonData	M	1	201 Created	Successful case. The resource has been successfully created and a response body is returned containing a representation of the resource. (NOTE 2)
NOTE 1: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply. NOTE 2: When the resource is successfully created, the UDR shall create a new entry within the "umData" attribute of the SessionManagementPolicyData resource for the corresponding "limitId" accordingly.				

Table 5.2.6.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/policy-data/ues/{ueId}/sm-data/{usageMonId}

5.2.6.3.2 DELETE

This method shall support the URI query parameters specified in table 5.2.6.3.2-1.

Table 5.2.6.3.2-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.6.3.2-2 and the response data structures and response codes specified in table 5.2.6.3.2-3.

Table 5.2.6.3.2-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.6.3.2-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful case. The resource has been successfully deleted. (NOTE 2)
NOTE 1: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply. NOTE 2: When the resource is successfully deleted, the UDR shall delete within the "umData" attribute of the SessionManagementPolicyData resource the corresponding "limitId" entry accordingly.				

5.2.6.3.3 GET

This method shall support the URI query parameters specified in table 5.2.6.3.3-1.

Table 5.2.6.3.3-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
supp-feat	SupportedFeatures	O	0..1	The features supported by the NF service consumer.

This method shall support the request data structures specified in table 5.2.6.3.3-2 and the response data structures and response codes specified in table 5.2.6.3.3-3.

Table 5.2.6.3.3-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.6.3.3-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
UsageMonData	M	1	200 OK	Successful case. The usage monitoring data is returned.
n/a			204 No Content	The resource was found but no usage monitoring data is available.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.7 Resource: SponsorConnectivityData

5.2.7.1 Description

The resource represents sponsored data connectivity related information stored in the UDR for a given sponsorId.

5.2.7.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/sponsor-connectivity-data/{sponsorId}**

This resource shall support the resource URI variables defined in table 5.2.7.2-1.

Table 5.2.7.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
sponsorId	string	Represents the sponsor identity.

5.2.7.3 Resource Standard Methods

5.2.7.3.1 GET

This method shall support the URI query parameters specified in table 5.2.7.3.1-1.

Table 5.2.7.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.7.3.1-2 and the response data structures and response codes specified in table 5.2.7.3.1-3.

Table 5.2.7.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.7.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
SponsorConnectivityData	M	1	200 OK	Upon success, a response body containing Sponsor Connectivity Data shall be returned.
n/a			204 No Content	The resource was found but no Sponsor Connectivity Data is available.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.8 Resource: BdtData

5.2.8.1 Description

The resource represents background data transfer data stored in the UDR.

5.2.8.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/bdt-data**

This resource shall support the resource URI variables defined in table 5.2.8.2-1.

Table 5.2.8.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

5.2.8.3 Resource Standard Methods

5.2.8.3.1 GET

This method shall support the URI query parameters specified in table 5.2.8.3.1-1.

Table 5.2.8.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
bdt-ref-ids	array(BdtReferenceId)	O	1..N	If included, this IE shall contain a list of the BDT reference identifiers. Each member of the array is an individual BDT reference identifier.
supp-feat	SupportedFeatures	O	0..1	The features supported by the NF service consumer.

This method shall support the request data structures specified in table 5.2.8.3.1-2 and the response data structures and response codes specified in table 5.2.8.3.1-3.

Table 5.2.8.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.8.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
array(BdtData)	M	0..N	200 OK	Upon success, a response body containing the BDT data shall be returned. Each member of the array is an individual BDT data.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.9 Resource: IndividualBdtData

5.2.9.1 Description

The resource represents individual background data transfer data stored in the UDR for a given "bdtReferenceId".

5.2.9.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/bdt-data/{bdtReferenceId}**

This resource shall support the resource URI variables defined in table 5.2.9.2-1.

Table 5.2.9.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
bdtReferenceId	BdtReferenceId	Represents the transfer policy of background data transfer for provided ASP. Data type BdtReferenceId is defined in 3GPP TS 29.122 [9].

5.2.9.3 Resource Standard Methods

5.2.9.3.1 GET

This method shall support the URI query parameters specified in table 5.2.9.3.1-1.

Table 5.2.9.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
supp-feat	SupportedFeatures	O	0..1	The features supported by the NF service consumer.

This method shall support the request data structures specified in table 5.2.9.3.1-2 and the response data structures and response codes specified in table 5.2.9.3.1-3.

Table 5.2.9.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.9.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
BdtData	M	1	200 OK	Upon success, a response body containing the BDT data shall be returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.9.3.2 PUT

This method shall support the URI query parameters specified in table 5.2.9.3.2-1.

Table 5.2.9.3.2-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.9.3.2-2 and the response data structures and response codes specified in table 5.2.9.3.2-3.

Table 5.2.9.3.2-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
BdtData	M	1	Create the BDT data for a given bdtReferenceId.

Table 5.2.9.3.2-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
BdtData	M	1	201 Created	Successful case. The resource has been successfully created.
ProblemDetails	O	0..1	403 Forbidden	The resource is not allowed to be updated. (NOTE 2)
NOTE 1: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply. NOTE 2: The "cause" attribute within the "ProblemDetails" data structure may be set to "MODIFICATION_NOT_ALLOWED" as defined in table 5.2.7.2-1 of 3GPP TS 29.500 [4].				

Table 5.2.9.3.2-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/policy-data/bdt-data/{bdtReferenceId}

5.2.9.3.3 DELETE

This method shall support the URI query parameters specified in table 5.2.9.3.3-1.

Table 5.2.9.3.3-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.9.3.3-2 and the response data structures and response codes specified in table 5.2.9.3.3-3.

Table 5.2.9.3.3-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.9.3.3-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful case. The resource has been successfully deleted.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.9.3.4 PATCH

This method shall support the URI query parameters specified in table 5.2.9.3.4-1.

Table 5.2.9.3.4-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.9.3.4-2 and the response data structures and response codes specified in table 5.2.9.3.4-3.

Table 5.2.9.3.4-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
BdtDataPatch	M	1	Modify the BDT data for a given bdtReferenceId.

Table 5.2.9.3.4-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
BdtData	M	1	200 OK	The resource has been successfully updated and a response body containing BDT data shall be returned.
n/a			204 No Content	Successful case. The resource has been successfully updated and no additional content is to be sent in the response message.
NOTE: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.10 Resource: PolicyDataSubscriptions

5.2.10.1 Description

This resource is used to represent subscriptions to notification of policy data modification.

5.2.10.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/subs-to-notify**

This resource shall support the resource URI variables defined in table 5.2.10.2-1.

Table 5.2.10.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

5.2.10.3 Resource Standard Methods

5.2.10.3.1 POST

This method shall support the URI query parameters specified in table 5.2.10.3.1-1.

Table 5.2.10.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.10.3.1-2 and the response data structures and response codes specified in table 5.2.10.3.1-3.

Table 5.2.10.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
PolicyDataSubscription	M	1	Identifies the subscription to notification of policy data modification.

Table 5.2.10.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
PolicyDataSubscription	M	1	201 Created	Upon success, a response body containing a representation of each Individual subscription resource shall be returned. The HTTP response shall include a "Location" HTTP header that contains the resource URI of the created resource.
NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 5.2.10.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/policy-data/subs-to-notify/{subsId}

5.2.11 Resource: IndividualPolicyDataSubscription

5.2.11.1 Description

This resource is used to represent an individual subscription to notification of policy data modification.

5.2.11.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/subs-to-notify/{subsId}**

This resource shall support the resource URI variables defined in table 5.2.11.2-1.

Table 5.2.11.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
subsId	string	The subsId identifies an individual policy data subscription to notifications. The value is allocated by the UDR during creation of the PolicyDataSubscriptions resource.

5.2.11.3 Resource Standard Methods

5.2.11.3.1 PUT

This method shall support the URI query parameters specified in table 5.2.11.3.1-1.

Table 5.2.11.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.11.3.1-2 and the response data structures and response codes specified in table 5.2.11.3.1-3.

Table 5.2.11.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
PolicyDataSubscription	M	1	Updates the subscription to notification of policy data modification.

Table 5.2.11.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
PolicyDataSubscription	M	1	200 OK	The individual subscription resource was updated successfully.
n/a			204 No Content	The individual subscription resource was updated successfully and no additional content is to be sent in the response message.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.11.3.2 DELETE

This method shall support the URI query parameters specified in table 5.2.11.3.2-1.

Table 5.2.11.3.2-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.11.3.2-2 and the response data structures and response codes specified in table 5.2.11.3.2-3.

Table 5.2.11.3.2-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			The request body shall be empty.

Table 5.2.11.3.2-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon success, an empty response body shall be returned.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.12 Resource: OperatorSpecificData

5.2.12.1 Description

This resource represents the UE's operator specific data.

This resource is modelled with the Document resource archetype (see clause C.1 of 3GPP TS 29.501 [5]).

5.2.12.2 Resource Definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/ues/{ueId}/operator-specific-data**

This resource shall support the resource URI variables defined in table 5.2.12.2-1.

Table 5.2.12.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
ueId	VarUeId	Represents the Subscription Identifier SUPI or GPSI. Data type VarUeId is defined in 3GPP TS 29.571 [7].

5.2.12.3 Resource Standard Methods

5.2.12.3.1 GET

This method is used to retrieve operator specific data from the UDR.

This method shall support the URI query parameters specified in table 5.2.12.3.1-1.

Table 5.2.12.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
fields	array(string)	C	1..N	When the NF consumer only retrieves a subset of the resource, the "fields" query parameter shall be included. The "fields" query parameter contains the pointers of the attribute(s) to be retrieved.
supp-feat	SupportedFeatures	O	0..1	Identifies the features supported by the NF service consumer.

This method shall support the request data structures specified in table 5.2.12.3.1-2 and the response data structures and response codes specified in table 5.2.12.3.1-3.

Table 5.2.12.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.12.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
map(OperatorSpecificDataContainer)	M	0..N	200 OK	Upon success, a response body containing a map shall be returned. The key of the map is an operator specific data element name and the value is the operator specific data of such data element, for the current UE (identified by the {ueId} variable in the resource path). If the resource does not contain any keys, the response consists of an empty JSON object (empty map).
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.12.3.2 PATCH

This method is used to modify operator specific data in the UDR.

This method shall support the URI query parameters specified in table 5.2.12.3.2-1.

Table 5.2.12.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.12.3.2-2 and the response data structures and response codes specified in table 5.2.12.3.2-3.

Table 5.2.12.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
array(PatchItem)	M	1..N	Contains the delta data to the operator specific data

Table 5.2.12.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon successful modification there is no body in the response message.(NOTE 2)
PatchResult	M	1	200 OK	Upon success, the execution report is returned. (NOTE 2)
ProblemDetails	O	0..1	403 Forbidden	The resource is not allowed to be updated. (NOTE 3)
NOTE 1: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				
NOTE 2: If all the modification instructions in the PATCH request have been implemented, the UDR shall respond with 204 No Content response; if some of the modification instructions in the PATCH request have been discarded, and the NF service consumer supported the "PatchReport" feature, the UDR shall respond with PatchResult.				
NOTE 3: The "cause" attribute within the "ProblemDetails" data structure may be set to "MODIFICATION_NOT_ALLOWED" as defined in table 5.2.7.2-1 of 3GPP TS 29.500 [4].				

5.2.12.3.3 PUT

This method is used to create or modify operator specific data in the UDR.

This method shall support the URI query parameters specified in table 5.2.12.3.3-1.

Table 5.2.12.3.3-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.12.3.3-2 and the response data structures and response codes specified in table 5.2.12.3.3-3.

Table 5.2.12.3.3-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
map(OperatorSpecificDataContainer)	M	1..N	A map containing the new representation of the operator specific data. The key of the map is operator specific data element name and the value is the operator specific data of the UE.

Table 5.2.12.3.3-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
map(OperatorSpecificDataContainer)	M	1	200 OK	Upon successful modification a response body containing a map shall be returned. The key of the map is operator specific data element name and the value is the operator specific data of the UE.
map(OperatorSpecificDataContainer)	M	1	201 Created	If the "OSDResource_Create_Delete" feature is supported and the resource has been successfully created, a response body is returned. The key of the map is operator specific data element name and the value is the operator specific data of the UE.
n/a			204 No Content	The resource has been successfully updated.
ProblemDetails	O	0..1	403 Forbidden	The resource is not allowed to be updated. (NOTE 2)
NOTE 1: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply. NOTE 2: The "cause" attribute within the "ProblemDetails" data structure may be set to "MODIFICATION_NOT_ALLOWED" as defined in table 5.2.7.2-1 of 3GPP TS 29.500 [4].				

Table 5.2.12.3.3-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/policy-data/ues/{ueId}/operator-specific-data.

5.2.12.3.4 DELETE

This method shall support the URI query parameters specified in table 5.2.12.3.4-1.

Table 5.2.12.3.4-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.12.3.4-2 and the response data structures and response codes specified in table 5.2.12.3.4-3.

Table 5.2.12.3.4-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.12.3.4-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful case, applicable when the "OSDResource_Create_Delete" feature is supported. The resource has been successfully deleted.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.13 Resource: PlmnUePolicySet

5.2.13.1 Description

The resource represents the UE policy set for a given "plmnId" representing an H-PLMN stored in the UDR of a V-PLMN.

5.2.13.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/plmns/{plmnId}/ue-policy-set**

This resource shall support the resource URI variables defined in table 5.2.13.2-1.

Table 5.2.13.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
plmnId	string	Concatenating a Mcc and Mnc as defined in 3GPP TS 29.505 [15].

5.2.13.3 Resource Standard Methods

5.2.13.3.1 GET

This method shall support the URI query parameters specified in table 5.2.13.3.1-1.

Table 5.2.13.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.13.3.1-2 and the response data structures and response codes specified in table 5.2.13.3.1-3.

Table 5.2.13.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.13.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
UePolicySet	M	1	200 OK	Upon success, a response body containing the UE policies shall be returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.14 Resource: PolicyDataForIndividualUe

5.2.14.1 Description

The resource represents all UE related policy data sets in the UDR for a given "ueId".

5.2.14.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/ues/{ueId}**

This resource shall support the resource URI variables defined in table 5.2.14.2-1.

Table 5.2.14.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1
ueId	VarUeId	Represents the Subscription Identifier SUPI or GPSI. Data type VarUeId is defined in 3GPP TS 29.571 [7].

5.2.14.3 Resource Standard Methods

5.2.14.3.1 GET

This method shall support the URI query parameters specified in table 5.2.14.3.1-1.

Table 5.2.14.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
supp-feat	SupportedFeatures	O	0..1	The features supported by the NF service consumer, as per clause 6.6 of 3GPP TS 29.500 [4].
data-subset-names	array(PolicyDataSubset)	O	2..N	The policy data subsets (e.g. AM policies, SM policies, etc.) requested by the NF service consumer. If this query parameter is not provided by the NF service consumer, this means that all the available policy data subsets shall be returned.
NOTE: When two or more policy data subsets sets shall be retrieved, it is recommended to use a single GET request with query parameters.				

This method shall support the request data structures specified in table 5.2.14.3.1-2 and the response data structures and response codes specified in table 5.2.14.3.1-3.

Table 5.2.14.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.14.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
PolicyDataForIndividualUE	M	1	200 OK	Upon success, a response body containing the policy data subsets requested by the NF service consumer or all the policy data subsets shall be returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.15 Resource: SlicePolicyControlData

5.2.15.1 Description

The resource represents the network slice specific policy control subscription data stored in the UDR for a given S-NSSAI.

5.2.15.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/slice-control-data/{snssai}**

This resource shall support the resource URI variables defined in table 5.2.15.2-1.

Table 5.2.15.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
snssai	Snssai	The S-NSSAI value used in the last segment of the resource URI is encoded as a string as defined in 3GPP TS 29.571 [7], clause 5.4.4.2.

5.2.15.3 Resource Standard Methods

5.2.15.3.1 GET

This method shall support the URI query parameters specified in table 5.2.15.3.1-1.

Table 5.2.15.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
supp-feat	SupportedFeatures	O	0..1	The features supported by the NF service consumer.

This method shall support the request data structures specified in table 5.2.15.3.1-2 and the response data structures and response codes specified in table 5.2.15.3.1-3.

Table 5.2.15.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.15.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
SlicePolicyData	M	1	200 OK	Upon success, a response body containing network slice specific policy control data shall be returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.15.3.2 PATCH

This method shall support the URI query parameters specified in table 5.2.15.3.2-1.

Table 5.2.15.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description

This method shall support the request data structures specified in table 5.2.15.3.2-2 and the response data structures and response codes specified in table 5.2.15.3.2-3.

Table 5.2.15.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
SlicePolicyDataPatch	M	1	Modify a network slice specific policy control data resource for a given S-NSSAI.

Table 5.2.15.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
SlicePolicyData	M	1	200 OK	The resource has been successfully updated and a response body containing network slice specific policy control data shall be returned.
n/a			204 No Content	The resource has been successfully updated and no additional content is to be sent in the response message.
NOTE: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.2.16 Resource: MBSSessionPolicyControlData

5.2.16.1 Description

The resource represents the MBS Session Policy Control Data for an MBS Session.

5.2.16.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/policy-data/mbs-session-pol-data/{polSessionId}**

This resource shall support the resource URI variables defined in table 5.2.16.2-1.

Table 5.2.16.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
polSessionId	MbsSessPolDataId	Represents the identifier of the MBS Session Policy Control Data.

5.2.16.3 Resource Standard Methods

5.2.16.3.1 GET

This method shall support the URI query parameters specified in table 5.2.16.3.1-1.

Table 5.2.16.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
n/a				

Table 5.2.16.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.16.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
MbsSessPolCtrlData	M	1	200 OK	Successful case. The requested MBS Session Policy Control Data is successfully returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.3 Notifications

5.3.1 General

Notifications shall comply with clause 6.2 of 3GPP TS 29.500 [4] and clause 4.6.2.3 of 3GPP TS 29.501 [5].

This clause describes the resources to provide notification to NF service consumers which have subscribed to be notified when policy data is changed.

Table 5.3.1-1: Notifications overview

Notification	Callback URI	HTTP method or custom operation	Description (service operation)
Policy Data Change Notification	{notificationUri}	POST	Used for policy data change notification

5.3.2 Policy Data Change Notification

The POST method shall be used for policy data change notification and the URI shall be provided during the subscription procedure.

Callback URI: **{notificationUri}**

This method shall support the URI query parameters specified in table 5.3.2-1.

Table 5.3.2-1: URI query parameters supported by the POST method

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.3.2-2 and the response data structures and response codes specified in table 5.3.2-3.

Table 5.3.2-2: Data structures supported by the POST Request Body

Data type	P	Cardinality	Description
array(PolicyDataChangeNotification)	M	1..N	Notification of policy data changes.

Table 5.3.2-3: Data structures supported by the POST Response Body

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The reception of the notification is acknowledged.
NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

5.4 Data Model

5.4.1 General

This clause specifies the application data model supported by the API.

Table 5.4.1-1 specifies the data types defined for the Nudr_DataRepository for Policy Data service based interface protocol.

Table 5.4.1-1: Nudr_DataRepository specific Data Types for Policy Data

Data type	Section defined	Description	Applicability
AmPolicyData	5.4.2.2	Contains the AM policy data for a given subscriber.	
BdtData	5.4.2.9	Contains the background data transfer data.	
BdtDataPatch	5.4.2.27	Contains the modifiable background data transfer data	
BdtPolicyStatus	5.4.3.5	Contains the validation status for a negotiated BDT policy.	
DnnRouteSelectionDescriptor	5.4.2.20	Contains the route selector parameters per DNN.	
UpdatedItem	5.4.2.26	An updated resource fragment, represented by its location in a resource and its data type.	ConditionalSubscriptionwithPartialNotification
LimitIdToMonitoringKey	5.4.2.16	Contains the limit identifier and the corresponding monitoring key for a given S-NSSAI and DNN.	
MbsSessPolCtrlData	5.4.2.31	Contains MBS Session Policy Control Data.	
MbsSessPolDataId	5.4.2.32	Represents the identifier used to access the MBS Session Policy Control Data.	
NotificationItem	5.4.2.25	Contains the list of resource fragments included in the notification triggered by the modification of a given resource fragment.	ConditionalSubscriptionwithPartialNotification
OsId	5.4.3.2	Operating System supported by the UE.	
Periodicity	5.4.3.4	Indicates a type of time period.	
PlmnRouteSelectionDescriptor	5.4.2.18	Contains the route selectors for a serving PLMN.	
PolicyDataChangeNotification	5.4.2.11	Contains changed policy data for which notification was requested.	
PolicyDataForIndividualUe	5.4.2.28	Contains policy data sets for a given subscriber.	
PolicyDataSubscription	5.4.2.10	Identifies a subscription to policy data change notification.	
PolicyDataSubset	5.4.3.6	Indicates a policy data subset (e.g. AM policy data, SM policy data).	
SlicePolicyData	5.4.2.29	Contains network slice specific policy control information for an S-NSSAI.	
SlicePolicyDataPatch	5.4.2.30	Contains modifiable network slice specific policy control information for an S-NSSAI.	
SmPolicyData	5.4.2.5	Contains SM policy data for a subscriber.	
SmPolicyDataPatch	5.4.2.21	Contains modifiable SM policy data for a subscriber.	SessionManagementPolicyDataPatch
SmPolicyDnnData	5.4.2.15	Contains SM policy data for a DNN and S-NSSAI.	
SmPolicyDnnDataPatch	5.4.2.23	Contains modifiable SM policy data for a DNN and S-NSSAI.	SessionManagementPolicyDataPatch
SmPolicySnssaiData	5.4.2.14	Contains SM policy data for an S-NSSAI.	
SmPolicySnssaiDataPatch	5.4.2.22	Contains modifiable SM policy data for an S-NSSAI.	SessionManagementPolicyDataPatch
SnssaiRouteSelectionDescriptor	5.4.2.19	Contains the route selector parameters per SNSSAI.	
SponsorConnectivityData	5.4.2.8	Contains the sponsored data connectivity related information for a sponsor identifier.	
ResourceItem	5.4.2.24	Represents list of fragments of a resource, i.e. a list of subsets of resource data monitored for notification of data changes.	ConditionalSubscriptionwithPartialNotification
ItemPath	5.4.3.2	Represents a fragment of a resource, i.e. a subset of resource data.	ConditionalSubscriptionwithPartialNotification
TimePeriod	5.4.2.12	Contains the periodicity for the defined usage monitoring data limits.	
UePolicySetPatch	5.4.2.17	Contains the UE policy data that can be modified by the PCF.	
UePolicySection	5.4.2.3	Contains the UE policy section.	

UePolicySet	5.4.2.4	Contains the UE policy set for a given subscriber.	
UsageMonLevel	5.4.3.3	Indicates the level of the usage monitoring instance (PDU Session level or per Service)	
UsageMonData	5.4.2.7	Contains remain allowed usage data for a subscriber.	
UsageMonDataLimit	5.4.2.6	Contains usage monitoring control data for a subscriber.	
UsageMonDataScope	5.4.2.13	Contains SNSSAI and DNN combinations to which the UsageMonData instance belongs to.	

Table 5.4.1-2 specifies data types re-used by the Nudr_DataRepository for Policy Data service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nudr_DataRepository for Policy Data service based interface.

Table 5.4.1-2: Nudr_DataRepository re-used Data Types for Policy Data

Data type	Reference	Comments	Applicability
5Qi	3GPP TS 29.571 [7]	Indicates the 5G QoS Identifier.	
ArpPriorityLevel	3GPP TS 29.571 [7]	Indicates the allocation and retention priority level.	
BdtReferenceId	3GPP TS 29.122 [9]	Indicates the background data transfer reference ID for the transfer policy.	
BdtReferenceIdRm	3GPP TS 29.122 [9]	This data type is defined in the same way as the "BdtReferenceId" data type, but with the "nullable: true" property.	EnhancedBackgroundDataTransfer
BitRate	3GPP TS 29.571 [7]	String representing a bit rate that shall be formatted as follows: pattern: "^\d+(\.\d+)?(bps|Kbps|Mbps|Gbps|Tbps)$" Examples: "125 Mbps", "0.125 Gbps", "125000 Kbps".	
Bytes	3GPP TS 29.571 [7]	String with format "byte" as defined in OpenAPI Specification [3], i.e., base64-encoded characters.	
ChargingInformation	3GPP TS 29.512 [12]	The address(es), and if available, the instance ID and the set ID of the Charging Function.	
DateTime	3GPP TS 29.571 [7]	String with format "date-time" as defined in OpenAPI Specification [3].	
Dnn	3GPP TS 29.571 [7]	Data Network Name.	
MbsSessionId	3GPP TS 29.571 [7]	Contains the MBS Session Identifier.	
NetworkAreaInfo	3GPP TS 29.554 [13]	Represents a network area information.	
OperatorSpecificDataContainer	3GPP TS 29.505 [15]	Container for operator specific data	
PatchResult	3GPP TS 29.571 [7]		
PatchItem	3GPP TS 29.571 [7]	Data structure used for JSON patch.	
PduSessionType	3GPP TS 29.571 [7]	PDU Session Type.	
Pei	3GPP TS 29.571 [7]	Personal Equipment Identifier.	
PlmnId	3GPP TS 29.571 [7]	PLMN Identity.	
PresenceInfo	3GPP TS 29.571 [7]	Presence Reporting Area Information.	
ProblemDetails	3GPP TS 29.571 [7]	Used in error responses to provide more detailed information about an error.	
SliceMbr	3GPP TS 29.571 [7]	Contains the slice Maximum Bit Rate including UL and DL.	NSAC
Snssai	3GPP TS 29.571 [7]	Identifies the S-NSSAI.	
SscMode	3GPP TS 29.571 [7]	SSC mode.	
SupportedFeatures	3GPP TS 29.571 [7]	Used to negotiate the applicability of the optional features	
TrafficDescriptor	3GPP TS 29.122 [9]	Identifies the traffic descriptor of the background data.	
TransferPolicy	3GPP TS 29.554 [13]	Represents a transfer policy.	
Uinteger	3GPP TS 29.571 [7]	Unsigned Integer, i.e. only value 0 and integers greater than 0 are allowed.	
Uri	3GPP TS 29.571 [7]	String providing an URI.	
UsageThreshold	3GPP TS 29.122 [9]	Usage Thresholds (a data volume expected to be transferred per UE and/or time duration in seconds).	
VarUeId	3GPP TS 29.571 [7]	String represents the SUPI or GPSI.	

5.4.2 Structured data types

5.4.2.1 Introduction

This clause defines the structures to be used in resource representations.

5.4.2.2 Type AmPolicyData

Table 5.4.2.2-1: Definition of type AmPolicyData

Attribute name	Data type	P	Cardinality	Description
praInfos	map(PresenceInfo)	O	1..N	Presence reporting area information. Each PresenceInfo element shall include the Presence Reporting Area Identifier within the "praId" attribute and, for a UE-dedicated presence reporting area, shall also include the list of elements composing the presence reporting area. A "praId" may indicate a Presence Reporting Area Set. The "praId" attribute within the PresenceInfo data type shall also be the key of the map. The attribute "presenceState" shall not be present.
subscCats	array(string)	O	1..N	List of categories associated with the subscriber

5.4.2.3 Type UePolicySection

Table 5.4.2.3-1: Definition of type UePolicySection

Attribute name	Data type	P	Cardinality	Description
uePolicySectionInfo	Bytes	M	1	Contains the UE Policy Section Information for the given UPSI as specified in clause D.6.2 of 3GPP TS 24.501 [11].
upsi	string	M	1	Identifier for UE Policy Section. The format is represented in 3GPP TS 24.501 [11] clause D.6.2.

5.4.2.4 Type UePolicySet

Table 5.4.2.4-1: Definition of type UePolicySet

Attribute name	Data type	P	Cardinality	Description
praInfos	map(PresenceInfo)	O	1..N	Presence reporting area information. Each PresenceInfo element shall include the Presence Reporting Area Identifier within the "praId" attribute and, for a UE-dedicated presence reporting area, shall also include the list of elements composing the presence reporting area. A "praId" may indicate a Presence Reporting Area Set. The "praId" attribute within the PresenceInfo data type shall also be the key of the map. The attribute "presenceState" shall not be present.
subscCats	array(string)	O	1..N	List of categories associated with the subscriber
uePolicySections	map(UePolicySection)	O	1..N	Contains the UE Policy Sections. The UPSI (UE Policy Section Identifier) is used as the key in the map.
upsis	array(string)	O	1..N	List of identifiers for the "uePolicySections". The format of the UPSI is represented in 3GPP TS 24.501 [11] clause D.6.2
allowedRouteSelDescs	map(PlmnRouteSelectionDescriptor)	O	1..N	Contains allowed route selection descriptors per serving PLMN for a UE. The serving PLMN identifier is the key of the map.
andspInd	boolean	O	0..1	Indication of UE supporting ANDSP. True: The UE supports ANDSP; False: The UE does not support ANDSP.
pei	Pei	O	0..1	Personal Equipment Identifier.
osIds	array(OsId)	O	1..N	Identification of the UE Operating System(s).
suppFeat	SupportedFeatures	C	0..1	This IE represents a list of Supported features used as described in clause 5.6. This attribute shall be provided in the PUT request and in the response of successful resource creation, and in the HTTP GET response if the "supp-feat" attribute query parameter is included in the HTTP GET request. (NOTE).
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.
NOTE: In the HTTP request, it represents the set of features supported by the NF service consumer. In the HTTP response, it represents the set of features supported by both the NF service consumer and the UDR.				

5.4.2.5 Type SmPolicyData

Table 5.4.2.5-1: Definition of type SmPolicyData

Attribute name	Data type	P	Cardinality	Description	Applicability
smPolicySnssaiData	map(SmPolicySnssaiData)	M	1..N	Session Management Policy data per S-NSSAI for all the SNSSAIs of the subscriber. The key of the map is the S-NSSAI. (NOTE 2)	
umDataLimits	map(UsageMonDataLimit)	O	1..N	Contains a list of usage monitoring profiles associated with the subscriber. The limit identifier is used as the key of the map.	
umData	map(UsageMonData)	O	1..N	Contains the remaining allowed usage data associated with the subscriber. The limit identifier is used as the key in the map.	
suppFeat	SupportedFeatures	C	0..1	Indicates the list of negotiated supported features. This parameter shall be supplied by the UDR in the response to the GET request when the GET request includes the "supp-feat" query parameter and the UDR supports feature negotiation for Session Management Policy Data.	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	
NOTE 1: In this Release of the specification the "limitId" corresponds to the Monitoring Key. NOTE 2: The S-NSSAI value used as key of the map is encoded as a string as defined in 3GPP TS 29.571[7], clause 5.4.4.2.					

5.4.2.6 Type UsageMonDataLimit

Table 5.4.2.6-1: Definition of type UsageMonDataLimit

Attribute name	Data type	P	Cardinality	Description
limitId	string	M	1	Identifies the limit, i.e. the usage monitoring control instance
scopes	map(UsageMonDataScope)	O	1..N	Identifies the SNSSAI and DNN combinations to which the usage monitoring data limit applies. The S-NSSAI is the key of the map. (NOTE 2) If the "scopes" attribute is omitted, the limit applies to every S-NSSAI and DNN.
umLevel	UsageMonLevel	O	0..1	Indicates the level of the usage monitoring instance (PDU Session level or per Service).
startDate	DateTime	C	0..1	Start date and time when the usage monitoring instance applies. It shall be provided when the resetPeriod attribute is provided.
endDate	DateTime	O	0..1	End date and time when the usage monitoring instance applies.
usageLimit	UsageThreshold	O	0..1	Maximum allowed traffic volume/time usage for a usage monitoring instance. If a resetPeriod is specified, this limit applies to every period separately and not for the entire time between the start and end dates. So for example, if the resetPeriod is daily, the usageLimit is applicable for each day until the end date is specified.
resetPeriod	TimePeriod	O	0..1	Time period to reset the remaining allowed usage for a periodic usage monitoring instance (postpaid subscriptions). This attribute along with the startDate determine the actual absolute time at which usage needs to be reset.

NOTE 1: In this Release of the specification the "limitId" attribute is the Monitoring Key, and the "scopes" attribute shall always be present including one distinct SNSSAI and DNN pair.
NOTE 2: The S-NSSAI value used as key of the map is encoded as a string as defined in 3GPP TS 29.571[7], clause 5.4.4.2.

5.4.2.7 Type UsageMonData

Table 5.4.2.7-1: Definition of type UsageMonData

Attribute name	Data type	P	Cardinality	Description	Applicability
limitId	string	M	1	Identifies the limit as defined in the "limitId" attribute of the UsageMonDataLimit data type (see clause 5.4.2.6).	
scopes	map(UsageMonDataScope)	O	1..N	Identifies the SNSSAI and DNN combinations to which the remaining usage monitoring data applies. The S-NSSAI is the key of the map. (NOTE 3)	
umLevel	UsageMonLevel	O	0..1	Indicates the level of the usage monitoring instance (PDU Session level or per Service).	
allowedUsage	UsageThreshold	O	0..1	Remaining allowed traffic volume usage and/or resource time usage.	
resetTime	DateTime	O	0..1	Time to reset the allowed usage back to the usageLimit of the corresponding UsageMonDataLimit. This is computed based on the corresponding startDate, resetPeriod and endDate of the corresponding UsageMonDataLimit. The allowedUsage is only applicable if the resetTime is in the future relative to the current time.	
suppFeat	SupportedFeatures	C	0..1	This IE represents a list of Supported features used as described in clause 5.6. This attribute shall be provided in the PUT request and in the response of successful resource creation, and in the HTTP GET response if the "supp-feat" attribute query parameter is included in the HTTP GET request. (NOTE 2).	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	

NOTE 1: In this Release of the specification the "limitId" attribute is the Monitoring Key, and the "scopes" attribute shall always be present including one distinct SNSSAI and DNN pair.
NOTE 2: In the HTTP request, it represents the set of features supported by the NF service consumer. In the HTTP response, it represents the set of features supported by both the NF service consumer and the UDR.
NOTE 3: The S-NSSAI value used as key of the map is encoded as a string as defined in 3GPP TS 29.571 [7], clause 5.4.4.2.

5.4.2.8 Type SponsorConnectivityData

Table 5.4.2.8-1: Definition of type SponsorConnectivityData

Attribute name	Data type	P	Cardinality	Description
aspIds	array(string)	M	1..N	Identifies a list of Application Service Provider Identifiers associated with the sponsor identity.

5.4.2.9 Type BdtData

Table 5.4.2.9-1: Definition of type BdtData

Attribute name	Data type	P	Cardinality	Description
aspId	string	M	1	This IE contains an identity of an application service provider.
transPolicy	TransferPolicy	M	1	This IE contains the transfer policy.
bdtRefId	BdtReferenceId	O	0..1	This IE indicates the background data transfer reference ID for the transfer policy.
nwAreaInfo	NetworkAreaInfo	O	0..1	This IE represents a network area information.
numOfUes	Uinteger	O	0..1	This IE contains the number of UEs using the indicated transfer policy. It shall be present when available.
volPerUe	UsageThreshold	O	0..1	This IE contains a data volume expected to be transferred per UE for the indicated transfer policy. It shall be present when available.
dnn	Dnn	O	0..1	This IE identifies a DNN.
snssai	Snssai	O	0..1	This IE identifies a slice.
trafficDes	TrafficDescriptor	O	0..1	Contains the traffic descriptor of the background data.
bdtpStatus	BdtPolicyStatus	O	0..1	Contains the validation status for a negotiated BDT policy. It shall be included when available. Default value is "VALID" if omitted.
suppFeat	SupportedFeatures	C	0..1	This IE represents a list of Supported features used as described in clause 5.6. This attribute shall be provided in the PUT request and in the response of successful resource creation, and in the HTTP GET response if the "supp-feat" attribute query parameter is included in the HTTP GET request. (NOTE).
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.
NOTE: In the HTTP request, it represents the set of features supported by the NF service consumer. In the HTTP response, it represents the set of features supported by both the NF service consumer and the UDR.				

5.4.2.10 Type PolicyDataSubscription

Table 5.4.2.10-1: Definition of type PolicyDataSubscription

Attribute name	Data type	P	Cardinality	Description	Applicability
notificationUri	Uri	M	1	URI provided by the NF service consumer indicating where to receive the subscribed notifications from the UDR.	
notifId	string	C	0..1	Notification Correlation ID assigned by the NF service consumer. It shall be included when the "ConditionalSubscriptionwithPartialNotification" or the "ConditionalSubscriptionWithExcludeNotification" feature is supported.	ConditionalSubscriptionwithPartialNotification ConditionalSubscriptionWithExcludeNotification
monitoredResourceUris	array(Uri)	M	1..N	A set of URIs that identify the resources as defined in table 5.2.2-1 for which a modification triggers a notification. (NOTE 1)	
monResItems	array(ResourceItem)	O	1..N	When present, this IE indicates the trigger of the notification is conditioned to the data change occurs in one or more of the attributes contained in a resource fragment as defined by the ResourceItem data type, and that the triggered notification shall only include the resource fragment as defined by the ResourceItem data type. (NOTE 2)	ConditionalSubscriptionwithPartialNotification
excludedResItems	array(ResourceItem)	O	1..N	When present, this IE indicates the trigger of the notification is conditioned to the data change occurs in any attribute contained in the resource(s) identified by "monitoredResourceUris", except when the changed data is in one or more of the attributes contained in a resource fragment as defined by the ResourceItem data type. (NOTE 2) (NOTE 3) (NOTE 4)	ConditionalSubscriptionWithExcludeNotification
expiry	DateTime	C	0..1	This IE shall be included in a subscription response if, based on operator policy and taking into account the expiry time included in the request, the UDR needs to include an expiry time. This IE may be included in a subscription request. When present, this IE shall represent the time after which the subscription becomes invalid. The absence of this attribute in the subscription response means the subscription to be valid without an expiry time.	
supportedFeatures	SupportedFeatures	C	0..1	Used to negotiate the applicability of the optional features. This attribute shall be provided in the POST request and in the response of successful resource creation.	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	

NOTE 1: Neither the resource URI of the IndividualPolicyDataSubscription resource nor the resource URI of the PolicyDataSubscriptions resource shall be included in the "monitoredResourceUris" attribute. NOTE 2: If both ConditionalSubscriptionwithPartialNotification and ConditionalSubscriptionWithExcludeNotification are supported, the fragments defined in "monResItems" attribute and in "excludedResItems" attribute shall refer to different resources defined in "monitoredResourceUris". NOTE 3: When "excludedResItems" attribute is included, and a change is detected in the monitored part of a resource (i.e., in those properties not included in the "excludedResItems" attribute), the triggered notification shall include the complete resource representation. NOTE 4: The "excludedResItems" attribute may only be used with the OperatorSpecificData resource. E.g. if the OperatorSpecificData resource contains a map with two elements, one with key "operatorSpecificData1" and the other one with key "operatorSpecificData2", to excude the notification of changes in the "operatorSpecificData1" element the "excludedResItems" attribute will contain the value "/operatorSpecificData1" within the "items" array.	

5.4.2.11 Type PolicyDataChangeNotification

Table 5.4.2.11-1: Definition of type PolicyDataChangeNotification

Attribute name	Data type	P	Cardinality	Description	Applicability
amPolicyData	AmPolicyData	O	0..1	Access and Mobility Policy Data, if changed and notification was requested.	
uePolicySet	UePolicySet	O	0..1	UE Policy Set, if changed and notification was requested.	
plmnUePolicySet	UePolicySet	O	0..1	PLMN UE Policy Set, if changed and notification was requested.	
smPolicyData	SmPolicyData	O	0..1	Session Management Policy Data, if changed and notification was requested.	
usageMonData	UsageMonData	O	0..1	Usage Monitoring Data, if changed and notification was requested.	
SponsorConnectivityData (NOTE 4)	SponsorConnectivityData	O	0..1	Sponsor data connectivity profile information, if changed and notification was requested.	
bdtData	BdtData	O	0..1	Background Data Transfer Data, if changed and notification was requested.	
opSpecData	OperatorSpecificDataContainer	O	0..1	Operator Specific Data, if changed and notification was requested. It may only be used when the receiver of the notification is able to univocally identify the changed operator specific data. (NOTE 3)	
opSpecDataMap	map(OperatorSpecificDataContainer)	O	1..N	Operator Specific Data resource data, if changed and notification was requested. The key of the map is operator specific data element name and the value is the operator specific data of the UE.	OpSpecDataMapNotification
ueId	VarUeId	C	0..1	Represents the UE subscription identifier SUPI or GPSI. It shall only be present when the "amPolicyData", "uePolicySet", "smPolicyData", "opSpecData" and/or "usageMonData" attribute is present.	
sponsorId	string	C	0..1	Represents the sponsor identity. It shall only be present when the "sponsorConnectivityData" attribute is present.	
bdtRefId	BdtReferenceId	C	0..1	Represents the BDT reference identifier. It shall only be present when the "bdtData" attribute is present.	
usageMonId	string	C	0..1	Represents the unique identifier of the individual SM Policy usage monitoring resource. It shall only be present when the "usageMonData" attribute is present.	
plmnId	PlmnId	C	0..1	Represents the PLMN identifier. It shall only be present when the "plmnUePolicySet" attribute is present.	
delResources	array(Uri)	O	1..N	The resources, as defined in table 5.2.2-1, if removed from UDR and notification on resource data change was requested. (NOTE 2)	ResourceRemovalNotificationPolicyData
notifId	string	C	0..1	Notification Correlation ID assigned by the NF service consumer. It shall be included when the "ConditionalSubscriptionwithPartialNotification" feature is supported.	ConditionalSubscriptionwithPartialNotification

reportedFragments	array(NotificationItem)	O	1..N	This attribute contains the resource fragments indicated in the "monResItems" attribute of the PolicyDataSubscription data type.	ConditionalSubscriptionwithPartialNotification
slicePolicyData	SlicePolicyData	O	0..1	Network slice specific policy control data for an S-NSSAI, if changed and notification was requested.	
snssai	Snssai	C	0..1	Represents the S-NSSAI identifier, the unique identifier of the SlicePolicyControlData resource. It shall only be present when the "slicePolicyData" attribute is present.	

NOTE 1: When the "ResourceRemovalNotificationPolicyData" feature and/or "ConditionalSubscriptionwithPartialNotification" feature is not supported or supported but the applicable attribute(s) are not provided, at least one of the "amPolicyData", "uePolicySet", "smPolicyData", "usageMonData", "SponsorConnectivityData", "bdtData", "opSpecData", "plmnUePolicySet" or "slicePolicyData" shall be present.
NOTE 2: When the "ResourceRemovalNotificationPolicyData" feature is supported, and the "delResources" attribute is present, the attributes that correspond with the values included in the "delResources" attribute shall be omitted. E.g., when the "delResources" attribute includes the value "../policy-data/ues/{ueId}/am-data" the attribute "amPolicyData" shall be omitted.
NOTE 3: This attribute should not be used if the receiver of the notification is not able to univocally identify the changed operator specific data, and "opSpecDataMap" should be used instead.
NOTE 4: The attribute does not follow the related naming convention (i.e. "lowerCamel") defined in clause 5.1.4 of 3GPP TS 29.501 [7]. This attribute is however kept as currently defined in this specification for backward compatibility considerations.

5.4.2.12 Type TimePeriod

Table 5.4.2.12-1: Definition of type TimePeriod

Attribute name	Data type	P	Cardinality	Description	Applicability
period	Periodicity	M	1	Indicates whether the periodicity is "YEARLY", "MONTHLY", "WEEKLY", "DAILY" or "HOURLY".	
maxNumPeriod	Uinteger	O	0..1	Indicates the maximum number of periods after which the usage monitoring instance does not apply. If omitted, there is no limit in the number of periods.	

5.4.2.13 Type UsageMonDataScope

Table 5.4.2.13-1: Definition of type UsageMonDataScope

Attribute name	Data type	P	Cardinality	Description	Applicability
snssai	Snssai	M	1	Represents the SNSSAI.	
dnn	array(Dnn)	O	1..N	Represents the Dnn. If omitted, the usage monitoring scope applies to all the DNNs.	
NOTE: In this Release of the specification the "dnn" attribute shall always be present with one and only one entry.					

5.4.2.14 Type SmPolicySnssaiData

Table 5.4.2.14-1: Definition of type SmPolicySnssaiData

Attribute name	Data type	P	Cardinality	Description	Applicability
snssai	Snssai	M	1	S-NSSAI associated with the data.	
smPolicyDnnData	map(SmPolicyDnn Data)	O	1..N	Session Management Policy data per DNN for all the DNNs of the indicated S-NSSAI. The key of the map is the DNN.	
ueSliceMbr	SliceMbr	O	0..1	Subscribed UE-Slice-MBR for the indicated S-NSSAI.	NSAC

5.4.2.15 Type SmPolicyDnnData

Table 5.4.2.15-1: Definition of type SmPolicyDnnData

Attribute name	Data type	P	Cardinality	Description	Applicability
dnn	Dnn	M	1	DNN associated with the data	
allowedServices	array(string)	O	1..N	List of subscriber's allowed service identifiers	
subscCats	array(string)	O	1..N	List of categories associated with the subscriber	
gbrUl	BitRate	O	0..1	Maximum aggregate UL bitrate that can be provided across all GBR QoS Flows in the DNN	
gbrDl	BitRate	O	0..1	Maximum aggregate DL bitrate that can be provided across all GBR QoS Flows in the DNN	
adcSupport	boolean	O	0..1	Indicates whether application detection and control is enabled for the PDU session. True: Application detection and control is enabled for the PDU session; False: Application detection and control is not enabled for the PDU session. The absence of this attribute means that ADC support is not provisioned for the UE and PDU session.	
subscSpendingLimits	boolean	O	0..1	Indicates whether the PCF must enforce policies based on subscriber spending limits. True: Spending limit control is enabled; False: Spending limit control is not enabled. The absence of this attribute means that spending limit control is not provisioned for the UE and PDU session.	
ipv4Index	IpIndex	O	0..1	Information that identifies which IP pool or external server is used to allocate the IPv4 address.	
ipv6Index	IpIndex	O	0..1	Information that identifies which IP pool or external server is used to allocate the IPv6 address.	
offline	boolean	O	0..1	Indicates whether the offline charging is applicable to the PDU session. True: Offline charging is applicable to the PDU session; False: Offline charging is not applicable to the PDU session. The absence of this attribute means that the charging method is not provisioned for the UE and PDU session.	
online	boolean	O	0..1	Indicates whether the online charging is applicable to the PDU session. True: Online charging is applicable to the PDU session; False: Online charging is not applicable to the PDU session. The absence of this attribute means that the charging method is not provisioned for the UE and PDU session.	
chfInfo	ChargingInformation	O	0..1	The address(es) and, if available, the CHF instance ID and the CHF set ID of the Charging Function. (NOTE)	
refUmDataLimitIds	map(LimitIdToMonitoringKey)	O	1..N	A reference to the "UsageMonDataLimit" or "UsageMonData" instances for this DNN and SNSSAI that may also include the related monitoring key(s). The key of the map is the limit identifier.	

mpsPriority	boolean	O	0..1	True: Indicates subscription to the MPS priority service; priority applies to all traffic on the PDU Session. False: MPS priority service is not subscribed. The absence of this attribute means that MPS priority is not provisioned for the UE and PDU session.	
mcsPriority	boolean	O	0..1	True: Indicates subscription to the MCS priority service; priority applies to all traffic on the PDU Session. False: MCS priority service is not subscribed. The absence of this attribute means that MCS priority is not provisioned for the UE and PDU session.	
imsSignallingPrio	boolean	O	0..1	True: Indicates subscription to the IMS signalling priority service; priority only applies to IMS signalling traffic. False: IMS signalling priority service is not subscribed. The absence of this attribute means that IMS signalling priority is not provisioned for the UE and PDU session.	
mpsPriorityLevel	integer	O	0..1	Relative priority level for the multimedia priority services	
mcsPriorityLevel	integer	O	0..1	Relative priority level for the mission critical services	
praInfos	map(PresenceInfo)	O	1..N	Presence reporting area information. Each PresenceInfo element shall include the Presence Reporting Area Identifier within the "praId" attribute and, for a UE-dedicated presence reporting area, shall also include the list of elements composing the presence reporting area. A "praId" may indicate a Presence Reporting Area Set. The "praId" attribute within the PresenceInfo data type shall also be the key of the map. The attribute "presenceState" shall not be present.	
bdtRefIds	map(BdtReferenceIdRm)	O	1..N	Identifies transfer policies of background data transfer. Any string value can be used as a key of the map.	EnhancedBackgroundDataTransfer
locRoutNotAllowed	boolean	O	0..1	Identifies whether AF influence on traffic routing is allowed or not. True: if no local routing is allowed; False: local routing is allowed. The absence of this attribute means that AF influence on traffic routing is not provisioned for the UE and PDU session.	
NOTE: When the feature "CHFsetSupport" is supported, the "secondaryChfAddress" may be omitted (see 3GPP TS 29.512 [12], clause 4.2.2.3.1).					

5.4.2.16 Type LimitIdToMonitoringKey

Table 5.4.2.16-1: Definition of type LimitIdToMonitoringKey

Attribute name	Data type	P	Cardinality	Description	Applicability
limitId	string	M	1	Limit Identifier.	
monkey	array(string)	O	1..N	Monitoring Keys.	
NOTE: In this Release of the specification there is only one Monitoring Key per Limit Identifier					

5.4.2.17 Type UePolicySetPatch

Table 5.4.2.17-1: Definiton of type UePolicySetPatch

Attribute name	Data type	P	Cardinality	Description
uePolicySections	map(UePolicySection)	O	1..N	Contains the UE Policy Sections. The UPSI (UE Policy Section Identifier) is used as the key in the map.
upsis	array(string)	O	1..N	List of identifiers for the "uePolicySections". The format of the UPSI is represented in 3GPP TS 24.501 [11] clause D.6.2.
andspInd	boolean	O	0..1	Indication of UE supporting ANDSP.
pei	Pei	O	0..1	Personal Equipment Identifier.
osIds	array(OsId)	O	1..N	Identification of the UE Operating System(s).

5.4.2.18 Type PlmnRouteSelectionDescriptor

Table 5.4.2.18-1: Definition of type PlmnRouteSelectionDescriptor

Attribute name	Data type	P	Cardinality	Description
servingPlmn	PlmnId	M	1	Identity of the serving PLMN.
snssaiRouteSelDescs	array(SnssaiRouteSelectionDescriptor)	O	1..N	Contains the route selection descriptor parameters (DNNs, PDU session types, SSC modes and the ATSSS information) per SNSSAI.

5.4.2.19 Type SnssaiRouteSelectionDescriptor

Table 5.4.2.19-1: Definition of type SnssaiRouteSelectionDescriptor

Attribute name	Data type	P	Cardinality	Description
snssai	Snssai	M	1	SNSSAI.
dnnRouteSelDescs	array(DnnRouteSelectionDescriptor)	O	1..N	Contains the route selection descriptor parameters (PDU session types, SSC modes and the ATSSS information) per DNN.

5.4.2.20 Type DnnRouteSelectionDescriptor

Table 5.4.2.20-1: Definition of type DnnRouteSelectionDescriptor

Attribute name	Data type	P	Cardinality	Description
dnn	Dnn	M	1	DNN.
sscModes	array(SscMode)	O	1..N	Contains the allowed SSC modes for the traffic to this DNN.
pduSessTypes	array(PduSessionType)	O	1..N	Contains the allowed PDU Session types for the traffic to this DNN.
atsssInfo	boolean	O	0..1	Indicates whether MA PDU session establishment is allowed for this DNN. True: MA PDU session establishment is allowed for this DNN. False: MA PDU session establishment is not allowed for this DNN. The absence of this attribute means that the ATSSS information is not provisioned for the UE and PDU session.

5.4.2.21 Type SmPolicyDataPatch

Table 5.4.2.21-1: Definition of type SmPolicyDataPatch

Attribute name	Data type	P	Cardinality	Description	Applicability
umData	map(UsageMonData)	O	1..N	Contains the remaining allowed usage data associated with the subscriber. The value of the limit identifier is used as the key in the map.	
smPolicySnssaiData	map(SmPolicySnssaiDataPatch)	O	1..N	Modifiable Session Management Policy data per S-NSSAI for all the SNSSAIs of the subscriber. The key of the map is the S-NSSAI. (NOTE 2)	
NOTE 1: In this Release of the specification the "limitId" corresponds to the Monitoring Key. NOTE 2: The S-NSSAI value used as key of the map is encoded as a string as defined in 3GPP TS 29.571[7], clause 5.4.4.2.					

5.4.2.22 Type SmPolicySnssaiDataPatch

Table 5.4.2.22-1: Definition of type SmPolicySnssaiDataPatch

Attribute name	Data type	P	Cardinality	Description	Applicability
snssai	Snssai	M	1	S-NSSAI associated with the data.	
smPolicyDnnData	map(SmPolicyDnnDataPatch)	O	1..N	Modifiable Session Management Policy data per DNN for all the DNNs of the indicated S-NSSAI. The key of the map is the DNN.	

5.4.2.23 Type SmPolicyDnnDataPatch

Table 5.4.2.23-1: Definition of type SmPolicyDnnDataPatch

Attribute name	Data type	P	Cardinality	Description	Applicability
dnn	Dnn	M	1	DNN associated with the data	
bdtRefIds	map(BdtReferenceIdRm)	O	1..N	Updated transfer policies of background data transfer. Any string value can be used as a key of the map.	EnhancedBackgroundDataTransfer

5.4.2.24 Type ResourceItem

Table 5.4.2.24-1: Definition of type ResourceItem

Attribute name	Data type	P	Cardinality	Description	Applicability
monResourceUri	Uri	M	1	Represents the URI that identifies the monitored resource as defined in table 5.2.2-1. (NOTE)	
items	array(ItemPath)	M	1..N	Represents fragments of the resource identified by the "monResourceUri" attribute, i.e. a subset of resource data, for which a modification triggers a notification.	
NOTE: The resource URI included in the "monResourceUri" attribute shall be one of the resource URIs included in the "monitoredResourceUris" attribute of the PolicyDataSubscription data type.					

5.4.2.25 Type NotificationItem

Table 5.4.2.25-1: Definition of type NotificationItem

Attribute name	Data type	P	Cardinality	Description	Applicability
resourceId	Uri	M	1	This IE contains the URI of the resource which has been changed.	
notifItems	array(UpdatedItem)	M	1..N	This IE contains the fragments of the resource identified by the "resourceId" attribute which have been updated or requested to be notified.	

5.4.2.26 Type UpdatedItem

Table 5.4.2.26-1: Definition of type UpdatedItem

Attribute name	Data type	P	Cardinality	Description	Applicability
item	ItemPath	M	1	This IE contains a JSON pointer value (as defined in IETF RFC 6901 [20]) that references a target location within the resource that identifies the resource fragment.	
value	Any type	M	1	This IE indicates the value of the resource fragment located at the target location within the resource specified in the "item" attribute. The data type of this attribute shall be the same as the type of the attribute or element located in the path. The null value shall be allowed.	

5.4.2.27 Type BdtDataPatch

Table 5.4.2.27-1: Definition of type BdtDataPatch

Attribute name	Data type	P	Cardinality	Description
bdtpStatus	BdtPolicyStatus	O	0..1	Contains the validation status for a negotiated BDT policy. It shall be included when the BDT policy is re-negotiated.
transPolicy	TransferPolicy	O	0..1	This IE contains the transfer policy.

5.4.2.28 Type PolicyDataForIndividualUe

Table 5.4.2.28-1: Definition of type PolicyDataForIndividualUe

Attribute name	Data type	P	Cardinality	Description	Applicability
uePolicyDataSet	UePolicySet	O	0..1	The IE contains UE policies.	
smPolicyDataSet	SmPolicyData	O	0..1	The IE contains SM policies.	
amPolicyDataSet	AmPolicyData	O	0..1	The IE contains AM policies.	
umData	map(UsageMonData)	C	1..N	The IE contains UM policies. The value of the limit identifier is used as the key of the map. (NOTE)	
operatorSpecificDataSet	map(OperatorSpecificDataContainer)	O	1..N	The IE contains operator specific data. The key of the map is operator specific data element name and the value is the operator specific data of the UE.	
NOTE: The "umData" attribute shall be absent when the smPolicyDataSet attribute is present and includes the "umData" attribute.					

5.4.2.29 Type SlicePolicyData

Table 5.4.2.29-1: Definition of type SlicePolicyData

Attribute name	Data type	P	Cardinality	Description	Applicability
mbrUl	BitRate	O	0..1	Maximum aggregate ULdata rate that can be provided across all GBR and Non-GBR QoS Flows in the S-NSSAI. (NOTE 1)	
mbrDl	BitRate	O	0..1	Maximum aggregate DL data rate that can be provided across all GBR and Non-GBR QoS Flows in the S-NSSAI. (NOTE 1)	
remainMbrUl	BitRate	O	0..1	The remaining maximum aggregate UL data rate that can be provided across all GBR and Non-GBR QoS Flows in the S-NSSAI. (NOTE 2) It is set to the maximum allowed aggregate UL data rate across all GBR and Non-GBR QoS Flows in the S-NSSAI when the system is initially configured based on operator policies.	
remainMbrDl	BitRate	O	0..1	The remaining maximum aggregate DL data rate that can be provided across all GBR and Non-GBR QoS Flows in the S-NSSAI. (NOTE 2) It is set to the maximum allowed aggregate DL data rate across all GBR and Non-GBR QoS Flows in the S-NSSAI when the system is initially configured based on operator policies.	
suppFeat	SupportedFeatures	C	0..1	Indicates the list of negotiated supported features. It shall be provided in the HTTP GET response if the "supp-feat" attribute query parameter is included in the HTTP GET request.	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	
NOTE 1: The "mbrUl" and "mbrDl" attributes are only used for limiting data rate per network slice with assistance of the NWDAF. NOTE 2: The "remainMbrUl" and "remainMbrDl" attributes are only used for limiting data rate per network slice with PCF based monitoring.					

5.4.2.30 Type SlicePolicyDataPatch

Table 5.4.2.30-1: Definition of type SlicePolicyDataPatch

Attribute name	Data type	P	Cardinality	Description	Applicability
remainMbrUl	BitRate	O	0..1	The remaining maximum aggregate UL bitrate that can be provided across all GBR and Non-GBR QoS Flows in the S-NSSAI.	
remainMbrDl	BitRate	O	0..1	The remaining maximum aggregate DL bitrate that can be provided across all GBR and Non-GBR QoS Flows in the S-NSSAI.	
NOTE: At least one of the attributes shall be included.					

5.4.2.31 Type MbsSessPolCtrlData

Table 5.4.2.31-1: Definition of type MbsSessPolCtrlData

Attribute name	Data type	P	Cardinality	Description	Applicability
5qis	array(5Qi)	O	1..N	Contains the list of allowed 5QI(s) for an MBS PCC rule of the MBS Session.	
maxMbsArpLevel	ArpPriorityLevel	O	0..1	Contains the highest ARP level value allowed for any MBS PCC rule of the MBS Session.	
maxMbsSessionAmbr	BitRate	O	0..1	Contains the maximum MBS Session-AMBR for all non-GBR QoS flows of an MBS session in the downlink.	
maxGbr	BitRate	O	0..1	Contains the maximum aggregated bitrate that can be provided across all the GBR QoS Flows for an MBS session in the downlink.	
suppFeat	SupportedFeatures	C	0..1	Indicates the list of negotiated supported features. This parameter shall be provided if feature negotiation needs to take place.	

5.4.2.32 Type MbsSessPolDataId

Table 5.4.2.32-1: Definition of type MbsSessPolDataId

Attribute name	Data type	P	Cardinality	Description	Applicability
mbsSessionId	MbsSessionId	C	0..1	Contains the source specific multicast address used as MBS session identifier. (NOTE)	
afAppId	string	C	0..1	Contains the AF application identifier. (NOTE)	
NOTE: Either the "mbsSessionId" attribute or the "afAppId" attribute shall be provided.					

5.4.3 Simple data types and enumerations

5.4.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

5.4.3.2 Simple data types

The simple data types defined in table 5.4.3.2-1 shall be supported.

Table 5.4.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability
IpIndex	integer	Information that identifies which IP pool or external server is used to allocate the IP address.	
ItemPath	string	It contains one JSON pointer value (as defined in IETF RFC 6901 [20]) that references a target location within the resource. E.g. "/smPolicySnssaiData/<snssai>" represents the subscription to data changes in the SmPolicyData data document, in particular to the element of the "smPolicySnssaiData" map whose key is identified by the <snssai> value. E.g. 2 "/smPolicySnssaiData/<snssai>/smPolicyDnnData/<dnn>" represents the subscription to data changes in the SmPolicyData data document, in particular to the element of the "smPolicySnssaiData" map whose key is identified by the <snssai> value and, within this one, to the element of the "smPolicyDnnData" map whose key is identified by the <dnn> value. (NOTE)	ConditionalSubscriptionwithPartialNotification
OsId	string	Operating System of the served UE. It contains a Universally Unique IDentifier (UUID) as specified in IETF RFC 4122 [17]. The format of the OS Id is represented in 3GPP TS 24.501 [11].	
NOTE: The S-NSSAI value used as key of the map is encoded as a string as defined in 3GPP TS 29.571[7], clause 5.4.4.2.			

5.4.3.3 Enumeration: UsageMonLevel

The enumeration represents the usage monitoring level. It shall comply with the provisions defined in table 5.4.3.3-1.

Table 5.4.3.3-1: Enumeration UsageMonLevel

Enumeration value	Description
SESSION_LEVEL	The usage monitoring is applicable for PDU session level.
SERVICE_LEVEL	The usage monitoring is applicable for service level.

5.4.3.4 Enumeration: Periodicity

The enumeration represents the time period. It shall comply with the provisions defined in table 5.4.3.4-1.

Table 5.4.3.4-1: Enumeration Periodicity

Enumeration value	Description
YEARLY	Time period is yearly
MONTHLY	Time period is monthly
WEEKLY	Time period is weekly
DAILY	Time period is daily
HOURLY	Time period is hourly

5.4.3.5 Enumeration: BdtPolicyStatus

The enumeration represents the validation status of the BDT policy. It shall comply with the provisions defined in table 5.4.3.5-1.

Table 5.4.3.5-1: Enumeration BdtPolicyStatus

Enumeration value	Description
INVALID	This value is used to indicate that the BDT policy is invalid while the BDT policy re-negotiation is ongoing because the network performance has degraded.
VALID	This value is used to indicate that the BDT policy is valid.

5.4.3.6 Enumeration: PolicyDataSubset

This enumeration indicates the policy data subsets (e.g. AM policy data, SM policy data, etc.). It shall comply with the provisions of table 5.4.3.6-1.

Table 5.4.3.6-1: Enumeration PolicyDataSubset

Enumeration value	Description
AM_POLICY_DATA	AM policy data.
SM_POLICY_DATA	SM policy data.
UE_POLICY_DATA	UE policy data.
UM_DATA	Usage Monitoring data.
OPERATOR_SPECIFIC_DATA	Operator specific data.

5.5 Error handling

5.5.1 General

For the Nudr_DataRepository service API for Policy Data, HTTP error responses shall be supported as specified in clause 4.8 of 3GPP TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of 3GPP TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of 3GPP TS 29.500 [4].

In addition, the requirements in the following clauses are applicable for the Nudr_DataRepository service API for Policy Data.

5.5.2 Protocol Errors

No specific procedures for the Nudr_DataRepository service API for Policy Data are specified.

5.5.3 Application Errors

The application errors defined for the Nudr_DataRepository service API for Policy Data are listed in table 5.5.3-1.

Table 5.5.3-1: Application errors

Application Error	HTTP status code	Description
NOTE: Including a "ProblemDetails" data structure with the "cause" attribute in the HTTP response is optional unless explicitly mandated by other clauses in this document.		

The application error handling shall follow 3GPP TS 29.504 [6] clause 6.1.6 unless explicitly specified in this document.

5.6 Feature negotiation

The optional features in table 6.1.8-1 of 3GPP TS 29.504 [6] are defined for the Nudr_DataRepository API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of 3GPP TS 29.500 [4].

6 Usage of Nudr_DataRepository Service API for Application Data

6.1 Introduction

The following clauses specify the usage of Nudr_DataRepository service for application data. The principles specified in 3GPP TS 29.504 [6] are followed unless explicitly specified otherwise in the following clauses.

6.2 Resources

6.2.1 Overview

This clause specifies the resources for the usage of the Unified Data Repository service for Application Data.

6.2.2 Resource Structure

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.2.2-1 depicts the resource URIs structure for the Nudr_DataRepository API for application data

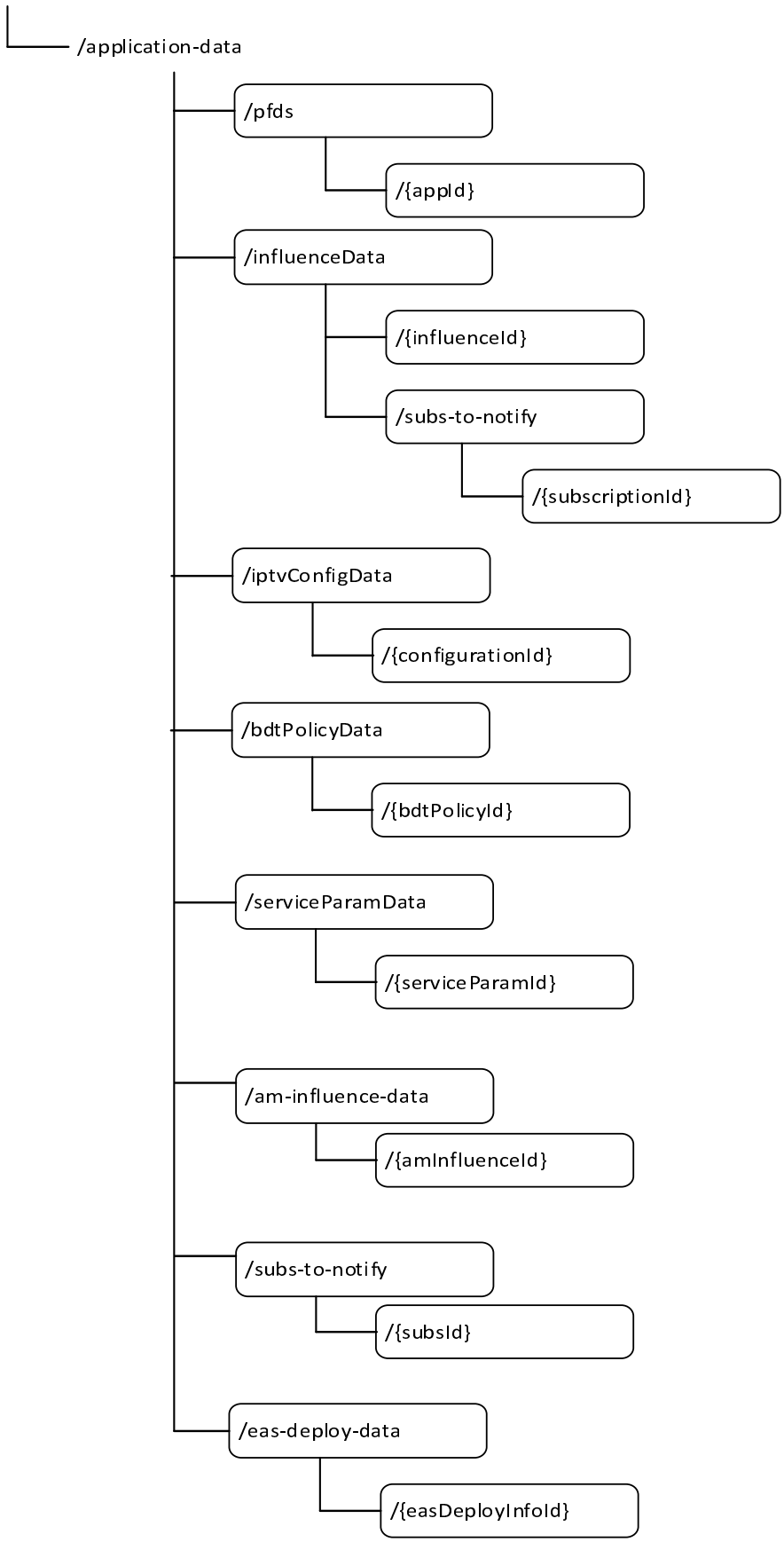

/application-data
/pfds
/{appId}
/influenceData
/{influenceId}
/subs-to-notify
/{subscriptionId}
/iptvConfigData
/{configurationId}
/bdtPolicyData
/{bdtPolicyId}
/serviceParamData
/{serviceParamId}
/am-influence-data
/{amInfluenceId}
/subs-to-notify
/{subsId}
/eas-deploy-data
/{easDeployInfoId}

Figure 6.2.2-1: Resource URI structure of the Nudr_DataRepository API for application data

Table 6.2.2-1 provides an overview of the resources and applicable HTTP methods.

Table 6.2.2-1: Resources and methods overview

Resource name	Resource URI	HTTP method or custom operation	Description
PFD Data	/application-data/pfds	GET	Retrieve PFDs for application identifier(s) identified by query parameter(s). Retrieve PFDs for all application identifier(s) if no query parameter is included in the Request URI.
Individual PFD Data	/application-data/pfds/{appId}	GET	Retrieve the corresponding PFDs of the specified application identifier.
		DELETE	Delete the corresponding PFDs of the specified application identifier.
		PUT	Create or update the corresponding PFDs for the specified application identifier.
Influence Data	/application-data/influenceData (NOTE)	GET	Retrieve the Session Influence Data of given services, S-NSSAIs and DNNs or Internal Group Identifiers or SUPIs.
Individual Influence Data	/application-data/influenceData/{influenceId} (NOTE)	PUT	Create an individual Influence Data resource identified by {influenceId}, or modify all of the properties of an individual Influence Data resource identified by {influenceId}.
		PATCH	Modify part of the properties of an individual Influence Data resource identified by {influenceId}.
		DELETE	Delete an individual Influence Data resource identified by {influenceId}.
Influence Data Subscription	/application-data/influenceData/subs-to-notify (NOTE)	POST	Create a new Individual Influence Data Subscription resource.
		GET	Read subscriptions for a given S-NSSAI and DNN or Internal Group Identifier or SUPI.
Individual Influence Data Subscription	/application-data/influenceData/subs-to-notify/{subscriptionId} (NOTE)	GET	Get an existing individual Influence Data Subscription resource identified by {subscriptionId}.
		PUT	Modify an existing individual Influence Data Subscription resource identified by {subscriptionId}.
		DELETE	Delete an individual Influence Data Subscription resource identified by {subscriptionId}.
Applied BDT Policy Data	/application-data/bdtPolicyData (NOTE)	GET	Retrieve the applied BDT policy data.
Individual Applied BDT Policy Data	/application-data/bdtPolicyData/{bdtPolicyId} (NOTE)	PUT	Create an individual applied BDT Policy Data resource identified by {bdtPolicyId}.
		PATCH	Modify BDT Reference Id of an individual applied BDT Policy Data resource identified by {bdtPolicyId}.
		DELETE	Delete an individual applied BDT Policy Data resource identified by {bdtPolicyId}.
IPTV Configurations	/application-data/iptvConfigData (NOTE)	GET	Retrieve IPTV configurations for configuration identifier(s), given S-NSSAI(s) and DNN(s), or SUPIs or Internal Group Identifiers
Individual IPTV Configuation	/application-data/iptvConfigData/{configurationId} (NOTE)	PUT	Create an Individual IPTV Configuration resource identified by {configurationId}, or modify all the properties of an Individual IPTV Configuration resource identified by {configurationId}.
		PATCH	Modify some properties of an Individual IPTV Configuration resource identified by {configurationId}.

		DELETE	Delete an Individual IPTV Configuration resource identified by {configurationId}
Service Parameter Data	/application-data/serviceParamData (NOTE)	GET	Retrieve the Service Parameter Data of given services, S-NSSAIs and DNNs or Internal Group Identifiers or SUPIs.
Individual Service Parameter Data	/application-data/serviceParamData/{serviceParamId} (NOTE)	PUT	Create an individual Service Parameter Data resource identified by {serviceParamId}, or modify all of the properties of an individual Service Parameter Data resource identified by {serviceParamId}.
		PATCH	Modify part of the properties of an individual Service Parameter Data resource identified by {serviceParamId}.
		DELETE	Delete an individual Service Parameter Data resource identified by {serviceParamId}.
AM Influence Data	/application-data/am-influence-data	GET	Retrieve the AM Influence Data of given S-NSSAIs and DNNs and/or Internal Group Identifiers or SUPIs.
Individual AM Influence Data	/application-data/am-influence-data/{amInfluenceId}	PUT	Create an individual AM Influence Data resource identified by {amInfluenceId}, or modify all of the properties of an individual AM Influence Data resource identified by {amInfluenceId}.
		PATCH	Modify part of the properties of an individual AM Influence Data resource identified by {amInfluenceId}.
		DELETE	Delete an individual AM Influence Data resource identified by {amInfluenceId}.
ApplicationDataSubscriptions	/application-data/subs-to-notify	POST	Create a subscription to receive notification of application data changes.
		GET	Read all the subscriptions, or subscriptions for given S-NSSAI and DNN or Internal Group Identifier or SUPI.
IndividualApplicationDataSubscription	/application-data/subs-to-notify/{subsId}	PUT	Modify a subscription to receive notification of application data changes identified by {subsId}.
		DELETE	Delete a subscription identified by {subsId}.
		GET	Read an existing individual Subscription resource identified by {subsId}.
EAS Deployment Information Data	/application-data/eas-deploy-data	GET	Retrieve the EAS Deployment Information Data, given DNN(s), S-NSSAI(s), Application ID or Internal Group Identifiers.
Individual EAS Deployment Information Data	/application-data/eas-deploy-data/{easDeployInfoId}	GET	Read an existing individual EAS Deployment Data identified by {easDeployInfoId}.
		PUT	Create an individual EAS deployment information Data resource identified by {easDeployInfoId} or Update an individual EAS Deployment Data resource identified by {easDeployInfoId}.
		DELETE	Delete an individual EAS Deployment Data resource identified by {easDeployInfoId}.
NOTE: The path segment does not follow the related naming convention defined in 3GPP TS 29.501 [5]. The path segment is kept though as defined in the current specification for backward compatibility considerations.			

6.2.3 Resource: PFD Data

6.2.3.1 Description

The PFD Data resource represents all the PFD Datas to the Nudr_DataRepository Service at a given UDR.

6.2.3.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/pfds**

Table 6.2.3.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

6.2.3.3 Resource Standard Methods

6.2.3.3.1 GET

This method shall support the URI query parameters specified in table 6.2.3.3.1-1.

Table 6.2.3.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
appId	array(ApplicationId)	O	1..N	Contains the information of the application identifier(s) for the querying PFD Data resource. If the appId query paremeter is not included in the URI, the query applies to all application identifier(s) for the querying PFD Data resource. If the appId array contains one or more ApplicationId entries, the UDR returns an Individual PFD Data resource representation for each matched ApplicationId.
supp-feat	SupportedFeatures	O	0..1	The features supported by the NF service consumer.

This method shall support the request data structures specified in table 6.2.3.3.1-2 and the response data structures and response codes specified in table 6.2.3.3.1-3.

Table 6.2.3.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.3.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
array(PfdDataForAppExt)	M	0..N	200 OK	A representation of PFDs for request applications is returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.4 Resource: Individual PFD Data

6.2.4.1 Description

The Individual PFD Data resource represents Individual PFD Datas to the Nudr_DataRepository Service at a given UDR.

6.2.4.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/pfds/{appId}**

This resource shall support the resource URI variables defined in table 6.2.4.2-1.

Table 6.2.4.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
appId	ApplicationId	Indicates the application identifier for the request pfd(s). Data type ApplicationId is defined in 3GPP TS 29.571 [7].

6.2.4.3 Resource Standard Methods

6.2.4.3.1 GET

This method shall support the URI query parameters specified in table 6.2.4.3.1-1.

Table 6.2.4.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
supp-feat	SupportedFeatures	O	0..1	The features supported by the NF service consumer.

This method shall support the request data structures specified in table 6.2.4.3.1-2 and the response data structures and response codes specified in table 6.2.4.3.1-3.

Table 6.2.4.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.4.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
PfdDataForAppExt	M	1	200 OK	A representation of PFDs for the request application identified by the application identifier is returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.4.3.2 DELETE

This method shall support the URI query parameters specified in table 6.2.4.3.2-1.

Table 6.2.4.3.2-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.4.3.2-2 and the response data structures and response codes specified in table 6.2.4.3.2-3.

Table 6.2.4.3.2-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.4.3.2-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No content	Successful case. The Individual PFD Data resource related to the application identifier was deleted.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.4.3.3 PUT

This method shall support the URI query parameters specified in table 6.2.4.3.3-1.

Table 6.2.4.3.3-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.4.3.3-2 and the response data structures and response codes specified in table 6.2.4.3.3-3.

Table 6.2.4.3.3-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
PfdDataForAppExt	M	1	Contains the information for the creation or modification of an individual PFD Data resource.

Table 6.2.4.3.3-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
PfdDataForAppExt	M	1	201 Created	The resource has been successfully created and a response body is returned containing a representation of the resource.
PfdDataForAppExt	M	1	200 OK	The resource has been successfully updated and a response body is returned containing a representation of the resource.
n/a			204 No Content	The resource has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 6.2.4.3.3-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/application-data/pfds/{appId}

6.2.5 Resource: Influence Data

6.2.5.1 Description

The Influence Data resource represents a Traffic Influence Data to the Nudr_DataRepository Service at a given UDR.

6.2.5.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/influenceData**

This resource shall support the resource URI variables defined in table 6.2.5.2-1.

Table 6.2.5.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

6.2.5.3 Resource Standard Methods

6.2.5.3.1 GET

This method shall support the URI query parameters specified in table 6.2.5.3.1-1.

Table 6.2.5.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
influence-Ids	array(string)	O	1..N	Each element identifies a influenceId, i.e. an Individual Influence Data resource.
dnns	array(Dnn)	O	1..N	Each element identifies a DNN. The UDR shall return only Individual Influence Data resource(s) that match one of the provided DNNs.
snssais	array(Snssai)	O	1..N	Each element identifies a slice. The UDR shall return only Individual Influence Data resource(s) that match one of the provided S-NSSAIs.
internal-Group-Ids	array(GroupId)	O	1..N	Each element identifies a group of users. The UDR shall return only Individual Influence Data resource(s) that match one of the provided internal group IDs.
supis	array(Supi)	O	1..N	Each element identifies a user. The UDR shall return only Individual Influence Data resource(s) that match one of the provided SUPIs.
supp-feat	SupportedFeatures	O	0..1	Identifies the features supported by the NF service consumer.
NOTE: At least one of the "influence-Ids", "dnns", "snssais", "internal-Group-Ids" or "supis" attributes shall be provided.				

NOTE: The "internal-Goup-Ids" and the "supis" query parameter represent mutually exclusive resource properties, i.e., if both query parameters are simultaneously present in the same request, the search will not match any resource.

When the request query contains more than one optional query parameters defined as an array, the UDR shall return the Individual Influence Data resources for each matching combination of the values of the elements of the array of the provided query parameters.

EXAMPLE: If "snssais" query parameter is included with two S-NSSAI entries (S-NSSAI_1 and S-NSSAI_2) and "dnns" is included with two DNN entries (e.g. DNN_A and DNN_B), and the other optional query parameters are not included, the UDR shall return the Individual Influence Data resources for each matching S-NSSAI and DNN combination (for all the possible values for supi and groupId), i.e., the resource(s) matching SNSSAI_1 and DNN_A, the resource(s) matching SNSSAI_1 and DNN_B, the resource(s) matching SNSSAI_2 and DNN_A, and the resource(s) matching SNSSAI_2 and DNN_B.

To query for the Individual Influence Data resource(s) that apply to any UE, an entry of the "internal-Group-Ids" shall be set to "AnyUE".

This method shall support the request data structures specified in table 6.2.5.3.1-2 and the response data structures and response codes specified in table 6.2.5.3.1-3.

Table 6.2.5.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.5.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
array(TrafficInfluData)	M	0..N	200 OK	The Traffic Influence Data stored in the UDR are returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.6 Resource: Individual Influence Data

6.2.6.1 Description

The Individual Influence Data resource represents an Individual Traffic Influence Data to the Nudr_DataRepository Service at a given UDR.

6.2.6.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/influenceData/{influenceId}**

This resource shall support the resource URI variables defined in table 6.2.6.2-1.

Table 6.2.6.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
influenceId	string	The Identifier of an Individual Influence Data to be created or updated. To enable the value to be used as part of a URI, the string shall only contain allowed characters according to the "lower-with-hyphen" naming convention defined in clause 5.1.3 of 3GPP TS 29.501 [5].

6.2.6.3 Resource Standard Methods

6.2.6.3.1 PUT

This method shall support the URI query parameters specified in table 6.2.6.3.1-1.

Table 6.2.6.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.6.3.1-2 and the response data structures and response codes specified in table 6.2.6.3.1-3.

Table 6.2.6.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
TrafficInfluData	M	1	The Traffic Influence Data to be stored in the UDR.

Table 6.2.6.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
TrafficInfluData	M	1	201 Created	The resource has been successfully created and a response body is returned containing a representation of the resource.
TrafficInfluData	M	1	200 OK	The resource has been successfully updated and a response body is returned containing a representation of the resource.
n/a			204 No Content	The resource has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 6.2.6.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/application-data/influenceData/{influenceId}

6.2.6.3.2 PATCH

This method shall support the URI query parameters specified in table 6.2.6.3.2-1.

Table 6.2.6.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.6.3.2-2 and the response data structures and response codes specified in table 6.2.6.3.2-3.

Table 6.2.6.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
TrafficInfluDataPatch	M	1	The Traffic Influence Data to be updated in the UDR.

Table 6.2.6.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
TrafficInfluData	M	1	200 OK	The update of an Individual Traffic Influence Data resource is confirmed and a response body containing Traffic Influence Data shall be returned.
n/a			204 No Content	The Individual Traffic Influence Data resource has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.6.3.3 DELETE

This method shall support the URI query parameters specified in table 6.2.6.3.3-1.

Table 6.2.6.3.3-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.6.3.3-2 and the response data structures and response codes specified in table 6.2.6.3.3-3.

Table 6.2.6.3.3-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.6.3.3-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The Individual Influence Data was deleted successfully.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.7 Resource: Influence Data Subscription

6.2.7.1 Description

The Influence Data Subscription resource represents all Influence Data subscriptions to the Nudr_DataRepository Service at a given UDR.

6.2.7.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/influenceData/subs-to-notify**

This resource shall support the resource URI variables defined in table 6.2.7.2-1.

Table 6.2.7.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

6.2.7.3 Resource Standard Methods

6.2.7.3.1 POST

This method shall support the URI query parameters specified in table 6.2.7.3.1-1.

Table 6.2.7.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.7.3.1-2 and the response data structures and response codes specified in table 6.2.7.3.1-3.

Table 6.2.7.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
TrafficInfluSub	M	1	Parameters to register a subscription to Traffic Influence Data.

Table 6.2.7.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
TrafficInfluSub	M	1	201 Created	The subscription was created successfully.
NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 6.2.7.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/application-data/influenceData/subs-to-notify/{subscriptionId}

6.2.7.3.2 GET

This method shall support the URI query parameters specified in table 6.2.7.3.2-1.

Table 6.2.7.3.2-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
dnn	Dnn	O	0..1	Identifies a DNN.
snssai	Snssai	O	0..1	Identifies a slice.
internal-Group-Id	GroupId	O	0..1	Identifies a group of users.(NOTE 2)
supi	Supi	O	0..1	Identifies a user.
NOTE 1: At least one of the above attribute shall be provided. NOTE 2: For the Traffic Influence Data applies to any UE, the "internal-Group-Id" sets to "AnyUE".				

This method shall support the request data structures specified in table 6.2.7.3.2-2 and the response data structures and response codes specified in table 6.2.7.3.2-3.

Table 6.2.7.3.2-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.7.3.2-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
array(TrafficInfluSub)	M	0..N	200 OK	The subscription information as request in the request URI query parameter(s) are returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.8 Resource: Individual Influence Data Subscription

6.2.8.1 Description

The Individual Influence Data Subscription resource represents a single Influence Data subscription to the Nudr_DataRepository Service at a given UDR.

6.2.8.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/influenceData/subs-to-notify/{subscriptionId}**

This resource shall support the resource URI variables defined in table 6.2.8.2-1.

Table 6.2.8.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
subscriptionId	string	Identifies an Individual Influence Data Subscription resource. To enable the value to be used as part of a URI, the string shall only contain allowed characters according to the "lower-with-hyphen" naming convention defined in clause 5.1.3 of 3GPP TS 29.501 [5].

6.2.8.3 Resource Standard Methods

6.2.8.3.1 GET

This method shall support the URI query parameters specified in table 6.2.8.3.1-1.

Table 6.2.8.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.8.3.1-2 and the response data structures and response codes specified in table 6.2.8.3.1-3.

Table 6.2.8.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.8.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
TrafficInfluSub	M	1	200 OK	The subscription information is returned.

6.2.8.3.2 PUT

This method shall support the URI query parameters specified in table 6.2.8.3.2-1.

Table 6.2.8.3.2-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.8.3.2-2 and the response data structures and response codes specified in table 6.2.8.3.2-3.

Table 6.2.8.3.2-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
TrafficInfluSub	M	1	Modify an existing subscription to Traffic Influence Data.

Table 6.2.8.3.2-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
TrafficInfluSub	M	1	200 OK	The subscription was updated successfully.
n/a			204 No Content	The subscription has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.8.3.3 DELETE

This method shall support the URI query parameters specified in table 6.2.8.3.3-1.

Table 6.2.8.3.3-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.8.3.3-2 and the response data structures and response codes specified in table 6.2.8.3.3-3.

Table 6.2.8.3.3-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.8.3.3-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The subscription was terminated successfully.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.9 Resource: Applied BDT Policy Data

6.2.9.1 Description

The Applied BDT Policy Data resource represents an applied BDT Policy Data to the Nudr_DataRepository Service at a given UDR.

6.2.9.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/bdtPolicyData**

This resource shall support the resource URI variables defined in table 6.2.9.2-1.

Table 6.2.9.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

6.2.9.3 Resource Standard Methods

6.2.9.3.1 GET

This method shall support the URI query parameters specified in table 6.2.9.3.1-1.

Table 6.2.9.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
bdt-policy-ids	array(string)	O	1..N	Each element identifies a bdtPolicyId, i.e., an identifier of an Individual BDT Policy Data resource. The UDR shall return an Individual Applied BDT Policy Data resource for each matched bdtPolicyId.
internal-group-ids	array(GroupId)	O	1..N	Each element identifies a group of users. The UDR shall return only Individual Applied BDT Policy Data resource(s) that match one of the provided internal group ID(s).
supis	array(Supi)	O	1..N	Each element identifies a user. The UDR shall return only Individual Applied BDT Policy Data resource(s) that match one of the provided SUPI(s).

NOTE: The "internal-group-ids" and the "supis" query parameters represent mutually exclusive resource properties, i.e., if both query parameters are simultaneously present in the same request, the search will not match any resource.

When the request contains the "bdt-policy-ids" and the "internal-group-ids" or "supis" query parameters, the UDR shall return the resource(s) that match any combination of the values of the "bdt-policy-ids" and the "internal-group-ids" or "supis" arrays.

EXAMPLE: If "bdt-policy-ids" query parameter is included with two bdtPolicyId entries (bdtPolicyId_1 and bdtPolicyId_2) and "supis" is included with two SUPI entries (e.g. SUPI_A and SUPI_B), the UDR shall return the Individual BDT Policy Data resource(s) for each matching bdtPolicyId and SUPI combination i.e., the resource(s) matching bdtPolicyId_1 and SUPI_A or SUPI_B, and the resource(s) matching bdtPolicyId_2 and SUPI A or SUPI_B (note that other combinations are not possible).

This method shall support the request data structures specified in table 6.2.9.3.1-2 and the response data structures and response codes specified in table 6.2.9.3.1-3.

Table 6.2.9.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.9.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
array(BdtPolicyData)	M	0..N	200 OK	The applied BDT Policy Data stored in the UDR are returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.10 Resource: Individual Applied BDT Policy Data

6.2.10.1 Description

The Individual Applied BDT Policy Data resource represents an Individual Applied BDT Policy Data to the Nudr_DataRepository Service at a given UDR.

6.2.10.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/bdtPolicyData/{bdtPolicyId}**

This resource shall support the resource URI variables defined in table 6.2.10.2-1.

Table 6.2.10.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
bdtPolicyId	string	The Identifier of an Individual BDT Policy Data to be created or updated. To enable the value to be used as part of a URI, the string shall only contain allowed characters according to the "lower-with-hyphen" naming convention defined in clause 5.1.3 of 3GPP TS 29.501 [5].

6.2.10.3 Resource Standard Methods

6.2.10.3.1 PUT

This method shall support the URI query parameters specified in table 6.2.10.3.1-1.

Table 6.2.10.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.10.3.1-2 and the response data structures and response codes specified in table 6.2.10.3.1-3.

Table 6.2.10.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
BdtPolicyData	M	1	The applied BDT Policy Data to be stored in the UDR.

Table 6.2.10.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
BdtPolicyData	M	1	201 Created	The resource has been successfully created and a response body is returned containing a representation of the resource.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 6.2.10.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/application-data/bdtPolicyData/{bdtPolicyId}

6.2.10.3.2 PATCH

This method shall support the URI query parameters specified in table 6.2.10.3.2-1.

Table 6.2.10.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.10.3.2-2 and the response data structures and response codes specified in table 6.2.10.3.2-3.

Table 6.2.10.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
BdtPolicyDataPatch	M	1	The applied BDT Policy Data to be updated in the UDR.

Table 6.2.10.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
BdtPolicyData	M	1	200 OK	The update of an applied BDT Policy Data resource is confirmed and a response body containing applied BDT Policy Data shall be returned.
n/a			204 No Content	The update of an applied BDT Policy Data resource is confirmed.
NOTE: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.10.3.3 DELETE

This method shall support the URI query parameters specified in table 6.2.10.3.3-1.

Table 6.2.10.3.3-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.10.3.3-2 and the response data structures and response codes specified in table 6.2.10.3.3-3.

Table 6.2.10.3.3-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.10.3.3-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The Individual Applied BDT Policy Data was deleted successfully.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.11 Resource: IPTV Configurations

6.2.11.1 Description

This resource allows the NF service consumer to read all active IPTV Configurations applicable for given configuration identifier(s), DNN(s) and/or S-NSSAI(s), SUPI(s) or Internal Group Identifier(s) stored at the UDR.

6.2.11.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/iptvConfigData**

This resource shall support the resource URI variables defined in table 6.2.11.2-1.

Table 6.2.11.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

6.2.11.3 Resource Standard Methods

6.2.11.3.1 GET

This method shall support the URI query parameters specified in table 6.2.11.3.1-1.

Table 6.2.11.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
config-ids	array(string)	O	1..N	Each element identifies a configurationId, i.e. an Individual IPTV Configuration resource. The UDR shall return an Individual IPTV Configuration resource for each matched configurationId.
dnns	array(Dnn)	O	1..N	Each element identifies a DNN. The UDR shall return only Individual IPTV Configuration resource(s) that match one of the provided DNN(s).
snssais	array(Snssai)	O	1..N	Each element identifies a slice. The UDR shall return only Individual IPTV Configuration resource(s) that match one of the provided S-NSSAI(s).
supis	array(Supi)	O	1..N	Each element identifies a SUPI. The UDR shall return only Individual IPTV Configuration resource(s) that match one of the provided SUPI(s).
inter-group-ids	array(GroupId)	O	1..N	Each element identifies a group of users. The UDR shall return only Individual IPTV Configuration resource(s) that match one of the provided internal group ID(s).
NOTE: At least one of the above attributes shall be provided.				

NOTE: The "inter-group-ids" and the "supis" query parameters represent mutually exclusive resource properties, i.e., if both query parameters are simultaneously present in the same request, the search will not match any resource.

When the request query contains more than one optional query parameters defined as an array, the UDR shall return the Individual IPTV Configuration resources for each matching combination of the values of the provided query parameters.

EXAMPLE: If "snssais" query parameter is included with two S-NSSAI entries (S-NSSAI_1 and S-NSSAI_2) and "dnns" is included with two DNN entries (e.g. DNN_A and DNN_B), and the other optional query parameters are not included, the UDR shall return IPTV Configuration resources for each matching (for all the possible values for supi, and groupId), i.e. the resource(s) matching SNSSAI_1 and DNN_A, the resource(s) matching SNSSAI_1 and DNN_B, the resource(s) matching SNSSAI_2 and DNN_A, and the resources matching SNSSAI_2 and DNN_B.

This method shall support the request data structures specified in table 6.2.11.3.1-2 and the response data structures and response codes specified in table 6.2.11.3.1-3.

Table 6.2.11.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.11.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
array(IptvConfigData)	M	0..N	200 OK	The IPTV configuration data stored in the UDR are returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.12 Resource: Individual IPTV Configuration

6.2.12.1 Description

The resource allows the NF sevice consumer to create an Individual IPTV configuration Data, modify or delete an existing Individual IPTV configuration Data.

6.2.12.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/iptvConfigData/{configurationId}**

This resource shall support the resource URI variables defined in table 6.2.12.2-1.

Table 6.2.12.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
configurationId	string	The Identifier of an Individual IPTV Configuration to be created or updated. To enable the value to be used as part of a URI, the string shall only contain allowed characters according to the "lower-with-hyphen" naming convention defined in clause 5.1.3 of 3GPP TS 29.501 [5].

6.2.12.3 Resource Standard Methods

6.2.12.3.1 PUT

This method shall support the URI query parameters specified in table 6.2.12.3.1-1.

Table 6.2.12.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.12.3.1-2 and the response data structures and response codes specified in table 6.2.12.3.1-3.

Table 6.2.12.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
IptvConfigData	M	1	The IPTV configuration data to be stored in the UDR.

Table 6.2.12.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
IptvConfigData	M	1	201 Created	The resource has been successfully created and a response body is returned containing a representation of the resource.
IptvConfigData	M	1	200 OK	The resource has been successfully updated and a response body is returned containing a representation of the resource.
n/a			204 No Content	The resource has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 6.2.12.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/application-data/iptvConfigData/{configurationId}

6.2.12.3.2 DELETE

This method shall support the URI query parameters specified in table 6.2.12.3.2-1.

Table 6.2.12.3.2-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.12.3.2-2 and the response data structures and response codes specified in table 6.2.12.3.2-3.

Table 6.2.12.3.2-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.12.3.2-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The resource was deleted successfully.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.12.3.3 PATCH

This method shall support the URI query parameters specified in table 6.2.12.3.3-1.

Table 6.2.12.3.3-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.12.3.3-2 and the response data structures and response codes specified in table 6.2.12.3.3-3.

Table 6.2.12.3.3-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
IptvConfigDataPatch	M	1	The IPTV configuration related data to be updated in the UDR.

Table 6.2.12.3.3-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
IptvConfigData	M	1	200 OK	The resource has been successfully updated.
n/a			204 No Content	The resource has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.13 Resource: ApplicationDataSubscriptions

6.2.13.1 Description

This resource is used to represent subscriptions to notification of application data modification, e.g. PFD data change, IPTV configuration data change, BDT policy data change, but not for Traffic Influence Data.

6.2.13.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/subs-to-notify**

This resource shall support the resource URI variables defined in table 6.2.13.2-1.

Table 6.2.13.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

6.2.13.3 Resource Standard Methods

6.2.13.3.1 POST

This method shall support the URI query parameters specified in table 6.2.13.3.1-1.

Table 6.2.13.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.13.3.1-2 and the response data structures and response codes specified in table 6.2.13.3.1-3.

Table 6.2.13.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
ApplicationDataSubs	M	1	Identifies the subscription to notification of application data modification.

Table 6.2.13.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
ApplicationDataSubs	M	1	201 Created	Upon success, a response body containing a representation of each Individual subscription resource shall be returned. The HTTP response shall include a "Location" HTTP header that contains the resource URI of the created resource.
NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 6.2.13.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/application-data/subs-to-notify/{subsId}

6.2.13.3.2 GET

This method shall support the URI query parameters specified in table 6.2.13.3.2-1.

Table 6.2.13.3.2-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
data-filter	DataFilter	O	0..1	Indicate the data filter (e.g. dnn) for the query. When present, the UDR shall return the IndividualApplicationDataSubscription resource(s) for which the indicated information in the "data-filter" query parameter matches an entry of the "dataFilters" property of the resource (i.e. within the resource representation in the ApplicationDataSubs data structure). If the values of the "dataFilters" entry match the values provided in the "data-filter" query parameter, then the entry is considered to be a match even if it contains further attributes, for which no value was provided in the "data-filter" query parameter.

This method shall support the request data structures specified in table 6.2.13.3.2-2 and the response data structures and response codes specified in table 6.2.13.3.2-3.

Table 6.2.13.3.2-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.13.3.2-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
array(ApplicationDataSubs)	M	0..N	200 OK	The subscription information as request in the request URI query parameter(s) are returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.14 Resource: IndividualApplicationDataSubscription

6.2.14.1 Description

This resource is used to represent an individual subscription to notification of application data modification.

6.2.14.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/subs-to-notify/{subsId}**

This resource shall support the resource URI variables defined in table 6.2.14.2-1.

Table 6.2.14.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
subsId	string	The subsId identifies an individual application data subscription to notifications. The value is allocated by the UDR during creation of the ApplicationDataSubscriptions resource.

6.2.14.3 Resource Standard Methods

6.2.14.3.1 PUT

This method shall support the URI query parameters specified in table 6.2.14.3.1-1.

Table 6.2.14.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.14.3.1-2 and the response data structures and response codes specified in table 6.2.14.3.1-3.

Table 6.2.14.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
ApplicationDataSubs	M	1	Updates the subscription to notification of application data modification.

Table 6.2.14.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
ApplicationDataSubs	M	1	200 OK	The individual subscription resource was updated successfully.
n/a			204 No Content	The individual subscription resource was updated successfully and no additional content is to be sent in the response message.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.14.3.2 DELETE

This method shall support the URI query parameters specified in table 6.2.14.3.2-1.

Table 6.2.14.3.2-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.14.3.2-2 and the response data structures and response codes specified in table 6.2.14.3.2-3.

Table 6.2.14.3.2-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			The request body shall be empty.

Table 6.2.14.3.2-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon success, an empty response body shall be returned.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.14.3.3 GET

This method shall support the URI query parameters specified in table 6.2.14.3.3-1.

Table 6.2.14.3.3-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.14.3.3-2 and the response data structures and response codes specified in table 6.2.14.3.3-3.

Table 6.2.14.3.3-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.14.3.3-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
ApplicationDataSubs	M	1	200 OK	The subscription information is returned.

6.2.15 Resource: Service Parameter Data

6.2.15.1 Description

The Service Parameter Data resource represents a Service Parameter Data to the Nudr_DataRepository Service at a given UDR.

6.2.15.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/serviceParamData**

This resource shall support the resource URI variables defined in table 6.2.15.2-1.

Table 6.2.15.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

6.2.15.3 Resource Standard Methods

6.2.15.3.1 GET

This method shall support the URI query parameters specified in table 6.2.15.3.1-1.

Table 6.2.15.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description	Applicability
service-param-ids	array(string)	O	1..N	Each element identifies a serviceParamId, i.e., an Individual Service Parameter Data. The UDR shall return an Individual Service Parameter Data resource for each matched serviceParamId.	
dnns	array(Dnn)	O	1..N	Each element identifies a DNN. The UDR shall return only Individual Service Parameter Data resource(s) that match one of the provided DNNs.	
snssais	array(Snssai)	O	1..N	Each element identifies a slice. The UDR shall return only Individual Service Parameter Data resource(s) that match one of the provided S-NSSAIs.	
internal-group-ids	array(GroupId)	O	1..N	Each element identifies a group of users. The UDR shall return only Individual Service Parameter Data resource(s) that match one of the provided internal group IDs.	
supis	array(Supi)	O	1..N	Each element identifies a user. The UDR shall return only Individual Service Parameter Data resource(s) that match one of the provided SUPIs.	
ue-ipv4s	array(Ipv4Addr)	O	1..N	Each element identifies a user. The UDR shall return only Individual Service Parameter Data resource(s) that match one of the provided IPv4 addresses.	
ue-ipv6s	array(Ipv6Addr)	O	1..N	Each element identifies a user. The UDR shall return only Individual Service Parameter Data resource(s) that match one of the provided IPv6 addresses.	
ue-macs	array(MacAddr48)	O	1..N	Each element identifies a user. The UDR shall return only Individual Service Parameter Data resource(s) that match one of the provided UE MAC addresses.	
any-ue	boolean	O	0..1	Indicates whether the request is for any UE: - "true": the request is for any UE; - "false"(default): the request is not for any UE. The UDR shall return only Individual Service Parameter Data resource(s) that match the any UE indication.	FilterAnyUE
supp-feat	SupportedFeatures	O	0..1	Identifies the features supported by the NF service consumer.	
NOTE 1: At least one of the "service-param-ids", "dnns", "snssais", "internal-group-ids", "supis", "any_ue", "ue-ipv4s", "ue-ipv6s" or "ue-macs", and if the feature FilterAnyUE is supported, or "any-ue" attributes shall be provided.					

NOTE: The "any-ue" query parameter is commonly related to the search of the service parameter data resource(s) that relate to any UE using the service identified by a combination of DNN and S-NSSAI, and thus, it is commonly present together with the "dnns" and "snssais" query parameters. Note that, if the "any-ue" query parameter is present together with the "internal-group-ids", "supis", "ue-ipv4s", "ue-ipv6s" and/or "ue-macs" query parameters the search will not match any resource, since according to clause 6.4.2.15 only one of the the "supi", "anyUeInd", "interGroupId", "ueIpv4", "ueIpv6" or "ueMac" properties are simultaneously present in the resource.

When the request query contains more than one optional query parameters defined as an array, the UDR shall return the Service Parameter Data resources for each matching combination of the values of the elements of the array of the provided query parameters.

EXAMPLE: If "snssais" query parameter is included with two S-NSSAI entries (S-NSSAI_1 and S-NSSAI_2) and "dnns" is included with two DNN entries (e.g. DNN_A and DNN_B), and the other optional query parameters are not included, the UDR shall return the Individual Service Parameter Data resources for each matching S-NSSAI and DNN, i.e., the resource(s) matching SNSSAI_1 and DNN_A, the resource(s) matching SNSSAI_1 and DNN_B, the resource(s) matching SNSSAI_2 and DNN_A, and the resources matching SNSSAI_2 and DNN_B.

This method shall support the request data structures specified in table 6.2.15.3.1-2 and the response data structures and response codes specified in table 6.2.15.3.1-3.

Table 6.2.15.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.15.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
array(ServiceParameterData)	M	0..N	200 OK	The Service Parameter Data stored in the UDR are returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.16 Resource: Individual Service Parameter Data

6.2.16.1 Description

The Individual Service Parameter Data resource represents an Individual Service Parameter Data to the Nudr_DataRepository Service at a given UDR.

6.2.16.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/serviceParamData/{serviceParamId}**

This resource shall support the resource URI variables defined in table 6.2.16.2-1.

Table 6.2.16.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
serviceParamId	string	The Identifier of an Individual Service Parameter Data to be created or updated. To enable the value to be used as part of a URI, the string shall only contain allowed characters according to the "lower-with-hyphen" naming convention defined in clause 5.1.3 of 3GPP TS 29.501 [5].

6.2.16.3 Resource Standard Methods

6.2.16.3.1 PUT

This method shall support the URI query parameters specified in table 6.2.16.3.1-1.

Table 6.2.16.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.16.3.1-2 and the response data structures and response codes specified in table 6.2.16.3.1-3.

Table 6.2.16.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
ServiceParameterData	M	1	The Service Parameter Data to be stored in the UDR.

Table 6.2.16.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
ServiceParameterData	M	1	201 Created	The resource has been successfully created and a response body is returned containing a representation of the resource.
ServiceParameterData	M	1	200 OK	The resource has been successfully updated and a response body is returned containing a representation of the resource.
n/a			204 No Content	The resource has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 6.2.16.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/application-data/serviceParamData/{serviceParamId}

6.2.16.3.2 PATCH

This method shall support the URI query parameters specified in table 6.2.16.3.2-1.

Table 6.2.16.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.6.3.2-2 and the response data structures and response codes specified in table 6.2.6.3.2-3.

Table 6.2.16.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
ServiceParameterDataPatch	M	1	The Service Parameter Data to be updated in the UDR.

Table 6.2.16.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
ServiceParameter Data	M	1	200 OK	The update of an Individual Service Parameter Data resource is confirmed and a response body containing Service Parameter Data shall be returned.
n/a			204 No Content	The Individual Service Parameter Data resource has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.16.3.3 DELETE

This method shall support the URI query parameters specified in table 6.2.16.3.3-1.

Table 6.2.16.3.3-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.16.3.3-2 and the response data structures and response codes specified in table 6.2.16.3.3-3.

Table 6.2.16.3.3-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.16.3.3-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The Individual Service Parameter Data was deleted successfully.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.17 Resource: AM Influence Data

6.2.17.1 Description

The AM Influence Data resource represents an AM Influence Data to the Nudr_DataRepository Service at a given UDR.

6.2.17.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/am-influence-data**

This resource shall support the resource URI variables defined in table 6.2.17.2-1.

Table 6.2.17.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

6.2.17.3 Resource Standard Methods

6.2.17.3.1 GET

This method shall support the URI query parameters specified in table 6.2.17.3.1-1.

Table 6.2.17.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
am-influence-ids	array(string)	O	1..N	Each element identifies a amInfluenceId, i.e., an identifier of an Individual AM Influence Data resource. The UDR shall return an Individual AM Influence Data resource that matches one of provided amInfluenceId(s).
dnns	array(Dnn)	O	1..N	Each element identifies a DNN. The UDR shall return only AM Influence Data resource(s) that match at least one of the DNN(s) of the array. (NOTE 2)
snssais	array(Snssai)	O	1..N	Each element identifies a slice. The UDR shall return only AM Influence Data resource(s) that match at least one of the S-NSSAI(s) of the array. (NOTE 2)
dnn-snssai-infos	array(DnnSnssaiInformation)	O	1..N	Each element identifies a combination of (DNN, S-NSSAI). The UDR shall return only AM Influence Data resource(s) that match at least one of the SNSSAI and DNN combination(s) of the array. (NOTE 2)
internal-group-ids	array(GroupId)	O	1..N	Each element identifies a group of users. The UDR shall return only AM Influence Data resource(s) that match one of the provided GroupId(s).
supis	array(Supi)	O	1..N	Each element identifies a user. The UDR shall return only AM Influence Data resource(s) that match one of the provided SUPI(s).
any-ue	boolean	O	0..1	Indicates whether the request is for any UE: - "true": the request is for any UE; - "false"(default): the request is not for any UE. The UDR shall return only AM Influence Data resource(s) that contain the Any UE indication.
supp-feat	SupportedFeatures	O	0..1	Identifies the features supported by the NF service consumer.
NOTE 1: At least one of the query parameters other than the "supp-feat" query parameter shall be provided. NOTE 2: Only one of the "dnns", "snssais" or "dnn-snssai-infos" attributes may be simultaneously provided.				

NOTE: The "any-ue" query parameter is commonly related to the search of the AM influence data resource(s) that relate to any UE using the service(s) identified by a combination of DNN and S-NSSAI, and thus, it is commonly present together with the "dnn-snssai-infos" query parameters. Note also that, if the "any-ue" query parameter is present together with the "internal-group-ids", and/or "supis" query parameters the search will not match any resource, since according to clause 6.4.2.16 only one of the the "supi", "anyUeInd" and "interGroupId" properties are simultaneously present in the resource.

When the request query contains more than one optional query parameters defined as an array, the UDR shall return the Individual AM Influence Data resources for each matching combination of the values of the elements of the array of the provided query parameters.

EXAMPLE 1: If the "dnn-snssai-infos" query parameter is included with two DNN and S-NSSAI combinations (comb_1 and comb_2) and the "supis" query parameter is included with two SUPI entries (e.g. SUPI_A and SUPI_B), and all of them are matching entries, the UDR shall return the Individual AM Influence Data resources for each matching S-NSSAI and DNN and SUPI combination, i.e. the resource(s) matching comb_1 and SUPI_A, comb_2 and SUPI_A, comb_1 and SUPI_B, and comb2 and SUPI_B.

EXAMPLE 2: If the "dnn-snssai-infos" query parameter is included with two DNN and S-NSSAI combination (comb_1 and comb_2) and the "supis" query parameter is included with two SUPI entries (e.g. SUPI_A and SUPI_B), and there are no matching entries for comb_1, the UDR shall return the Individual AM Influence Data resources for the matching S-NSSAI and DNN and SUPI combination, i.e., the resource(s) matching comb_2 and SUPI_A, and comb2 and SUPI_B.

This method shall support the request data structures specified in table 6.2.17.3.1-2 and the response data structures and response codes specified in table 6.2.17.3.1-3.

Table 6.2.17.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.17.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
array(AmInfluData)	M	0..N	200 OK	The AM Influence Data stored in the UDR are returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.18 Resource: Individual AM Influence Data

6.2.18.1 Description

The Individual AM Influence Data resource represents an Individual AM Influence Data to the Nudr_DataRepository Service at a given UDR.

6.2.18.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/am-influence-data/{amInfluenceId}**

This resource shall support the resource URI variables defined in table 6.2.18.2-1.

Table 6.2.18.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
amInfluenceId	string	The identifier of an Individual AM Influence Data to be created or updated. To enable the value to be used as part of a URI, the string shall only contain allowed characters according to the "lower-with-hyphen" naming convention defined in clause 5.1.3 of 3GPP TS 29.501 [5].

6.2.18.3 Resource Standard Methods

6.2.18.3.1 PUT

This method shall support the URI query parameters specified in table 6.2.18.3.1-1.

Table 6.2.18.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.18.3.1-2 and the response data structures and response codes specified in table 6.2.18.3.1-3.

Table 6.2.18.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
AmInfluData	M	1	The AM Influence Data to be stored in the UDR.

Table 6.2.18.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AmInfluData	M	1	201 Created	The resource has been successfully created and a response body is returned containing a representation of the resource.
AmInfluData	M	1	200 OK	The resource has been successfully updated and a response body is returned containing a representation of the resource.
n/a			204 No Content	The resource has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 6.2.18.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/application-data/am-influence-data/{amInfluenceId}

6.2.18.3.2 PATCH

This method shall support the URI query parameters specified in table 6.2.18.3.2-1.

Table 6.2.18.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.18.3.2-2 and the response data structures and response codes specified in table 6.2.18.3.2-3.

Table 6.2.18.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
AmInfluDataPatch	M	1	The AM Influence Data to be updated in the UDR.

Table 6.2.18.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AmInfluData	M	1	200 OK	The update of an Individual AM Influence Data resource is confirmed and a response body containing the AM Influence Data shall be returned.
n/a			204 No Content	The Individual AM Influence Data resource has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.18.3.3 DELETE

This method shall support the URI query parameters specified in table 6.2.18.3.3-1.

Table 6.2.18.3.3-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.18.3.3-2 and the response data structures and response codes specified in table 6.2.18.3.3-3.

Table 6.2.18.3.3-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.18.3.3-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The Individual AM Influence Data was deleted successfully.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.19 Resource: EAS Deployment Information Data

6.2.19.1 Description

The EAS Deployment Information Data resource represents all EAS Deployment Information Data subscriptions to the Nudr_DataRepository Service at a given UDR.

6.2.19.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/eas-deploy-data**

This resource shall support the resource URI variables defined in table 6.2.19.2-1.

Table 6.2.19.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

6.2.19.3 Resource Standard Methods

6.2.19.3.1 GET

This method shall support the URI query parameters specified in table 6.2.19.3.1-1.

Table 6.2.19.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
dnn	Dnn	C	0..1	Identifies a DNN.
snssai	Snssai	C	0..1	Identifies a slice.
internal-group-id	GroupId	C	0..1	Identifies a group of users.
appId	string	C	0..1	Identifies an application.
NOTE: At least one of the above attributes shall be provided.				

This method shall support the request data structures specified in table 6.2.19.3.1-2 and the response data structures and response codes specified in table 6.2.19.3.1-3.

Table 6.2.19.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.19.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
array(EasDeployInfoData)	M	1..N	200 OK	The subscribed EAS Deployment information as request in the request URI query parameter(s) are returned.
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.2.20 Resource: Individual EAS Deployment Information Data

6.2.20.1 Description

The Individual EAS Deployment Information Data resource represents a single EAS Deployment Information data subscription to the Nudr_DataRepository Service at a given UDR.

6.2.20.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/application-data/eas-deploy-data/{easDeployInfoId}**

This resource shall support the resource URI variables defined in table 6.2.20.2-1.

Table 6.2.20.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
easDeployInfoId	string	Identifies an Individual EAS Deployment Information Data resource.

6.2.20.3 Resource Standard Methods

6.2.20.3.1 GET

This method shall support the URI query parameters specified in table 6.2.20.3.1-1.

Table 6.2.20.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.20.3.1-2 and the response data structures and response codes specified in table 6.2.20.3.1-3.

Table 6.2.20.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.20.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
EasDeployInfoData	M	1	200 OK	The subscription information is returned.

6.2.20.3.2 PUT

This method shall support the URI query parameters specified in table 6.2.20.3.1-1.

Table 6.2.20.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				n/a

This method shall support the request data structures specified in table 6.2.20.3.1-2 and the response data structures and response codes specified in table 6.2.20.3.1-3.

Table 6.2.20.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
EasDeployInfoData	M	1	The EAS Deployment Information Data to be stored in the UDR.

Table 6.2.20.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
EasDeployInfoData	M	1	201 Created	The resource has been successfully created and a response body is returned containing a representation of the resource.
EasDeployInfoData	M	1	200 OK	The resource has been successfully updated and a response body is returned containing a representation of the resource.
n/a			204 No Content	The resource has been successfully updated.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 6.2.20.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/application-data/eas-deploy-data/{easDeployInfoId}

6.2.20.3.3 DELETE

This method shall support the URI query parameters specified in table 6.2.20.3.3-1.

Table 6.2.20.3.3-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.20.3.3-2 and the response data structures and response codes specified in table 6.2.20.3.3-3.

Table 6.2.20.3.3-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.20.3.3-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The subscription was terminated successfully.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.3 Notifications

6.3.1 General

Notifications shall comply with clause 6.2 of 3GPP TS 29.500 [4] and clause 4.6.2.3 of 3GPP TS 29.501 [5].

Table 6.3.1-1: Notifications overview

Notification	Callback URI	HTTP method or custom operation	Description (service operation)
Influence Data Update Notification	{notificationUri}	POST	One of notification reference provided by the service consumer during the subscription of the AF Influence Data.
Application Data Change Notification	{notificationUri}	POST	Application Data Change Notification

6.3.2 Void

6.3.3 Influence Data Update Notification

6.3.3.1 Description

The Influence Update Notification is used by the UDR to report one or several observed Influence Data(s) to a NF service consumer that has subscribed to such Notifications.

6.3.3.2 Operation Definition

The POST method shall be used for traffic influence data change notification and the URI shall be provided during the subscription procedure.

Callback URI: **{notificationUri}**

This method shall support the URI query parameters specified in table 6.3.3.2-1.

Table 6.3.3.2-1: URI query parameters supported by the POST method

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.3.3.2-2 and the response data structures and response codes specified in table 6.3.3.2-3.

Table 6.3.3.2-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
array(TrafficInfluData or TrafficInfluDataNotif)	M	1..N	Provides Information about observed Influence Data. If the EnhancedInfluDataNotification feature is supported, the TrafficInfluDataNotif data type shall be sent.

Table 6.3.3.2-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The receipt of the Notification is acknowledged.
NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.3.4 Application Data Change Notification

The POST method shall be used for application data change notification and the URI shall be provided during the subscription procedure.

Callback URI: **{notificationUri}**

This method shall support the URI query parameters specified in table 6.3.4-1.

Table 6.3.4-1: URI query parameters supported by the POST method

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.3.4-2 and the response data structures and response codes specified in table 6.3.4-3.

Table 6.3.4-2: Data structures supported by the POST Request Body

Data type	P	Cardinality	Description
array(ApplicationDataChangeNotif)	M	1..N	Notification of application data changes.

Table 6.3.4-3: Data structures supported by the POST Response Body

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The reception of the notification is acknowledged.
NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.4 Data Model

6.4.1 General

This clause specifies the application data model supported by the API.

Table 6.4.1-1 specifies the data types defined for the Nudr_DataRepository Service API for Application Data service based interface protocol.

Table 6.4.1-1: Nudr_DataRepository specific Data Types for Application Data

Data type	Section defined	Description	Applicability
AmInfluData	6.4.2.16	Contains AM influence data.	DCAMP
AmInfluDataPatch	6.4.2.17	Contains AM influence data that can be updated.	DCAMP
ApplicationDataSubs	6.4.2.10	Contains application data subscription data.	
ApplicationDataChangeNotif	6.4.2.11	Contains the new or updated application data or removed indication.	
BdtPolicyData	6.4.2.7	Contains applied BDT policy data.	EnhancedBackgroundDataTransfer
BdtPolicyDataPatch	6.4.2.8	Contains modification instructions to be performed on the applied BDT policy data.	EnhancedBackgroundDataTransfer
DataInd	6.4.3.3	Indicates the type of data.	
DataFilter	6.4.2.12	Indicates an application data filter.	
IptvConfigData	6.4.2.9	Represents IPTV configuration data information.	
PfdDataForAppExt	6.4.2.6	The PFDs and related data for the application	
ServiceParameterData	6.4.2.15	Contains the service parameter data.	
ServiceParameterDataPatch	6.4.2.15A	Contains the service parameter data that can be updated.	
TrafficInfluData	6.4.2.2	Contains traffic influence data.	
TrafficInfluDataPatch	6.4.2.3	Contains modification instructions to be performed on the traffic influence data.	
TrafficInfluDataNotif	6.4.2.14	Contains traffic influence data for notification.	EnhancedInfluDataNotification
TrafficInfluSub	6.4.2.4	Contains traffic influence subscription data.	

Table 6.4.1-2 specifies data types re-used by the Nudr_DataRepository Service API for Application Data service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nudr_DataRepository Service API for Application Data service based interface.

Table 6.4.1-2: Nudr_DataRepository re-used Data Types for Application Data

Data type	Reference	Comments	Applicability
AmInfluEvent	3GPP TS 29.522 [19]	Identifies the type of AM related events of which the AF requests to be notified.	
ApplicationId	3GPP TS 29.571 [7]	Indicates an application identifier.	
BdtReferenceId	3GPP TS 29.122 [9]	Identifies a selected policy of background data transfer.	EnhancedBackgroundDataTransfer
DateTime	3GPP TS 29.571 [7]	Indicates a date and time.	
DateTimeRm	3GPP TS 29.571 [7]	Indicates a date and time that can be updated.	
DnaiChangeType	3GPP TS 29.571 [7]	Describes the types of DNAI change.	
Dnn	3GPP TS 29.571 [7]	Identifies a Data Network Name.	
DnnSnssaiInformation	3GPP TS 29.522 [19]	Represents a DNN, S-NSSAI combination.	DCAMP
DurationSec	3GPP TS 29.571 [7]	Represents a duration in seconds.	DCAMP
DurationSecRm	3GPP TS 29.571 [7]	Represents a removable duration in seconds.	DCAMP
EasDeployInfoData	3GPP TS 29.591 [23]	Represnts the EAS Deployment Information.	EasDeployment
EthFlowDescription	3GPP TS 29.514 [16]	Contains the Ethernet data flow information.(NOTE)	
Event	3GPP TS 29.522 [19]	Contains the outcome of the UE Policy Delivery related to the invocation of AF provisioned service parameters.	DeliveryOutcome
FlowInfo	3GPP TS 29.122 [9]	Contains the flow information.	
GroupId	3GPP TS 29.571 [7]	Identifies a group of users.	EasDeployment
IptvConfigDataPatch	3GPP TS 29.522 [19]	Contains the IPTV configuration data used for PATCH.	
Ipv4Addr	3GPP TS 29.571 [7]	Identifies an IPv4 address.	
Ipv6Addr	3GPP TS 29.571 [7]	Identifies an IPv6 address.	
MacAddr48	3GPP TS 29.571 [7]	MAC Address.	
MulticastAccessControl	3GPP TS 29.522 [19]	Represents the multicast access control information.	
NetworkAreaInfo	3GPP TS 29.554 [13]	Describes a network area information.	
ParameterOverPc5	3GPP TS 29.522 [19]	Contains the V2X service parameters data provisioned over PC5.	
ParameterOverUu	3GPP TS 29.522 [19]	Contains the V2X service parameters data provisioned over Uu.	
ParamForProSeDd	3GPP TS 29.522 [19]	Contains the service parameters for 5G ProSe direct discovery.	ProSe
ParamForProSeDc	3GPP TS 29.522 [19]	Contains the service parameters for 5G ProSe direct communications.	ProSe
ParamForProSeU2NRelUe	3GPP TS 29.522 [19]	Contains the service parameters for 5G ProSe UE-to-network relay UE.	ProSe
ParamForProSeRemUe	3GPP TS 29.522 [19]	Contains the service parameters for 5G ProSe remote UE.	ProSe
PfdChangeNotification	3GPP TS 29.551 [8]	Describes the PFD change.	
PfdContent	3GPP TS 29.551 [8]	Represents the content of a PFD for an application identifier.	
RouteToLocation	3GPP TS 29.571 [7]	Identifies the N6 traffic routing requirement.	
ServiceAreaCoverageInfo	3GPP TS 29.534 [22]	Contains service area coverage information.	DCAMP
Snssai	3GPP TS 29.571 [7]	Identifies a Single Network Slice Selection Assistance Information.	
SubscribedEvent	3GPP TS 29.522 [19]	Identified the type of UP path management events of which the AF requests to be notified.	
Supi	3GPP TS 29.571 [7]	Identifies a SUPI that shall contain either an IMSI or an NAI.	
SupportedFeatures	3GPP TS 29.571 [7]	Used to negotiate the applicability of the optional features.	
TemporalValidity	3GPP TS 29.514 [16]	Indicates the time interval during which the AF request is to be applied.	MultiTemporalCondition
Uinteger	3GPP TS 29.571 [7]	Unsigned Integer, i.e. only value 0 and integers greater than 0 are allowed.	
UintegerRm	3GPP TS 29.571 [7]	This data type is defined in the same way as the "Uinteger" data type, but with the OpenAPI "nullable: true" property.	
Uri	3GPP TS 29.571 [7]	Identifies a URI.	
UriRm	3GPP TS 29.571 [7]	Identifies a removable URI.	DCAMP

UrspRuleRequest	3GPP TS 29.522 [19]	Contains service parameter data used to guide the URSP.	AfGuideURSP
NOTE: In order to support a set of MAC addresses with a specific range in the traffic filter, feature MacAddressRange as specified in clause 6.1.8 of TS 29.504 [6] shall be supported.			

6.4.2 Structured data types

6.4.2.1 Introduction

This clause defines the structures to be used in resource representations.

6.4.2.2 Type TrafficInfluData

Table 6.4.2.2-1: Definition of type TrafficInfluData

Attribute name	Data type	P	Cardinality	Description	Applicability
upPathChgNotifCorreId	string	C	0..1	Contains the Notification Correlation Id allocated by the NEF for the UP path change notification. It shall be included when the NEF requests the UP path change notification.	
appReloInd	boolean	O	0..1	Identifies whether an application can be relocated once a location of the application has been selected. True: the application cannot be relocated; False (default): the application can be relocated.	
afAppId	string	O	0..1	Identifies an application. (NOTE 1) (NOTE 3)	
dnn	Dnn	O	0..1	Identifies a DNN	
ethTrafficFilters	array(EthFlowDescription)	O	1..N	Identifies Ethernet packet filters. (NOTE 1) (NOTE 3)	
snssai	Snssai	O	0..1	The identification of slice.	
interGroupId	GroupId	O	0..1	Identifies a group of users. (NOTE 2) (NOTE 3)(NOTE 5)	
supi	Supi	O	0..1	Identifies a user. (NOTE 2) (NOTE 3)	
trafficFilters	array(FlowInfo)	O	1..N	Identifies IP packet filters. (NOTE 1) (NOTE 3)	
trafficRoutes	array(RouteToLocation)	O	1..N	Identifies the N6 traffic routing requirement.	
traffCorreInd	boolean	O	0..1	Indication of traffic correlation. May only be included when "interGroupId" attribute is included and not set to "AnyUE". It is used to indicate that for the group of UEs, the targeted PDU sessions should be correlated by a common DNAI. Set to "true" if it should be correlated; otherwise set to "false". Default value is "false" if omitted. (NOTE 7)	
validStartTime	DateTime	O	0..1	Identifies when the traffic routings start to be applicable. (NOTE 4)	
validEndTime	DateTime	O	0..1	Identifies when the traffic routings are not applicable. (NOTE 4)	
tempValidities	array(TemporalValidity)	O	1..N	Indicates the time interval(s) during which the AF request is to be applied. (NOTE 4)	MultiTemporalCondition
nwAreaInfo	NetworkAreaInfo	O	0..1	Identifies a network area information that the request applies only to the traffic of UE(s) located in this specific zone.	
upPathChgNotifUri	Uri	C	0..1	Contains the URI where the NEF receives the UP path change notification. It shall be included when the NEF requests the UP path change notification.	
headers	array(string)	O	1..N	Headers provisioned by the NEF to be used by other NFs to interact with the NEF. E.g. 3gpp-Sbi-Binding header with the binding indication for the URI where the NEF receives UP path change notification and whose information is used by the SMF. The encoding of the header shall comply with clause 3.2 of IETF RFC 7230 [21]. (NOTE 6)	
subscribedEvents	array(SubscribedEvent)	O	1..N	Identifies the requirement to be notified of the event(s).	
dnaiChgType	DnaiChangeType	O	0..1	Identifies a type of notification regarding UP path management event.	

afAckInd	boolean	O	0..1	Identifies whether the AF acknowledgement of UP path event notification is expected. Set to "true" if the AF acknowledgement is expected; otherwise set to "false". Default value is "false" if omitted.	URLLC
addrPreserInd	boolean	O	0..1	Indicates whether UE IP address should be preserved. This attribute shall set to "true" if preserved, otherwise, set to "false". Default value is false if omitted.	URLLC
maxAllowedUpLat	UintegerRm	O	0..1	Indicates the target user plane latency in units of milliseconds. The SMF may use this value to decide whether edge relocation is needed to ensure that the user plane latency does not exceed the value.	AF_latency
simConnInd	boolean	O	0..1	Indication of simultaneous connectivity temporarily maintained for the source and target PSA. It is used to indicate whether the simultaneous connectivity should be temporarily mantained for the source and target PSA. It is set to "true" if the temporary simultaneous connectivity should be temporarily maintained; otherwise, it is set to "false". Default value is "false" if omitted.	SimultConnectivity
simConnTerm	DurationSec	C	0..1	Indication of the minimum time interval to be considered for inactivity of the traffic routed via the source PSA during the edge re-location procedure before removing the source PSA. It may be included when the "simConnInd" attribute is set to true.	SimultConnectivity
supportedFeatures	SupportedFeatures	C	0..1	Indicates the list of negotiated supported features. This attribute shall be supplied by the UDR in the response to the PUT request when it was present in the PUT request and the UDR supports feature negotiation for Influence Data. This attribute shall be supplied by the UDR in the response to the GET request when the GET request includes the "supp-feat" query parameter and the UDR supports feature negotiation for Influence Data.	
resUri	Uri	C	0..1	Represents the URI of Individual Influence Data. It shall only be included in the HTTP GET response.	EnhancedInfluDataNotification
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	

NOTE 1: One of "afAppId", "trafficFilters" or "ethTrafficFilters" shall be included.
NOTE 2: Either "supi" or "interGroupId" shall be included.
NOTE 3: If the EnhancedInfluDataNotification feature is not supported, to indicate the deletion of a Traffic Individual Influence Data resource, only the appropriate combination of "supi" or "interGroupId", and "afAppId", "trafficFilters" or "ethTrafficFilters" that identify the resource shall be included. The rest of attributes shall be omitted.
NOTE 4: Properties "validStartTime" and "validEndTime" shall only be included for single temporal validity condition. Property "tempValidities" shall only be included for multiple temporal validity conditions when the feature MultiTemporalCondition is supported.
NOTE 5: If the Traffic Influence Data applies to any UE, then the "interGroupId" sets to "AnyUE".
NOTE 6: In this release of the specification, the property "headers" only includes the 3gpp-Sbi-Binding header with the binding indication for the URI included in the property "upPathChgNotifUri".
NOTE 7: The indication of traffic correlation shall be provided only when the AF indicated that all the PDU sessions related to the 5G VN group member UEs should be correlated by a common DNAI in the user plane for the traffic as described in 3GPP TS 29.522 [19].

6.4.2.3 Type TrafficInfluDataPatch

Table 6.4.2.3-1: Definition of type TrafficInfluDataPatch

Attribute name	Data type	P	Cardinality	Description	Applicability
upPathChgNotifCorrId	string	O	0..1	Contains the Notification Correlation Id allocated by the NEF for the UP path change notification.	
appReloInd	boolean	O	0..1	Identifies whether an application can be relocated once a location of the application has been selected.	
ethTrafficFilters	array(EthFlowDescription)	O	1..N	Identifies Ethernet packet filters.	
trafficFilters	array(FlowInfo)	O	1..N	Identifies IP packet filters	
trafficRoutes	array(RouteToLocation)	O	1..N	Identifies the N6 traffic routing requirement.	
traffCorreInd	boolean	O	0..1	Indication of traffic correlation. May only be included when "interGroupId" attribute was previously included in the "TrafficInfluData" data type and not set to "AnyUE". It is used to indicate that for the group of UEs, the targeted PDU sessions should be correlated by a common DNAI.	
validStartTime	DateTime	O	0..1	Identifies when the traffic routings start to be applicable.	
validEndTime	DateTime	O	0..1	Identifies when the traffic routings are not applicable.	
tempValidities	array(TemporalValidity)	O	1..N	Indicates the time interval(s) during which the AF request is to be applied.	MultiTemporalCondition
nwAreaInfo	NetworkAreaInfo	O	0..1	Identifies a network area information that the request applies only to the traffic of UE(s) located in this specific zone.	
upPathChgNotifUri	Uri	O	0..1	Contains the URI where the NEF receives the UP path change notification.	
headers	array(string)	O	1..N	Headers provisioned by the NEF. E.g. 3gpp-Sbi-Binding header with the binding indication for the URI where the NEF receives UP path change notification. The encoding of the header shall comply with clause 3.2 of IETF RFC 7230 [21]	
afAckInd	boolean	O	0..1	Identifies whether the AF acknowledgement of UP path event notification is expected.	URLLC
addrPreserInd	boolean	O	0..1	Indicates UE IP address should be preserved.	URLLC
maxAllowedUpLat	Uinteger	O	0..1	Indicates the target user plane latency in units of milliseconds. The SMF may use this value to decide whether edge relocation is needed to ensure that the user plane latency does not exceed the value.	AF_latency
simConnInd	boolean	O	0..1	Indication of simultaneous connectivity temporarily maintained for the source and target PSA. It is set to "true" if the temporary connectivity should be temporarily maintained; otherwise, it is set to "false".	SimultConnectivity
simConnTerm	DurationSecRm	C	0..1	Indication of the minimum time interval to be considered for inactivity of the traffic routed via the source PSA during the edge re-location procedure before removing the source PSA.	SimultConnectivity

6.4.2.4 Type TrafficInfluSub

Table 6.4.2.4-1: Definition of type TrafficInfluSub

Attribute name	Data type	P	Cardinality	Description	Applicability
dnns	array(Dnn)	O	1..N	Each element identifies a DNN.	
snssais	array(Snssai)	O	1..N	Each element identifies a slice.	
internalGroupIds	array(GroupId)	O	1..N	Each element identifies a group of users.(NOTE 2)	
supis	array(Supi)	O	1..N	Each element identifies the user.	
notificationUri	Uri	M	1	URI provided by the NF service consumer indicating where to receive the subscribed notifications from the UDR.	
expiry	DateTime	C	0..1	This IE shall be included in a subscription response if, based on operator policy and taking into account the expiry time included in the request, the UDR needs to include an expiry time. This IE may be included in a subscription request. When present, this IE shall represent the time after which the subscription becomes invalid. The absence of this attribute in the subscription response means the subscription to be valid without an expiry time.	
supportedFeatures	SupportedFeatures	C	0..1	Used to negotiate the applicability of the optional features. This attribute shall be provided in the POST request and in the response of successful resource creation.	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	
NOTE 1: At least one of "dnns", "snssais", "internalGroupIds" or "supis" shall be provided. NOTE 2: For the Traffic Influence Data applies to any UE, the "internalGroupIds" needs to have an entry set to "AnyUE".					

6.4.2.5 Void

6.4.2.6 Type: PfdDataForAppExt

Table 6.4.2.6-1: Definition of type PfdDataForAppExt

Attribute name	Data type	P	Cardinality	Description	Applicability
suppFeat	SupportedFeatures	O	0..1	Used to negotiate the applicability of the optional features. This attribute shall be provided in the PUT request and in the response of successful resource creation. This attribute shall be provided in the HTTP GET response if the "supp-feat" attribute query parameter is included in the HTTP GET request.	
applicationId	ApplicationId	M	1	Identifier of an application.	
pfds	array(PfdContent)	M	1..N	PFDs for the application identifier.	
cachingTime	DateTime	O	0..1	Caching time for an application identifier.	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	
allowedDelay	DurationSec	O	0..1	Indicates that the list of PFDs in this request should be deployed within the time interval indicated by the Allowed Delay.	

6.4.2.7 Type BdtPolicyData

Table 6.4.2.7-1: Definition of type BdtPolicyData

Attribute name	Data type	P	Cardinality	Description	Applicability
interGroupId	GroupId	O	0..1	Identifies a group of users. (NOTE)	
supi	Supi	O	0..1	Identifies a user. (NOTE)	
bdtRefId	BdtReferenceId	M	1	Identifies a selected policy of background data transfer.	
dnn	Dnn	O	0..1	Identifies a DNN	
snssai	Snssai	O	0..1	The identification of slice.	
resUri	Uri	C	0..1	Represents the URI of Individual Applied BDT Policy Data. It shall only be included in the HTTP GET response.	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	
NOTE: Either "supi" or "interGroupId" shall be included.					

6.4.2.8 Type BdtPolicyDataPatch

Table 6.4.2.8-1: Definition of type BdtPolicyDataPatch

Attribute name	Data type	P	Cardinality	Description	Applicability
bdtRefId	BdtReferenceId	M	1	Identifies a selected policy of background data transfer.	

6.4.2.9 Type IptvConfigData

Table 6.4.2.9-1: Definition of type IptvConfigData

Attribute name	Data type	P	Cardinality	Description	Applicability
supi	Supi	C	0..1	Identifies a user. (NOTE)	
interGroupId	GroupId	C	0..1	Identifies a group of users. (NOTE)	
dnn	Dnn	O	0..1	Identifies a DNN	
snssai	Snssai	O	0..1	The identification of slice.	
afAppId	string	M	1	Identifies an application.	
multiAccCtrls	map(MulticastAccessControl)	M	1..N	Identifies a list of multicast address access control information. Any string value can be used as a key of the map.	
suppFeat	SupportedFeatures	C	0..1	Indicates the negotiated supported features. This attribute shall be provided in the PUT request and in the response of successful resource creation.	
resUri	Uri	C	0..1	Represents the URI of Individual IPTV Configuration. It shall only be included in the HTTP GET response.	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	
NOTE: Either "supi" or "interGroupId" shall be included.					

6.4.2.10 Type ApplicationDataSubs

Table 6.4.2.10-1: Definition of type ApplicationDataSubs

Attribute name	Data type	P	Cardinality	Description	
notificationUri	Uri	M	1	URI provided by the NF service consumer indicating where to receive the subscribed notifications from the UDR.	Applicability
dataFilters	array(DataFilter)	O	1..N	Each element identifies a data filter for subscription to data change.notification. If absent, the subscription applies to all data change notification except traffic influence data.	
expiry	DateTime	C	0..1	This IE shall be included in a subscription response if, based on operator policy and taking into account the expiry time included in the request, the UDR needs to include an expiry time. This IE may be included in a subscription request. When present, this IE shall represent the time after which the subscription becomes invalid. The absence of this attribute in the subscription response means the subscription to be valid without an expiry time.	
immRep	boolean	O	0..1	This attribute is applicable only if the "dataFilters" attribute of the request includes an element with the "dataInd" attribute set to "AM". If provided and set to "true", it indicates that existing matching entries shall be immediately reported within the "amInfluEntries" attribute in the response.	DCAMP
amInfluEntries	array(AmInfluData)	O	1..N	The AM Influence Data stored in the UDR that match this subscription. It may be included only in the POST response body of a subscription creation or modification, and only if the request included the "immRep" attribute set to true and a "dataFilters" attribute that contained an element with the "dataInd" attribute set to "AM".	DCAMP
supportedFeatures	SupportedFeatures	C	0..1	Used to negotiate the applicability of the optional features. This attribute shall be provided in the POST request and in the response of successful resource creation.	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	

When multiple attributes defined as an array are included within an entry of the "dataFilters" attribute, the UDR shall determine that the subscription is for all application data of type "dataInd" that contain a matching combination of the values of the elements of the array of the provided attributes.

EXAMPLE: For a subscription to notification on changes of Service Parameter Data resources ("dataInd" is set to "SVC_PARAM"), if "snssais" attribute is included with two S-NSSAI entries (S-NSSAI_1 and S-NSSAI_2) and "dnns" is included with two DNN entries (e.g. DNN_A and DNN_B), and the other optional attributes are not included, the UDR shall notify about changes of Service Parameter Data resources that contain a matching S-NSSAI and DNN combination, i.e. the resource(s) matching SNSSAI_1 and DNN_A, the resource(s) matching SNSSAI_1 and DNN_B, the resource(s) matching SNSSAI_2 and DNN_A, and the resources matching SNSSAI_2 and DNN_B.

6.4.2.11 Type ApplicationDataChangeNotif

Table 6.4.2.11-1: Definition of type ApplicationDataChangeNotif

Attribute name	Data type	P	Cardinality	Description	Applicability
iptvConfigData	IptvConfigData	O	0..1	IPTV Configuration Data. (NOTE)	
pfdData	PfdChangeNotification	O	0..1	PFD Data. (NOTE)	
bdtPolicyData	BdtPolicyData	O	0..1	BDT Policy Data. (NOTE)	
serParamData	ServiceParameterData	O	0..1	Service Parameter Data, if changed and notification was requested. (NOTE)	
amInfluData	AmInfluData	O	0..1	AM Influence Data (NOTE)	
resUri	Uri	M	1	Identifies the resource in the corresponding data change. For notifying deletion, only resUri shall be provided in the ApplicationDataChangeNotif data type.	
NOTE: Only one among those attributes shall be provided in notifying data creation or update.					

6.4.2.12 Type DataFilter

Table 6.4.2.12-1: Definition of type DataFilter

Attribute name	Data type	P	Cardinality	Description	Applicability
dataInd	DataInd	M	1	Indicate the type of application data. When used in subscription procedures, the UDR shall notify only about changes of application data of type "dataInd", and match also the filters indicated in the rest of the attributes of DataFilter (if any).	
dnns	array(Dnn)	O	1..N	Each element identifies a DNN. When used in subscription procedures, the UDR shall notify only about changes of application data of type "dataInd" that contain at least one of the DNNs included in the array, and match also the filters indicated in the rest of the attributes of DataFilter (if any). (NOTE 3)	
snssais	array(Snssai)	O	1..N	Each element identifies a slice. When used in subscription procedures, the UDR shall notify only about changes of application data of type "dataInd" that contain at least one of the S-NSSAIs included in the array, and match also the filters indicated in the rest of the attributes of DataFilter (if any). (NOTE 3)	
internalGroupIds	array(GroupId)	O	1..N	Each element identifies a group of users. When used in subscription procedures, the UDR shall notify only about changes of application data of type "dataInd" that contain at least one of the Group IDs included in the array, and match also the filters indicated in the rest of the attributes of DataFilter (if any).	
supis	array(Supi)	O	1..N	Each element identifies the user. When used in subscription procedures, the UDR shall notify only about changes of application data of type "dataInd" that contain at least one of the SUPIs included in the array, and match also the filters indicated in the rest of the attributes of DataFilter (if any).	
appIds	array(ApplicationId)	O	1..N	Each element identifies an application. When used in subscription procedures, the UDR shall notify only about changes of application data of type "dataInd" that contain at least one of the ApplicationIds included in the array, and match also the filters indicated in the rest of the attributes of DataFilter (if any).	
ueIpv4s	array(Ipv4Addr)	O	1..N	Each element identifies the user. When used in subscription procedures, the UDR shall notify only about changes of application data of type "dataInd" that contain at least one of the IPv4 address(es) included in the array, and match also the filters indicated in the rest of the attributes of DataFilter (if any).	

ueIpv6s	array(Ipv6Addr)	O	1..N	Each element identifies the user. When used in subscription procedures, the UDR shall notify only about changes of application data of type "dataInd" that contain at least one of the IPv6 addresses included in the array, and match also the filters indicated in the rest of the attributes of DataFilter (if any).	
ueMacs	array(MacAddr48)	O	1..N	Each element identifies the user. When used in subscription procedures, the UDR shall notify only about changes of application data of type "dataInd" that contain at least one of the UE MAC address(es) included in the array, and match also the filters indicated in the rest of the attributes of DataFilter (if any).	
anyUeInd	boolean	O		Indicates whether the request is for any UE: - "true": the request is for any UE; - "false"(default): the request is not for any UE. When used in subscription procedures, the UDR shall notify only about changes of application data of type "dataInd" that contain an any UE indication, and match also the filters indicated in the rest of the attributes of DataFilter (if any).	FilterAnyUE
dnnSnssaiInfos	array(DnnSnssaiInformation)	O	1..N	Each element identifies a combination of (DNN, S-NSSAI). When used in subscription procedures, the UDR shall only notify about changes in resource(s) with "dataInd" value AM that contain at least one DNN and S-NSSAI combination indicated in the array, and match also the rest of the filters (if any). (NOTE 3)	DCAMP
NOTE 1: The "dnns", "snssais", "internalGroupIds" and "supis" attributes are applicable to the BDT Policy Data, IPTV Configuration Data, Service Parameter Data, and AM Influence Data, the "ueIpv4s" "ueIpv6s" and "ueMacs" attributes are applicable to the Service Parameter Data, the "appIds" attribute is applicable to the PFD Data, Service Parameter Data and IPTV Configuration Data, the "anyUeInd" attribute is applicable to Service Parameter Data and AM Influence Data, and the "dnnSnssaiInfos" is applicable to AM Influence Data. NOTE 2: At application datat changes, subscriptions to changes based on attributes of this table are considered to be matched, and the corresponding notifications are sent, no matter if the matching application data matches the provided values before the change, after the change, or both before and after the change. NOTE 3: For AM Influence Data, only one of the "dnns", "snssais", or "dnnSnssaiInfos" attributes may be simultaneously provided.					

NOTE: When the "anyUeInd" attribute is related to the subscription to notification of changes of the service parameter data resource(s), and it refers to any UE using the service identified by a combination of DNN and S-NSSAI, the "anyUeInd" attribute is present together with the "dnns" and "snssais" properties. Note also that, if the "anyUeInd" attribute is present together with the "internalGroupIds", "supis", "ueIpv4s", "ueIpv6s" and/or "ueMacs" attributes the subscription will not match any resource, since according to clause 6.4.2.15 only one of the the "supi", "anyUeInd", "interGroupId", "ueIpv4", "ueIpv6" or "ueMac" properties are simultaneously present in the resource.
When the "anyUeInd" attribute is related to the subscription to notification of changes of the AM Influence data resource(s), and it refers to any UE using the service(s) identified by the corresponding DNN and S-NSSAI, the "anyUeInd" attribute is present together with the "dnnSnssaiInfos" property. Note also that, if the "anyUeInd" attribute is present together with the "internalGroupIds" and/or "supis" attributes the subscription will not match any resource, since according to clause 6.4.2.16 only one of the the "supi", "anyUeInd", or "interGroupId" properties are simultaneously present in the resource.

6.4.2.13 Void

6.4.2.14 Type TrafficInfluDataNotif

Table 6.4.2.14-1: Definition of type TrafficInfluDataNotif

Attribute name	Data type	P	Cardinality	Description	Applicability
resUri	Uri	M	1	Represents the URI of Individual Influence Data. For notifying deletion, only resUri shall be provided in the TrafficInfluDataNotif data type.	
trafficInfluData	TrafficInfluData	C	0..1	Traffic Influence Data. It shall be present for notifying Individual Influence Data creation or update.	

6.4.2.15 Type ServiceParameterData

Table 6.4.2.15-1: Definition of type ServiceParameterData

Attribute name	Data type	P	Cardinality	Description	Applicability
dnn	Dnn	O	0..1	Identifies a DNN. (NOTE 2)	
snssai	Snssai	O	0..1	Identifies an S-NSSAI. (NOTE 2)	
appId	string	O	0..1	Identifies an application identifier. (NOTE 2)	
supi	Supi	O	0..1	Identifies a user. (NOTE1)	
ueIpv4	Ipv4Addr	O	0..1	The IPv4 address of the served UE. (NOTE1)	
ueIpv6	Ipv6Addr	O	0..1	The IPv6 address of the served UE. (NOTE1)	
ueMac	MacAddr48	O	0..1	The MAC address of the served UE. (NOTE1)	
interGroupId	GroupId	O	0..1	Identifies a group of users. (NOTE1)	
anyUeInd	boolean	O	0..1	Identifies whether the service parameters applies to any UE. This attribute shall set to "true" if applicable for any UE, otherwise, set to "false". (NOTE1)	
paramOverPc5	ParameterOverPc5	O	0..1	Contains the V2X service parameters used over PC5	
paramOverUu	ParameterOverUu	O	0..1	Contains the V2X service parameters used over Uu	
urspGuidance	array(UrspRuleRequest)	O	1..N	Contains the service parameter used to guide the URSP.	AfGuideURSP
paramForProSeDd	ParamForProSeDd	O	0..1	Contains the service parameters for 5G ProSe direct discovery.	ProSe
paramForProSeDc	ParamForProSeDc	O	0..1	Contains the service parameters for 5G ProSe direct communications.	ProSe
paramForProSeU2NRelUe	ParamForProSeU2NRelUe	O	0..1	Contains the service parameters for 5G ProSe UE-to-network relay UE.	ProSe
paramForProSeRemUe	ParamForProSeRemUe	O	0..1	Contains the service parameters for 5G ProSe remote UE.	ProSe
deliveryEvents	array(Event)	O	1..N	Contains the events related to the outcome of UE policy delivery.	DeliveryOutcome
policDelivNotifCorreId	string	C	0..1	Contains the Notification Correlation Id allocated by the NEF for the notification of UE Policy delivery outcome. It shall be included when the NEF requests the notification of UE Policy delivery outcome.	DeliveryOutcome
policDelivNotifUri	Uri	C	0..1	Contains the URI where the NEF receives the notification of UE Policy delivery outcome. It shall be included when the NEF requests the notification of UE policy delivery outcome.	DeliveryOutcome
headers	array(string)	O	1..N	Headers provisioned by the NEF. E.g. 3gpp-Sbi-Binding header (as specified in 3GPP TS 29.500 [4]) with the binding indication for the URI included in the policDelivNotifUri attribute. The encoding of the header shall comply with clause 3.2 of IETF RFC 7230 [21]	DeliveryOutcome

suppFeat	SupportedFeatures	C	0..1	Indicates the list of Supported features used as described in clause 5.8. This parameter shall be supplied in the PUT request that requested the creation of an individual Service parameter resource, and in the PUT response.	
resUri	Uri	C	0..1	Represents the URI of Individual Service Parameter Data. It shall only be included in the HTTP GET response.	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	
NOTE 1: Only one of the "supi", "anyUeInd", "interGroupId", "ueIpv4", "ueIpv6" or "ueMac" attribute shall be provided. NOTE 2: Only the combination of "dnn" and "snssai" or "appId" attribute shall be provided.					

6.4.2.15A Type ServiceParameterDataPatch

Table 6.4.2.15A-1: Definition of type ServiceParameterDataPatch

Attribute name	Data type	P	Cardinality	Description	Applicability
paramOverPc5	ParameterOverPc5	O	0..1	Contains the V2X service parameters used over PC5	
paramOverUu	ParameterOverUu	O	0..1	Contains the V2X service parameters used over Uu	
urspInfluence	array(UrspRuleRequest)	O	1..N	Contains the service parameter used to influence the URSP. This attribute is deprecated by the "urspGuidance" attribute that should be used instead.	AfGuideURSP
urspGuidance	array(UrspRuleRequest)	O	1..N	Contains the service parameter used to guide the URSP.	AfGuideURSPPatchCorrection
paramForProSeDd	ParamForProSeDd	O	0..1	Contains the service parameters for 5G ProSe direct discovery.	ProSe
paramForProSeDc	ParamForProSeDc	O	0..1	Contains the service parameters for 5G ProSe direct communications.	ProSe
paramForProSeU2NRelUe	ParamForProSeU2NRelUe	O	0..1	Contains the service parameters for 5G ProSe UE-to-network relay UE.	ProSe
paramForProSeRemUe	ParamForProSeRemUe	O	0..1	Contains the service parameters for 5G ProSe remote UE.	ProSe
deliveryEvents	array(Event)	O	1..N	Contains the events related to the outcome of UE policy delivery.	DeliveryOutcome
policDelivNotifUri	Uri	C	0..1	Contains the URI where the NEF receives the notification of UE Policy delivery outcome. It shall be included when the NEF requests the notification of UE policy delivery outcome.	DeliveryOutcome
headers	array(string)	O	1..N	Headers provisioned by the NEF. E.g. 3gpp-Sbi-Binding header (as specified in 3GPP TS 29.500 [4]) with the binding indication for the URI included in the policDelivNotifUri IE. The encoding of the header shall comply with clause 3.2 of IETF RFC 7230 [21]	DeliveryOutcome

NOTE: In this release of the specification, the "paramOverPc5", "paramOverUu", "paramForProSeDd", "paramForProSeDc", "paramForProSeU2NRelUe", "paramForProSeRemUe" and "deliveryEvents" attributes are defined as not nullable. The removal of these attributes cannot be performed using HTTP PATCH method and can only be performed using the HTTP PUT method.

6.4.2.16 Type AmInfluData

Table 6.4.2.16-1: Definition of type AmInfluData

Attribute name	Data type	P	Cardinality	Description	Applicability
appIds	array(string)	O	1..N	Identifies one or more applications.	
dnnSnssaiInfos	array(DnnSnssaiInformation)	O	1..N	Each element identifies a combination of (DNN, S-NSSAI).	
interGroupId	GroupId	O	0..1	Identifies a group of users. (NOTE 1)	
supi	Supi	O	0..1	Identifies a user. (NOTE 1)	
anyUeInd	boolean	O	0..1	Indicates whether the data applies to any UE. If present and set to "true" the data is applicable for any UE. (NOTE 1)	
policyDuration	DurationSec	O	0..1	Indicates the time duration that the policy shall last.	
evSubs	array(AmInfluEvent)	O	1..N	List of AM related events for which a subscription is required.	
headers	array(string)	O	1..N	Headers provisioned by the NEF. E.g. 3gpp-Sbi-Binding header (as specified in 3GPP TS 29.500 [4]) with the binding indication for the URI included in the notifUri IE. The encoding of the header shall comply with clause 3.2 of IETF RFC 7230 [21]	
notifUri	Uri	C	0..1	Contains the Callback URI to receive notifications. It shall be present if the "evSubs" attribute is present.	
notifCorrId	string	O	0..1	Notification correlation identifier.	
thruReq	boolean	O	0..1	If present and set to "true", it indicates that high throughput is desired for the indicated UE traffic. (NOTE 2)	
covReq	array(ServiceAreaCoverageInfo)	O	1..N	Identifies service area coverage requirements. (NOTE 2)	
supportedFeatures	SupportedFeatures	C	0..1	Indicates the list of negotiated supported features. This attribute shall be supplied by the UDR in the response to the PUT request when it was present in the PUT request and the UDR supports feature negotiation for AM Influence Data. This attribute shall be supplied by the UDR in the response to the GET request when the GET request includes the "supp-feat" query parameter and the UDR supports feature negotiation for AM Influence Data.	
resUri	Uri	C	0..1	Represents the URI of Individual AM Influence Data. It shall only be included in the HTTP GET response.	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	
NOTE 1: One of "supi", "interGroupId", and "anyUeInd" shall be included. NOTE 2: At least one of these attributes shall be included.					

6.4.2.17 Type AmInfluDataPatch

Table 6.4.2.17-1: Definition of type AmInfluDataPatch

Attribute name	Data type	P	Cardinality	Description	Applicability
appIds	array(string)	O	1..N	Identifies one or more applications.	
dnnSnssaiInfos	array(DnnSnssaiInformation)	O	1..N	Each element identifies a combination of (DNN, S-NSSAI).	
policyDuration	DurationSecRm	O	0..1	Indicates the time duration that the policy shall last.	
evSubs	array(AmInfluEvent)	O	1..N	List of AM related events for which a subscription is required.	
headers	array(string)	O	1..N	Headers provisioned by the NEF. E.g. 3gpp-Sbi-Binding header (as specified in 3GPP TS 29.500 [4]) with the binding indication for the URI included in the notifUri IE. The encoding of the header shall comply with clause 3.2 of IETF RFC 7230 [21]	
notifUri	UriRm	C	0..1	Contains the Callback URL to receive notifications.	
notifCorrId	string	O	0..1	Notification correlation identifier.	
thruReq	boolean	O	0..1	If present and set to "true", it indicates that high throughput is desired for the indicated UE traffic.	
covReq	array(ServiceAreaCoverageInfo)	O	1..N	Identifies service area coverage requirements.	

6.4.3 Simple data types and enumerations

6.4.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

6.4.3.2 Simple data types

The simple data types defined in table 6.4.3.2-1 shall be supported.

Table 6.4.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability

6.4.3.3 Enumeration: DataInd

Table 6.4.3.3-1: Enumeration DataInd

Enumeration value	Description	Applicability
PFD	PFD data	
IPTV	IPTV configuration data	
BDT	BDT data	
SVC_PARAM	Service parameter data	
AM	AM influence data	DCAMP

6.5 Error handling

6.5.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of 3GPP TS 29.500 [4].

For the Nudr_DataRepository Service API for Application Data, HTTP error responses shall be supported as specified in clause 4.8 of 3GPP TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of 3GPP TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of 3GPP TS 29.500 [4]. In addition, the requirements in the following clauses shall apply.

6.5.2 Protocol Errors

In this Release of the specification, there are no additional protocol errors applicable for the Nudr_DataRepository Service API for Application Data.

6.5.3 Application Errors

The application errors defined for the Nudr_DataRepository Service API for Application Data are listed in table 6.5.3-1.

Table 6.5.3-1: Application errors

Application Error	HTTP status code	Description
NOTE: Including a "ProblemDetails" data structure with the "cause" attribute in the HTTP response is optional unless explicitly mandated by other clauses in this document.		

The application error handling shall follow 3GPP TS 29.504 [6] clause 6.1.6 unless explicitly specified in this document.

6.6 Feature negotiation

The optional features in table 6.1.8-1 of 3GPP TS 29.504 [6] are defined for the Nudr_DataRepository API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of 3GPP TS 29.500 [4].

7 Usage of Nudr_DataRepository Service API for Structured Data for Exposure

7.1 Introduction

The following clauses specify the usage of Nudr_DataRepository service for exposure data. The principles specified in 3GPP TS 29.504 [6] are followed unless explicitly specified otherwise in the following clauses.

7.2 Resources

7.2.1 Overview

This clause specifies the resources for the usage of the Unified Data Repository service for Structured Data for Exposure.

7.2.2 Resource Structure

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 7.2.2-1 depicts the resource URIs structure for the Nudr_DataRepository API for exposure data data.

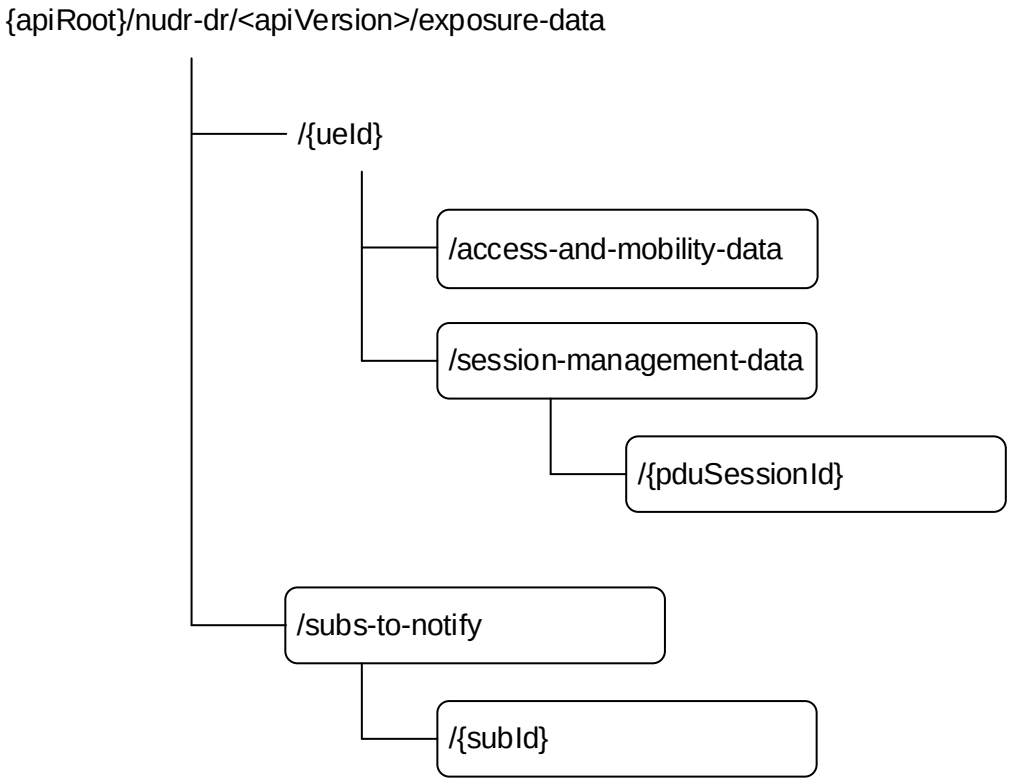


Figure 7.2.2-1: Resource URI structure of the Nudr_DataRepository API for exposure data

Table 7.2.2-1 provides an overview of the resources and applicable HTTP methods.

Table 7.2.2-1: Resources and methods overview

Resource name	Resource URI	HTTP method	Description
AccessAndMobility Data	/exposure-data/{ueId}/ access-and-mobility-data	PUT	Create and update the access and mobility exposure data for a UE
		PATCH	Update the access and mobility exposure data for a UE.
		GET	Retrieve the access and mobility exposure data for a UE
		DELETE	Delete the access and mobility exposure data for a UE
PduSessionManag ementData	/exposure-data/{ueId}/ session-management-data/ {pduSessionId}	PUT	Create and update the session management data for a UE and for an individual PDU session
		GET	Retrieve the session management data for a UE and for an individual PDU session
		DELETE	Delete the session management data for a UE and for an individual PDU session
ExposureDataSubs criptions	/exposure-data/subs-to-notify	POST	Create a subscription to receive notifications on exposure data changes.
IndividuaExposure DataSubscription	/exposure-data/subs-to-notify/ {subId}	PUT	Modify a subscription to receive notifications on exposure data changes.
		DELETE	Delete a subscription identified by {subId}.

7.2.3 Resource: AccessAndMobilityData

7.2.3.1 Description

This resource represents the exposure data related to access and mobility.

7.2.3.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/exposure-data/{ueId}/access-and-mobility-data**

This resource shall support the resource URI variables defined in table 7.2.3.2-1.

Table 7.2.3.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
ueId	VarUeId	Represents the SUPI or GPSI. Data type VarUeId is defined in 3GPP TS 29.571 [7].

7.2.3.3 Resource Standard Methods

7.2.3.3.1 PUT

This method shall support the URI query parameters specified in table 7.2.3.3.1-1.

Table 7.2.3.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 7.2.3.3.1-2 and the response data structures and response codes specified in table 7.2.3.3.1-3.

Table 7.2.3.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
AccessAndMobilityData	M	1	Access and mobility data for the UE is created or updated.

Table 7.2.3.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AccessAndMobilityData	M	1	201 Created	Successful case. The resource has been successfully created and a response body containing a representation of the access and mobility data shall be returned.
AccessAndMobilityData	M	1	200 OK	Successful case. The resource has been successfully updated and a response body containing a representation of the access and mobility data shall be returned.
n/a			204 No Content	Successful case. The resource has been successfully updated and no additional content is to be sent in the response message.
NOTE: The HTTP status code for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 7.2.3.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/exposure-data/{ueId}/access-and-mobility-data

7.2.3.3.2 GET

This method shall support the URI query parameters specified in table 7.2.3.3.2-1.

Table 7.2.3.3.2-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
supp-feat	SupportedFeatures	O	0..1	The features supported by the NF service consumer. See 3GPP TS 29.500 [4] clause 6.6.2 and 3GPP TS 29.571 [7].

This method shall support the request data structures specified in table 7.2.3.3.2-2 and the response data structures and response codes specified in table 7.2.3.3.2-3.

Table 7.2.3.3.2-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 7.2.3.3.2-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AccessAndMobilityData	M	1	200 OK	The response body contains the access and mobility data.
NOTE: The HTTP status code for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

7.2.3.3.3 DELETE

This method shall support the URI query parameters specified in table 7.2.3.3.3-1.

Table 7.2.3.3.3-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 7.2.3.3.3-2 and the response data structures and response codes specified in table 7.2.3.3.3-3.

Table 7.2.3.3.3-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 7.2.3.3.3-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon success, an empty response body shall be returned.
NOTE: The HTTP status code for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

7.2.3.3.4 PATCH

This method shall support the URI query parameters specified in table 7.2.3.3.4-1.

Table 7.2.3.3.4-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 7.2.3.3.4-2 and the response data structures and response codes specified in table 7.2.3.3.4-3.

Table 7.2.3.3.4-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
AccessAndMobilityData	M	1	Access and mobility data for the UE.

Table 7.2.3.3.4-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon successful modification there is no body in the response message.
NOTE: The HTTP status code for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

7.2.4 Resource: PduSessionManagementData

7.2.4.1 Description

This resource represents the exposure data related to session and management data for a given subscriber and a given PDU session.

7.2.4.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/exposure-data/{ueId}/session-management-data/{pduSessionId}**

This resource shall support the resource URI variables defined in table 7.2.4.2-1.

Table 7.2.4.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
ueId	VarUeId	Represents the SUPI or GPSI. Data type VarUeId is defined in 3GPP TS 29.571 [7].
pduSessionId	PduSessionId	Identifies an individual PDU session. Data type PduSessionId is defined in 3GPP TS 29.571 [7].

7.2.4.3 Resource Standard Methods

7.2.4.3.1 PUT

This method shall support the URI query parameters specified in table 7.2.4.3.1-1.

Table 7.2.4.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 7.2.4.3.1-2 and the response data structures and response codes specified in table 7.2.4.3.1-3.

Table 7.2.4.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
PduSessionManagementData	M	1	Session management data for a UE and a PDU session is created or updated.

Table 7.2.4.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response Codes	Description
PduSessionManagementData	M	1	201 Created	Successful case. The resource has been successfully created and a response body containing a representation of the session management data shall be returned.
PduSessionManagementData	M	1	200 OK	Successful case. The resource has been successfully updated and a response body containing a representation of the session management data shall be returned.
n/a			204 No Content	Successful case. The resource has been successfully updated and no additional content is to be sent in the response message.
NOTE: The HTTP status code for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 7.2.4.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/exposure-data/{ueId}/session-management-data/{pduSessionId}

7.2.4.3.2 GET

This method shall support the URI query parameters specified in table 7.2.4.3.2-1.

Table 7.2.4.3.2-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
ipv4-addr	Ipv4Addr	O	0..1	UE IPv4 address.
ipv6-prefix	Ipv6Prefix	O	0..1	UE IPv6 prefix.
dnn	Dnn	O	0..1	Identifies a DNN.
fields	array(string)	C	1..N	When the NF consumer only retrieves a subset of the resource, the "fields" query parameter shall be included. The "fields" query parameter contains the pointers of the attribute(s) to be retrieved.
supp-feat	SupportedFeatures	O	0..1	The features supported by the NF service consumer. See 3GPP TS 29.500 [4] clause 6.6.2 and 3GPP TS 29.571 [7].

This method shall support the request data structures specified in table 7.2.4.3.2-2 and the response data structures and response codes specified in table 7.2.4.3.2-3.

Table 7.2.4.3.2-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 7.2.4.3.2-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response Codes	Description
PduSessionManagement Data	M	1	200 OK	The response body contains the session management data.
NOTE: The HTTP status code for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

7.2.4.3.3 DELETE

This method shall support the URI query parameters specified in table 7.2.4.3.3-1.

Table 7.2.4.3.3-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 7.2.4.3.3-2 and the response data structures and response codes specified in table 7.2.4.3.3-3.

Table 7.2.4.3.3-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 7.2.4.3.3-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon success, an empty response body shall be returned.
NOTE: The HTTP status code for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

7.2.5 Resource: ExposureDataSubscriptions

7.2.5.1 Description

This resource is used to represent subscriptions to notifications of exposure data modifications.

7.2.5.2 Resource Definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/exposure-data/subs-to-notify**

This resource shall support the resource URI variables defined in table 7.2.5.2-1.

Table 7.2.5.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.

7.2.5.3 Resource Standard Methods

7.2.5.3.1 POST

This method shall support the URI query parameters specified in table 7.2.5.3.1-1.

Table 7.2.5.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 7.2.5.3.1-2 and the response data structures and response codes specified in table 7.2.5.3.1-3.

Table 7.2.5.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
ExposureDataSubscription	M	1	Identifies an individual subscription to notifications of exposure data modifications.

Table 7.2.5.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
ExposureDataSubscription	M	1	201 Created	Upon success, a response body containing a representation of the Individual subscription resource shall be returned. The HTTP response shall include a "Location" HTTP header that contains the resource URI of the created resource.
NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Table 7.2.5.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/exposure-data/subs-to-notify/{subId}

7.2.6 Resource: IndividualExposureDataSubscription

7.2.6.1 Description

This resource represents an individual subscription to notifications of exposure data modifications.

7.2.6.2 Resource Definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/exposure-data/subs-to-notify/{subId}**

This resource shall support the resource URI variables defined in table 7.2.6.2-1.

Table 7.2.6.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See 3GPP TS 29.504 [6] clause 6.1.1.
subId	string	Identifies an individual subscription to notifications. The value is allocated by the UDR during creation of the IndividualExposureDataSubscription resource.

7.2.6.3 Resource Standard Methods

7.2.6.3.1 PUT

This method shall support the URI query parameters specified in table 7.2.6.3.1-1.

Table 7.2.6.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 7.2.6.3.1-2 and the response data structures and response codes specified in table 7.2.6.3.1-3.

Table 7.2.6.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
ExposureDataSubscription	M	1	Identifies an individual subscription to notifications of exposure data modifications.

Table 7.2.6.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
ExposureData Subscription	M	1	200 OK	The individual subscription resource was updated successfully and a body with the modified subscription to notifications about exposure data is returned.
n/a			204 No Content	The individual subscription resource was successfully modified.
NOTE: The mandatory HTTP error status codes for the PUT method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

7.2.6.3.2 DELETE

This method shall support the URI query parameters specified in table 7.2.6.3.2-1.

Table 7.2.6.3.1-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 7.2.6.3.2-2 and the response data structures and response codes specified in table 7.2.6.3.2-3.

Table 7.2.6.3.2-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			The request body shall be empty.

Table 7.2.6.3.2-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon success, an empty response body shall be returned.
NOTE: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

7.3 Data Model

7.3.1 General

This clause specifies the application data model supported by the API.

Table 7.3.1-1 specifies the data types defined for the Nudr_DataRepository for Exposure Data service based interface protocol.

Table 7.3.1-1: Nudr_DataRepository specific Data Types for Exposure Data

Data type	Section defined	Description	Applicability
AccessAndMobilityData	7.3.2.2		
ExposureDataChangeNotification	7.3.2.5	Contains changed exposure data for one UE for which Notification was requested.	
ExposureDataSubscription	7.3.2.4	Subscription to Notifications about Exposure Data.	
PduSessionManagementData	7.3.2.3		
PduSessionStatus	7.3.3.3	Represents the state of the PDU session, i.e. active or released.	

Table 7.3.1-2 specifies data types re-used by the Nudr_DataRepository for Exposure Data service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nudr_DataRepository for Exposure Data service based interface.

Table 7.3.1-2: Nudr_DataRepository re-used Data Types for Exposure Data

Data type	Reference	Comments	Applicability
AccessType	3GPP TS 29.571 [7]	Identifies the access type, i.e. 3GPP access or Non-3GPP access.	
CmInfo	3GPP TS 29.518 [10]	Describes the connectivity management state of a UE.	
DateTime	3GPP TS 29.571 [7]		
Dnai	3GPP TS 29.571 [7]	Identifies a Data network access identifier.	
Dnn	3GPP TS 29.571 [7]	Identifies a Data Network Name.	
Ipv4Addr	3GPP TS 29.571 [7]	The IPv4 Address.	
Ipv6Addr	3GPP TS 29.571 [7]	The Ipv6 Address.	
Ipv6Prefix	3GPP TS 29.571 [7]	The Ipv6 Address Prefix.	
PduSessionId	3GPP TS 29.571 [7]	PDU Session Identifier	
PduSessionType	3GPP TS 29.571 [7]	PDU Session type.	
PlmnId	3GPP TS 29.571 [7]	Identifies a PLMN.	
RatType	3GPP TS 29.571 [7]	Identifies a RAT type.	
RmInfo	3GPP TS 29.518 [10]	Describes the registration management state of a UE.	
RouteToLocation	3GPP TS 29.571 [7]	A traffic route to/from a DNAI.	
TimeZone	3GPP TS 29.571 [7]	Identifies a time zone.	
SmsSupport	3GPP TS 29.518 [10]	Indicates the supported SMS delivery of a UE.	
SupportedFeatures	3GPP TS 29.571 [7]	Used to negotiate the applicability of the optional features.	
UeReachability	3GPP TS 29.518 [10]	Describes the reachability of the UE.	
Uri	3GPP TS 29.571 [7]		
UserLocation	3GPP TS 29.571 [7]	Identifies the user location.	
VarUeId	3GPP TS 29.571 [7]	String represents the SUPI or GPSI.	

7.3.2 Structured data types

7.3.2.1 Introduction

This clause defines the structures to be used in resource representations.

7.3.2.2 Type AccessAndMobilityData

Table 7.3.2.2-1: Definition of type AccessAndMobilityData

Attribute name	Data type	P	Cardinality	Description	Applicability
location	UserLocation	O	0..1	Current location for the UE, which can be a combination of E-UTRA, NR and non-3GPP access location.	
locationTs	DateTime	O	0..1	Indicates when the location was last updated.	
timeZone	TimeZone	O	0..1	Current time zone for the UE.	
timeZoneTs	DateTime	O	0..1	Indicates when the timeZone was last updated.	
accessType	AccessType	O	0..1	Current access type for the UE (3GPP access or non-3GPP access).	
accessTypeTs	DateTime	O	0..1	Indicates when the accessType was last updated.	
regStates	array(RmInfo)	O	1..N	Describes the registration management state of the UE.	
regStatesTs	DateTime	O	0..1	Indicates when the regStates were last updated.	
connStates	array(CmInfo)	O	1..N	Describes the connectivity management state of the UE.	
connStatesTs	DateTime	O	0..1	Indicates when the connStates were last updated.	
reachabilityStatus	UeReachability	O	0..1	Describes the reachability of the UE.	
reachabilityStatusTs	DateTime	O	0..1	Indicates when the reachabilityStatus was last updated.	
smsOverNasStatus	SmsSupport	O	0..1	Indicates the supported SMS delivery of the UE.	
smsOverNasStatusTs	DateTime	O	0..1	Indicates when thesmsOverNasStatus was last updated.	
roamingStatus	boolean	O	0..1	True: The serving PLMN of the UE is different from the HPLMN of the UE; False: The serving PLMN of the UE is the HPLMN of the UE.	
roamingStatusTs	DateTime	O	0..1	Indicates when the roamingStatus was last updated.	
currentPlmn	PlmnId	O	0..1	The identity of the PLMN serving the UE.	
currentPlmnTs	DateTime	O	0..1	Indicates when the currentPlmn was last updated.	
ratType	array(RatType)	O	1..N	RAT type(s)	
ratTypesTs	DateTime	O	0..1	Indicates when the ratTypes was last updated.	
suppFeat	SupportedFeatures	C	0..1	This IE represents a list of Supported features used as described in clause 7.5. Shall be present in the HTTP PUT request/response; or in the HTTP GET response if the "supp-feat" attribute query parameter is included in the HTTP GET request. (NOTE)	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	
NOTE: In the HTTP request, it represents the set of features supported by the NF service consumer. In the HTTP response, it represents the set of features supported by both the NF service consumer and the UDR.					

7.3.2.3 Type PduSessionManagementData

Table 7.3.2.3-1: Definition of type PduSessionManagementData

Attribute name	Data type	P	Cardinality	Description	Applicability
pduSessionStatus	PduSessionStatus	O	0..1	Status of the PDU session.	
pduSessionStatusTs	DateTime	O	0..1	Indicates when the pduSessionStatus was last updated.	
dnai	Dnai	O	0..1	DNAI	
dnaiTs	DateTime	O	0..1	Indicates when the dnai was last updated.	
n6TrafficRoutingInfo	array(RouteToLocation)	O	1..N	N6 traffic routing information.	
n6TrafficRoutingInfoTs	DateTime	O	0..1	Indicates when the n6TrafficRoutingInfo was last updated.	
ipv4Addr	Ipv4Addr	O	0..1	UE IPv4 address.	
ipv6Prefix	array(Ipv6Prefix)	O	1..N	UE IPv6 prefix.	
ipv6Addrs	array(Ipv6Addr)	O	1..N	IPv6 addresses.	
pduSessType	PduSessionType	O	0..1	PDU session type.	
ipAddrTs	DateTime	O	0..1	Indicates when the ipv4Addr and Ipv6Prefix were last updated.	
dnn	Dnn	O	0..1	DNN	
pduSessionId	PduSessionId	O	0..1	The identifier of a PduSessionManagement resource. (NOTE 1).	ResourceNotificationExposureDataFix
suppFeat	SupportedFeatures	C	0..1	This IE represents a list of Supported features used as described in clause 7.5. Shall be present in the HTTP PUT request/response; or in the HTTP GET response if the "supp-feat" attribute query parameter is included in the HTTP GET request. (NOTE 2)	
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.	

NOTE 1: The "pduSessionId" attribute shall be present when the ResourceNotificationExposureDataFix is supported.
NOTE 2: In the HTTP request, it represents the set of features supported by the NF service consumer. In the HTTP response, it represents the set of features supported by both the NF service consumer and the UDR.

7.3.2.4 Type ExposureDataSubscription

Table 7.3.2.4-1: Definition of type ExposureDataSubscription

Attribute name	Data type	P	Cardinality	Description
notificationUri	Uri	M	1	URI provided by the NF service consumer indicating where to receive to subscribed notifications from the UDR.
monitoredResourceUris	array(Uri)	M	1..N	A set of URIs that identify the resources for which a modification triggers a notification.
expiry	DateTime	C	0..1	This IE shall be included in a subscription response if, based on operator policy and taking into account the expiry time included in the request, the UDR needs to include an expiry time. This IE may be included in a subscription request. When present, this IE shall represent the time after which the subscription becomes invalid. The absence of this attribute in the subscription response means the subscription to be valid without an expiry time.
supportedFeatures	SupportedFeatures	C	0..1	Used to negotiate the applicability of the optional features. This attribute shall be provided in the POST request and in the response of successful resource creation.
resetIds	array(string)	O	1..N	This IE uniquely identifies a part of temporary data in UDR that contains the created resource. This attribute may be provided in the response of successful resource creation.

7.3.2.5 Type ExposureDataChangeNotification

Table 7.3.2.5-1: Definition of type ExposureDataChangeNotification

Attribute name	Data type	P	Cardinality	Description	Applicability
ueId	VarUeId	O	0..1	User identity of modified data.	
accessAndMobilityData	AccessAndMobility Data	O	0..1	Access and Mobility Data for the UE.	
pduSessionManagement Data	array(PduSession ManagementData)	O	1..N	PDU Session Management Data for one or several PDU sessions of the UE.	
delResources	array(Uri)	O	1..N	The resources, as defined in table 7.2.2-1, if removed from UDR and notification on resource data change was requested. It may be present when the ResourceNotificationExposureData Fix is supported (NOTE).	
NOTE: When the "delResources" attribute is present, the attributes that correspond with the values included in the "delResources" attribute shall be omitted. I.e., when the "delResources" attribute includes the value "../{ueId}/access-and-mobility-data" the attribute "accessAndMobilityData" shall be omitted, and when the "delResources" attribute includes all the existing "../{ueId}/session-management-data/{pduSessionId}" resource URIs, the attribute "pduSessionManagementData" shall be omitted.					

7.3.3 Simple data types and enumerations

7.3.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

7.3.3.2 Simple data types

The simple data types defined in table 7.3.3.2-1 shall be supported.

Table 7.3.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability

7.3.3.3 Enumeration: PduSessionStatus

Table 7.3.3.3-1: Enumeration PduSessionStatus

Enumeration value	Description	Applicability
"ACTIVE"	The PDU session is active.	
"RELEASED"	The PDU session is released.	

7.4 Error handling

7.4.1 General

For the Nudr_DataRepository service API for Exposure Data, HTTP error responses shall be supported as specified in clause 4.8 of 3GPP TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of 3GPP TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of 3GPP TS 29.500 [4].

In addition, the requirements in the following clauses are applicable for the Nudr_DataRepository service API for Exposure Data.

7.4.2 Protocol Errors

No specific procedures for the Nudr_DataRepository service API for Exposure Data are specified.

7.4.3 Application Errors

The application errors defined for the Nudr_DataRepository service API for Exposure Data are listed in table 7.4.3-1.

Table 7.4.3-1: Application errors

Application Error	HTTP status code	Description

The application error handling shall follow 3GPP TS 29.504 [6] clause 6.1.6 unless explicitly specified in this document.

7.5 Feature negotiation

The optional features in table 6.1.8-1 of 3GPP TS 29.504 [6] are defined for the Nudr_DataRepository API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of 3GPP TS 29.500 [4].

7.6 Notifications

7.6.1 General

Notifications shall comply with clause 6.2 of 3GPP TS 29.500 [4] and clause 4.6.2.3 of 3GPP TS 29.501 [5].

This clause describes the resources to provide Notification to NF service consumers which have subscribed to be notified when exposure data is changed.

Table 7.6.1-1: Notifications overview

Notification	Callback URI	HTTP method or custom operation	Description (service operation)
Exposure Data Change Notification	{notificationUri}	POST	Exposure Data Change Notification

7.6.2 Exposure Data Change Notification

The POST method shall be used for Data Change Notifications and the URI shall be as provided during the subscription procedure.

Callback URI: {**notificationUri**}

Support of URI query parameters is specified in table 7.6.2-1.

Table 7.6.2-1: URI query parameters supported by the POST method

Name	Data type	P	Cardinality	Description
n/a				

Support of request data structures is specified in table 7.6.2-2 and of response data structures and response codes is specified in table 7.6.2-3.

Table 7.6.2-2: Data structures supported by the POST Request Body

Data type	P	Cardinality	Description
array(ExposureDataChange Notification)	M	1..N	Each element in the array provides the changed exposure data for one user.

Table 7.6.2-3: Data structures supported by the POST Response Body

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon success, an empty response body shall be returned.
NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

Annex A (normative): OpenAPI specification

A.1 General

The present Annex contains an OpenAPI [3] specification of HTTP messages and content bodies used by the Nudr_DataRepository service for Policy Data, Application Data and Exposure Data.

This Annex shall take precedence when being discrepant to other parts of the specification with respect to the encoding of information elements and methods.

NOTE: The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

Informative copies of the OpenAPI specification files contained in this 3GPP Technical Specification are available on a Git-based repository that uses the GitLab software version control system (see clause 5B of the 3GPP TR 21.900 [18] and clause 5.3.1 of the 3GPP TS 29.501 [5] for further information).

A.2 Nudr_DataRepository API for Policy Data

For the purpose of referencing entities in the Open API file defined in this Annex, it shall be assumed that this Open API file is contained in a physical file named "TS29519_Policy_Data.yaml".

```
openapi: 3.0.0

info:
  version: '-'
  title: Unified Data Repository Service API file for policy data
  description: |
    The API version is defined in 3GPP TS 29.504
    © 2024, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: >
    3GPP TS 29.519 V17.14.0; 5G System; Usage of the Unified Data Repository Service for Policy
Data,
    Application Data and Structured Data for Exposure.
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.519/'

paths:
  /policy-data/ues/{ueId}:
    parameters:
     - name: ueId
       in: path
       required: true
       schema:
         $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
    get:
      summary: Retrieve the policy data for a subscriber
      operationId: ReadPolicyData
      tags:
        - PolicyDataForIndividualUe (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:ues:read
      parameters:
```

```
      - name: supp-feat
        in: query
        description: Supported Features
        required: false
        schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      - name: data-subset-names
        in: query
        style: form
        explode: false
        description: List of policy data subset names
        required: false
        schema:
          type: array
          items:
            $ref: '#/components/schemas/PolicyDataSubset'
          minItems: 2
    responses:
      '200':
        description: Upon success, a response body containing policy data shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PolicyDataForIndividualUe'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/policy-data/ues/{ueId}/am-data:
  parameters:
   - name: ueId
     in: path
     required: true
     schema:
       $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
  get:
    summary: Retrieves the access and mobility policy data for a subscriber
    operationId: ReadAccessAndMobilityPolicyData
    tags:
      - AccessAndMobilityPolicyData (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:ues:am-data:read
    responses:
      '200':
        description: >
          Upon success, a response body containing access and mobility policies shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AmPolicyData'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
```

```
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /policy-data/ues/{ueId}/ue-policy-set:
    parameters:
     - name: ueId
       in: path
       required: true
       schema:
         $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
    get:
      summary: Retrieves the UE policy set data for a subscriber
      operationId: ReadUEPolicySet
      tags:
        - UEPolicySet (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:ues:ue-policy-set:read
      parameters:
        - name: supp-feat
          in: query
          description: Supported Features
          required: false
          schema:
             $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      responses:
        '200':
          description: Upon success, a response body containing UE policies shall be returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/UePolicySet'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    put:
      summary: Create or modify the UE policy set data for a subscriber
      operationId: CreateOrReplaceUEPolicySet
      tags:
        - UEPolicySet (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
```

```
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:ues:ue-policy-set:create
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/UePolicySet'
    responses:
      '201':
        description: >
          Successful case. The resource has been successfully created and a response body
          containing a representation of the created UEPolicySet resource shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UePolicySet'
        headers:
          Location:
            description: 'Contains the URI of the newly created resource'
            required: true
            schema:
              type: string
      '200':
        description: >
          Successful case. The resource has been successfully created and a response body
          containing UE policies shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UePolicySet'
      '204':
        description: >
          Successful case. The resource has been successfully updated and no additional content
          is to be sent in the response message.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  patch:
    summary: Modify the UE policy set data for a subscriber
    operationId: UpdateUEPolicySet
    tags:
      - UEPolicySet (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
```

```
        - nudr-dr:policy-data:ues:ue-policy-set:modify
    requestBody:
      required: true
      content:
        application/merge-patch+json:
          schema:
            $ref: '#/components/schemas/UePolicySetPatch'
    responses:
      '204':
        description: >
          Successful case. The resource has been successfully updated and no additional content is
          to be sent in the response message.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /policy-data/ues/{ueId}/sm-data:
    get:
      summary: Retrieves the session management policy data for a subscriber
      operationId: ReadSessionManagementPolicyData
      tags:
        - SessionManagementPolicyData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:ues:sm-data:read
      parameters:
       - name: ueId
         in: path
         required: true
         schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
       - name: snssai
         in: query
         required: false
         content:
           application/json:
             schema:
               $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
       - name: dnn
         in: query
         required: false
         schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
       - name: fields
         in: query
         description: attributes to be retrieved
         required: false
         schema:
           type: array
           items:
             type: string
```

```
        minItems: 1
    - name: supp-feat
      in: query
      description: Supported Features
      required: false
      schema:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
  responses:
    '200':
      description: Upon success, a response body containing SmPolicyData shall be returned.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/SmPolicyData'
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '406':
      $ref: 'TS29571_CommonData.yaml#/components/responses/406'
    '414':
      $ref: 'TS29571_CommonData.yaml#/components/responses/414'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    default:
      $ref: 'TS29571_CommonData.yaml#/components/responses/default'
patch:
  summary: Modify the session management policy data for a subscriber
  operationId: UpdateSessionManagementPolicyData
  tags:
    - SessionManagementPolicyData (Document)
  security:
    - {}
    - oAuth2ClientCredentials:
      - nudr-dr
    - oAuth2ClientCredentials:
      - nudr-dr
      - nudr-dr:policy-data
    - oAuth2ClientCredentials:
      - nudr-dr
      - nudr-dr:policy-data
      - nudr-dr:policy-data:ues:sm-data:modify
  parameters:
    - name: ueId
      in: path
      required: true
      schema:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
  requestBody:
    required: true
    content:
      application/merge-patch+json:
        schema:
          $ref: '#/components/schemas/SmPolicyDataPatch'
  responses:
    '204':
      description: >
        Successful case. The resource has been successfully updated and no
        additional content is to be sent in the response message.
    '200':
      description: Expected response to a valid request
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/SmPolicyData'
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/policy-data/ues/{ueId}/sm-data/{usageMonId}:
  get:
    summary: Retrieve a usage monitoring resource
    operationId: ReadUsageMonitoringInformation
    tags:
      - UsageMonitoringInformation (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:ues:sm-data:read
    parameters:
      - name: ueId
        in: path
        required: true
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
      - name: usageMonId
        in: path
        required: true
        schema:
          type: string
      - name: supp-feat
        in: query
        description: Supported Features
        required: false
        schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    responses:
      '200':
        description: Successful case. The usage monitoring data is returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UsageMonData'
      '204':
        description: The resource was found but no usage monitoring data is available.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  put:
    summary: Create a usage monitoring resource
    operationId: CreateUsageMonitoringResource
    tags:
      - UsageMonitoringInformation (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:ues:sm-data:create
    parameters:
     - name: ueId
       in: path
       required: true
       schema:
         $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
     - name: usageMonId
       in: path
       required: true
       schema:
         type: string
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/UsageMonData'
    responses:
      '201':
        description: >
          Successful case. The resource has been successfully created and a response body is
          returned containing a representation of the resource.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UsageMonData'
        headers:
          Location:
            description: 'Contains the URI of the newly created resource'
            required: true
            schema:
              type: string
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Delete a usage monitoring resource
    operationId: DeleteUsageMonitoringInformation
```

```
    tags:
      - UsageMonitoringInformation (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:ues:sm-data:modify
    parameters:
     - name: ueId
       in: path
       required: true
       schema:
         $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
     - name: usageMonId
       in: path
       required: true
       schema:
         type: string
    responses:
      '204':
        description: Successful case. The resource has been successfully deleted.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /policy-data/sponsor-connectivity-data/{sponsorId}:
    parameters:
     - name: sponsorId
       in: path
       required: true
       schema:
         type: string
    get:
      summary: Retrieves the sponsored connectivity information for a given sponsorId
      operationId: ReadSponsorConnectivityData
      tags:
        - SponsorConnectivityData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:sponsor-connectivity-data:read
      responses:
        '200':
          description: >
            Upon success, a response body containing Sponsor Connectivity Data shall be returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/SponsorConnectivityData'
        '204':
          description: The resource was found but no Sponsor Connectivity Data is available.
        '400':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/policy-data/bdt-data:
  get:
    summary: Retrieves the BDT data collection
    operationId: ReadBdtData
    tags:
      - BdtData (Store)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:bdt-data:read
    parameters:
      - name: bdt-ref-ids
        in: query
        description: List of the BDT reference identifiers.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29122_CommonData.yaml#/components/schemas/BdtReferenceId'
          minItems: 1
        style: form
        explode: false
      - name: supp-feat
        in: query
        description: Supported Features
        required: false
        schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    responses:
      '200':
        description: Upon success, a response body containing the BDT data shall be returned.
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/BdtData'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
```

```
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /policy-data/bdt-data/{bdtReferenceId}:
    parameters:
     - name: bdtReferenceId
       in: path
       required: true
       schema:
         type: string
    get:
      summary: Retrieves the BDT data information associated with a BDT reference Id
      operationId: ReadIndividualBdtData
      tags:
        - IndividualBdtData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:bdt-data:read
      parameters:
        - name: supp-feat
          in: query
          description: Supported Features
          required: false
          schema:
             $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      responses:
        '200':
          description: Upon success, a response body containing the BDT data shall be returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/BdtData'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    put:
      summary: Creates an BDT data resource associated with an BDT reference Id
      operationId: CreateIndividualBdtData
      tags:
        - IndividualBdtData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:bdt-data:create
      requestBody:
        required: true
        content:
          application/json:
```

```
          schema:
            $ref: '#/components/schemas/BdtData'
      responses:
        '201':
          description: Successful case. The resource has been successfully created.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/BdtData'
          headers:
            Location:
              description: 'Contains the URI of the newly created resource'
              required: true
              schema:
                type: string
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '414':
          $ref: 'TS29571_CommonData.yaml#/components/responses/414'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    patch:
      summary: Modifies an BDT data resource associated with an BDT reference Id
      operationId: UpdateIndividualBdtData
      tags:
        - IndividualBdtData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:bdt-data:modify
      requestBody:
        required: true
        content:
          application/merge-patch+json:
            schema:
              $ref: '#/components/schemas/BdtDataPatch'
      responses:
        '200':
          description: Expected response to a valid request
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/BdtData'
        '204':
          description: >
            Successful case. The resource has been successfully updated and no additional content
            is to be sent in the response message.
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
```

```
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Deletes an BDT data resource associated with an BDT reference Id
    operationId: DeleteIndividualBdtData
    tags:
      - IndividualBdtData (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:bdt-data:modify
    responses:
      '204':
        description: Successful case. The resource has been successfully deleted.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/policy-data/subs-to-notify:
  post:
    summary: Create a subscription to receive notification of policy data changes
    operationId: CreateIndividualPolicyDataSubscription
    tags:
      - PolicyDataSubscriptions (Collection)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:subs-to-notify:create
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/PolicyDataSubscription'
    responses:
      '201':
        description: >
```

```
          Upon success, a response body containing a representation of each Individual
          subscription resource shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PolicyDataSubscription'
        headers:
          Location:
            description: 'Contains the URI of the newly created resource'
            required: true
            schema:
              type: string
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    callbacks:
      policyDataChangeNotification:
        '{$request.body#/notificationUri}':
          post:
            requestBody:
              required: true
              content:
                application/json:
                  schema:
                    type: array
                    items:
                      $ref: '#/components/schemas/PolicyDataChangeNotification'
                    minItems: 1
            responses:
              '204':
                description: No Content, Notification was successful
              '400':
                $ref: 'TS29571_CommonData.yaml#/components/responses/400'
              '401':
                $ref: 'TS29571_CommonData.yaml#/components/responses/401'
              '403':
                $ref: 'TS29571_CommonData.yaml#/components/responses/403'
              '404':
                $ref: 'TS29571_CommonData.yaml#/components/responses/404'
              '411':
                $ref: 'TS29571_CommonData.yaml#/components/responses/411'
              '413':
                $ref: 'TS29571_CommonData.yaml#/components/responses/413'
              '415':
                $ref: 'TS29571_CommonData.yaml#/components/responses/415'
              '429':
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
              default:
                $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /policy-data/subs-to-notify/{subsId}:
    parameters:
     - name: subsId
       in: path
       required: true
```

```
      schema:
        type: string
    put:
      summary: Modify a subscription to receive notification of policy data changes
      operationId: ReplaceIndividualPolicyDataSubscription
      tags:
        - IndividualPolicyDataSubscription (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:subs-to-notify:modify
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PolicyDataSubscription'
      responses:
        '200':
          description: The individual subscription resource was updated successfully.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/PolicyDataSubscription'
        '204':
          description: >
            The individual subscription resource was updated successfully and no
            additional content is to be sent in the response message.
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    delete:
      summary: Delete the individual Policy Data subscription
      operationId: DeleteIndividualPolicyDataSubscription
      tags:
        - IndividualPolicyDataSubscription (Document)
      responses:
        '204':
          description: Upon success, an empty response body shall be returned.
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
```

```
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /policy-data/ues/{ueId}/operator-specific-data:
    get:
      summary: Retrieve the operator specific policy data of an UE
      operationId: ReadOperatorSpecificData
      tags:
        - OperatorSpecificData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:ues:operator-specific-data:read
      parameters:
        - name: ueId
          in: path
          description: UE Id
          required: true
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
        - name: fields
          in: query
          description: attributes to be retrieved
          required: false
          schema:
            type: array
            items:
              type: string
            minItems: 1
        - name: supp-feat
          in: query
          description: Supported Features
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      responses:
        '200':
          description: Expected response to a valid request
          content:
            application/json:
              schema:
                type: object
                additionalProperties:
                  $ref:
'TS29505_Subscription_Data.yaml#/components/schemas/OperatorSpecificDataContainer'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '414':
          $ref: 'TS29571_CommonData.yaml#/components/responses/414'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

    patch:
      summary: Modify the operator specific policy data of a UE
      operationId: UpdateOperatorSpecificData
      tags:
```

```
      - OperatorSpecificData (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:ues:operator-specific-data:modify
    parameters:
      - name: ueId
        in: path
        description: UE Id
        required: true
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
    requestBody:
      content:
        application/json-patch+json:
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchItem'
      required: true
    responses:
      '204':
        description: No content. Response to successful modification.
      '200':
        description: Expected response to a valid request
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchResult'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  put:
    summary: Create or modify the operator specific policy data of a UE
    operationId: ReplaceOperatorSpecificData
    tags:
      - OperatorSpecificData (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:ues:operator-specific-data:create
    parameters:
      - name: ueId
        in: path
        description: UE Id
```

```
          required: true
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              additionalProperties:
                $ref:
'TS29505_Subscription_Data.yaml#/components/schemas/OperatorSpecificDataContainer'
      responses:
        '200':
          description: OK
          content:
            application/json:
              schema:
                type: object
                additionalProperties:
                  $ref:
'TS29505_Subscription_Data.yaml#/components/schemas/OperatorSpecificDataContainer'
        '201':
          description: >
            Successful case. When the feature OSDResource_Create_Delete is supported
            and the resource has been successfully created, a response body containing a
            representation of the created OperatorSpecificData resource shall be returned.
          content:
            application/json:
              schema:
                type: object
                additionalProperties:
                  $ref:
'TS29505_Subscription_Data.yaml#/components/schemas/OperatorSpecificDataContainer'
          headers:
            Location:
              description: 'Contains the URI of the newly created resource'
              required: true
              schema:
                type: string
        '204':
          description: The resource has been successfully updated.
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    delete:
      summary: When the feature OSDResource_Create_Delete is supported, delete OperatorSpecificData
resource
      operationId: DeleteOperatorSpecificData
      tags:
        - OperatorSpecificData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
```

```
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:ues:operator-specific-data:modify
    parameters:
     - name: ueId
       in: path
       required: true
       schema:
         $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
    responses:
      '204':
        description: Successful case. The resource has been successfully deleted.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/policy-data/plmns/{plmnId}/ue-policy-set:
  parameters:
   - name: plmnId
     in: path
     required: true
     schema:
       $ref: 'TS29505_Subscription_Data.yaml#/components/schemas/VarPlmnId'
  get:
    summary: Retrieve the UE policy set data for an H-PLMN
    operationId: ReadPlmnUePolicySet
    tags:
      - PlmnUePolicySet (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:policy-data
        - nudr-dr:policy-data:plmns:ue-policy-set:read
    responses:
      '200':
        description: Upon success, a response body containing UE policies shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UePolicySet'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '412':
        $ref: 'TS29571_CommonData.yaml#/components/responses/412'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /policy-data/slice-control-data/{snssai}:
    parameters:
     - name: snssai
       in: path
       required: true
       schema:
         $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
    get:
      summary: Retrieves a network Slice specific policy control data resource
      operationId: ReadSlicePolicyControlData
      tags:
        - SlicePolicyControlData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:slice-control-data:read
      parameters:
        - name: supp-feat
          in: query
          description: Supported Features
          required: false
          schema:
             $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      responses:
        '200':
          description: >
            Successful case. The network slice specific policy control data shall be returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/SlicePolicyData'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    patch:
      summary: Modify a network Slice specific policy control data resource
      operationId: UpdateSlicePolicyControlData
      tags:
        - SlicePolicyControlData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:slice-control-data:modify
      requestBody:
        required: true
        content:
          application/merge-patch+json:
```

```
          schema:
            $ref: '#/components/schemas/SlicePolicyDataPatch'
      responses:
        '200':
          description: >
            The resource has been successfully updated and a response body containing network
            slice specific policy control data shall be returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/SlicePolicyData'
        '204':
          description: >
            The resource has been successfully updated and no additional content is
            to be sent in the response message.
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /policy-data/mbs-session-pol-data/{polSessionId}:
    parameters:
      - name: polSessionId
        description: >
          Represents the identifier of the MBS Session Policy Control Data.
        in: path
        required: true
        schema:
          $ref: '#/components/schemas/MbsSessPolDataId'

    get:
      summary: Retrieve MBS Session Policy Control Data for an MBS Session.
      operationId: GetMBSSessPolCtrlData
      tags:
        - MBSSessionPolicyControlData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:policy-data
          - nudr-dr:policy-data:mbs-session-pol-data:read
      responses:
        '200':
          description: >
            OK. The requested MBS Session Policy Control Data is successfully returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/MbsSessPolCtrlData'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
```

```
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '414':
          $ref: 'TS29571_CommonData.yaml#/components/responses/414'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  schemas:

    PolicyDataForIndividualUe:
      description: Contains policy data for a given subscriber.
      type: object
      properties:
        uePolicyDataSet:
          $ref: '#/components/schemas/UePolicySet'
        smPolicyDataSet:
          $ref: '#/components/schemas/SmPolicyData'
        amPolicyDataSet:
          $ref: '#/components/schemas/AmPolicyData'
        umData:
          type: object
          additionalProperties:
            $ref: '#/components/schemas/UsageMonData'
          minProperties: 1
          description: >
            Contains UM policies. The value of the limit identifier is used as the key of the map.
        operatorSpecificDataSet:
          type: object
          additionalProperties:
            $ref: 'TS29505_Subscription_Data.yaml#/components/schemas/OperatorSpecificDataContainer'
          minProperties: 1
          description: >
            Contains Operator Specific Data resource data. The key of the map is operator
            specific data element name and the value is the operator specific data of the UE.

    AmPolicyData:
      description: Contains the AM policy data for a given subscriber.
      type: object
      properties:
        praInfos:
          type: object
          additionalProperties:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceInfo'
          minProperties: 1
          description: >
            Contains Presence reporting area information. The praId attribute within the
            PresenceInfo data type is the key of the map.
        subscCats:
          type: array
          items:
            type: string
          minItems: 1

    UePolicySet:
      description: Contains the UE policy data for a given subscriber.
      type: object
      properties:
        praInfos:
          type: object
          additionalProperties:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceInfo'
          minProperties: 1
          description: >
            Contains Presence reporting area information. The praId attribute within the
            PresenceInfo data type is the key of the map.
        subscCats:
          type: array
          items:
```

```
          type: string
        minItems: 1
      uePolicySections:
        type: object
        additionalProperties:
          $ref: '#/components/schemas/UePolicySection'
        minProperties: 1
        description: >
          Contains the UE Policy Sections. The UE Policy Section Identifier is used as
          the key of the map.
      upsis:
        type: array
        items:
          type: string
        minItems: 1
      allowedRouteSelDescs:
        type: object
        additionalProperties:
          $ref: '#/components/schemas/PlmnRouteSelectionDescriptor'
        minProperties: 1
        description: >
          Contains allowed route selection descriptors per serving PLMN for a UE.
          The serving PLMN identifier is the key of the map.
      andspInd:
        type: boolean
      pei:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Pei'
      osIds:
        type: array
        items:
          $ref: '#/components/schemas/OsId'
        minItems: 1
      suppFeat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1

  UePolicySetPatch:
    description: Contains the UE policy set for a given subscriber.
    type: object
    properties:
      uePolicySections:
        type: object
        additionalProperties:
          $ref: '#/components/schemas/UePolicySection'
        minProperties: 1
        description: >
          Contains the UE Policy Sections. The UE Policy Section Identifier is used
          as the key of the map.
      upsis:
        type: array
        items:
          type: string
        minItems: 1
      andspInd:
        type: boolean
      pei:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Pei'
      osIds:
        type: array
        items:
          $ref: '#/components/schemas/OsId'
        minItems: 1

  UePolicySection:
    description: Contains the UE policy section.
    type: object
    properties:
      uePolicySectionInfo:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'
      upsi:
        type: string
    required:
      - uePolicySectionInfo
      - upsi
```

```
    SmPolicyData:
      description: Contains the SM policy data for a given subscriber.
      type: object
      properties:
        smPolicySnssaiData:
          type: object
          additionalProperties:
            $ref: '#/components/schemas/SmPolicySnssaiData'
          minProperties: 1
          description: >
            Contains Session Management Policy data per S-NSSAI for all the SNSSAIs
            of the subscriber. The key of the map is the S-NSSAI.
        umDataLimits:
          type: object
          additionalProperties:
            $ref: '#/components/schemas/UsageMonDataLimit'
          minProperties: 1
          description: >
            Contains a list of usage monitoring profiles associated with the subscriber.
            The limit identifier is used as the key of the map.
        umData:
          type: object
          additionalProperties:
            $ref: '#/components/schemas/UsageMonData'
          minProperties: 1
          description: >
            Contains the remaining allowed usage data associated with the subscriber.
            The limit identifier is used as the key of the map.
        suppFeat:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        resetIds:
          type: array
          items:
            type: string
          minItems: 1
      required:
        - smPolicySnssaiData

    SmPolicySnssaiData:
      description: Contains the SM policy data for a given subscriber and S-NSSAI.
      type: object
      properties:
        snssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        smPolicyDnnData:
          type: object
          additionalProperties:
            $ref: '#/components/schemas/SmPolicyDnnData'
          minProperties: 1
          description: >
            Session Management Policy data per DNN for all the DNNs of the indicated S-NSSAI.
            The key of the map is the DNN.
        ueSliceMbr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SliceMbr'
      required:
        - snssai

    SmPolicyDnnData:
      description: Contains the SM policy data for a given DNN (and S-NSSAI).
      type: object
      properties:
        dnn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        allowedServices:
          type: array
          items:
            type: string
          minItems: 1
        subscCats:
          type: array
          items:
            type: string
          minItems: 1
        gbrUl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        gbrDl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
```

```
      adcSupport:
        type: boolean
      subscSpendingLimits:
        type: boolean
      ipv4Index:
        $ref: '#/components/schemas/IpIndex'
      ipv6Index:
        $ref: '#/components/schemas/IpIndex'
      offline:
        type: boolean
      online:
        type: boolean
      chfInfo:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/ChargingInformation'
      refUmDataLimitIds:
        type: object
        additionalProperties:
          $ref: '#/components/schemas/LimitIdToMonitoringKey'
        minProperties: 1
        description: >
          A reference to the UsageMonDataLimit or UsageMonData instancesfor this DNN and SNSSAI
          that may also include the related monitoring key(s).
          The key of the map is the limit identifier.
      mpsPriority:
        type: boolean
      mcsPriority:
        type: boolean
      imsSignallingPrio:
        type: boolean
      mpsPriorityLevel:
        type: integer
      mcsPriorityLevel:
        type: integer
      praInfos:
        type: object
        additionalProperties:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceInfo'
        minProperties: 1
        description: >
          Contains Presence reporting area information. The praId attribute within the
          PresenceInfo data type is the key of the map.
      bdtRefIds:
        type: object
        additionalProperties:
          $ref: 'TS29122_CommonData.yaml#/components/schemas/BdtReferenceIdRm'
        minProperties: 1
        description: >
          Identifies transfer policies of background data transfer. Any string value can
          be used as a key of the map.
        nullable: true
      locRoutNotAllowed:
        type: boolean
    required:
      - dnn

  UsageMonDataLimit:
    description: Contains usage monitoring control data for a subscriber.
    type: object
    properties:
      limitId:
        type: string
      scopes:
        type: object
        additionalProperties:
          $ref: '#/components/schemas/UsageMonDataScope'
        minProperties: 1
        description: >
          Identifies the SNSSAI and DNN combinations to which the usage monitoring data
          limit applies. The S-NSSAI is the key of the map.
      umLevel:
        $ref: '#/components/schemas/UsageMonLevel'
      startDate:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      endDate:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      usageLimit:
        $ref: 'TS29122_CommonData.yaml#/components/schemas/UsageThreshold'
      resetPeriod:
```

```
        $ref: '#/components/schemas/TimePeriod'
    required:
      - limitId

  UsageMonData:
    description: Contains remain allowed usage data for a subscriber.
    type: object
    properties:
      limitId:
        type: string
      scopes:
        type: object
        additionalProperties:
          $ref: '#/components/schemas/UsageMonDataScope'
        minProperties: 1
        description: >
          Identifies the SNSSAI and DNN combinations for remain allowed usage data
          for a subscriber. The S-NSSAI is the key of the map.
      umLevel:
        $ref: '#/components/schemas/UsageMonLevel'
      allowedUsage:
        $ref: 'TS29122_CommonData.yaml#/components/schemas/UsageThreshold'
      resetTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      suppFeat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
    required:
      - limitId

  LimitIdToMonitoringKey:
    description: >
      Contains the limit identifier and the corresponding monitoring key for a given
      S-NSSAI and DNN.
    type: object
    properties:
      limitId:
        type: string
      monkey:
        type: array
        items:
          type: string
        minItems: 1
    required:
      - limitId
    nullable: true

  UsageMonDataScope:
    description: >
      Contains a SNSSAI and DNN combinations to which the UsageMonData instance belongs to.
    type: object
    properties:
      snssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      dnn:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        minItems: 1
    required:
      - snssai

  TimePeriod:
    description: Contains the periodicity for the defined usage monitoring data limits.
    type: object
    properties:
      period:
        $ref: '#/components/schemas/Periodicity'
      maxNumPeriod:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
    required:
      - period

  SponsorConnectivityData:
```

```
    description: >
      Contains the sponsored data connectivity related information for a sponsor identifier.
    type: object
    properties:
      aspIds:
        type: array
        items:
          type: string
    required:
      - aspIds

  BdtData:
    description: Contains the background data transfer data.
    type: object
    properties:
      aspId:
        type: string
      transPolicy:
        $ref: 'TS29554_Npcf_BDTPolicyControl.yaml#/components/schemas/TransferPolicy'
      bdtRefId:
        $ref: 'TS29122_CommonData.yaml#/components/schemas/BdtReferenceId'
      nwAreaInfo:
        $ref: 'TS29554_Npcf_BDTPolicyControl.yaml#/components/schemas/NetworkAreaInfo'
      numOfUes:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
      volPerUe:
        $ref: 'TS29122_CommonData.yaml#/components/schemas/UsageThreshold'
      dnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      snssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      trafficDes:
        $ref: 'TS29122_ResourceManagementOfBdt.yaml#/components/schemas/TrafficDescriptor'
      bdtpStatus:
        $ref: '#/components/schemas/BdtPolicyStatus'
      suppFeat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
    required:
      - aspId
      - transPolicy

  PolicyDataSubscription:
    description: Identifies a subscription to policy data change notification.
    type: object
    properties:
      notificationUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      notifId:
        type: string
      monitoredResourceUris:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      monResItems:
        type: array
        items:
          $ref: '#/components/schemas/ResourceItem'
        minItems: 1
      excludedResItems:
        type: array
        items:
          $ref: '#/components/schemas/ResourceItem'
        minItems: 1
      expiry:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
    required:
```

```
      - notificationUri
      - monitoredResourceUris

    PolicyDataChangeNotification:
      description: Contains changed policy data for which notification was requested.
      type: object
      properties:
        amPolicyData:
          $ref: '#/components/schemas/AmPolicyData'
        uePolicySet:
          $ref: '#/components/schemas/UePolicySet'
        plmnUePolicySet:
          $ref: '#/components/schemas/UePolicySet'
        smPolicyData:
          $ref: '#/components/schemas/SmPolicyData'
        usageMonData:
          $ref: '#/components/schemas/UsageMonData'
        SponsorConnectivityData:
          $ref: '#/components/schemas/SponsorConnectivityData'
        bdtData:
          $ref: '#/components/schemas/BdtData'
        opSpecData:
          $ref: 'TS29505_Subscription_Data.yaml#/components/schemas/OperatorSpecificDataContainer'
        opSpecDataMap:
          type: object
          additionalProperties:
            $ref: 'TS29505_Subscription_Data.yaml#/components/schemas/OperatorSpecificDataContainer'
          minProperties: 1
          description: >
            Operator Specific Data resource data, if changed and notification was requested.
            The key of the map is operator specific data element name and the value is the
            operator specific data of the UE.
        ueId:
         $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
        sponsorId:
          type: string
        bdtRefId:
          $ref: 'TS29122_CommonData.yaml#/components/schemas/BdtReferenceId'
        usageMonId:
          type: string
        plmnId:
         $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnId'
        delResources:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
          minItems: 1
        notifId:
          type: string
        reportedFragments:
          type: array
          items:
            $ref: '#/components/schemas/NotificationItem'
          minItems: 1
        slicePolicyData:
          $ref: '#/components/schemas/SlicePolicyData'
        snssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'

    PlmnRouteSelectionDescriptor:
      description: >
        Contains the route selection descriptors (combinations of SNSSAI, DNNs, PDU session types,
        SSC modes and ATSSS information) allowed by subscription to the UE for a serving PLMN
      type: object
      properties:
        servingPlmn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnId'
        snssaiRouteSelDescs:
          type: array
          items:
            $ref: '#/components/schemas/SnssaiRouteSelectionDescriptor'
          minItems: 1
      required:
        - servingPlmn

    SnssaiRouteSelectionDescriptor:
      description: >
        Contains the route selector parameters (DNNs, PDU session types, SSC modes and ATSSS
```

```
      information) per SNSSAI
    type: object
    properties:
      snssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      dnnRouteSelDescs:
        type: array
        items:
          $ref: '#/components/schemas/DnnRouteSelectionDescriptor'
        minItems: 1
    required:
      - snssai

  DnnRouteSelectionDescriptor:
    description: >
      Contains the route selector parameters (PDU session types, SSC modes and ATSSS
      information) per DNN
    type: object
    properties:
      dnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      sscModes:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SscMode'
        minItems: 1
      pduSessTypes:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionType'
        minItems: 1
      atsssInfo:
        description: >
          Indicates whether MA PDU session establishment is allowed for this DNN.
          When set to value true MA PDU session establishment is allowed for this DNN.
        type: boolean
        default: false
    required:
      - dnn

  SmPolicyDataPatch:
    description: Contains the SM policy data for a given subscriber.
    type: object
    properties:
      umData:
        type: object
        additionalProperties:
          $ref: '#/components/schemas/UsageMonData'
        minProperties: 1
        description: >
          Contains the remaining allowed usage data associated with the subscriber.
          The value of the limit identifier is used as the key of the map.
        nullable: true
      smPolicySnssaiData:
        type: object
        additionalProperties:
          $ref: '#/components/schemas/SmPolicySnssaiDataPatch'
        minProperties: 1
        description: >
          Modifiable Session Management Policy data per S-NSSAI for all the SNSSAIs
          of the subscriber. The key of the map is the S-NSSAI.

  SmPolicySnssaiDataPatch:
    description: Contains the SM policy data for a given subscriber and S-NSSAI.
    type: object
    properties:
      snssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      smPolicyDnnData:
        type: object
        additionalProperties:
          $ref: '#/components/schemas/SmPolicyDnnDataPatch'
        minProperties: 1
        description: >
          Modifiable Session Management Policy data per DNN for all the DNNs of the
          indicated S-NSSAI. The key of the map is the DNN.
    required:
      - snssai
```

```
    SmPolicyDnnDataPatch:
      description: Contains the SM policy data for a given DNN (and S-NSSAI).
      type: object
      properties:
        dnn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        bdtRefIds:
          type: object
          additionalProperties:
            $ref: 'TS29122_CommonData.yaml#/components/schemas/BdtReferenceIdRm'
          minProperties: 1
          description: >
            Contains updated transfer policies of background data transfer.
            Any string value can be used as a key of the map.
          nullable: true
      required:
        - dnn

    ResourceItem:
      description: >
        Identifies a subscription to policy data change notification when the change occurs
        in a fragment (subset of resource data) of a given resource.
      type: object
      properties:
        monResourceUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        items:
          type: array
          items:
            $ref: '#/components/schemas/ItemPath'
          minItems: 1
      required:
        - monResourceUri
        - items

    NotificationItem:
      description: >
        Identifies a data change notification when the change occurs in a fragment
        (subset of resource data) of a given resource.
      type: object
      properties:
        resourceId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        notifItems:
          type: array
          items:
            $ref: '#/components/schemas/UpdatedItem'
          minItems: 1
      required:
        - resourceId
        - notifItems

    UpdatedItem:
      description: Identifies a fragment of a resource.
      type: object
      properties:
        item:
          $ref: '#/components/schemas/ItemPath'
        value: {}
      required:
        - item
        - value

    BdtDataPatch:
      description: Contains the modified background data transfer data.
      type: object
      properties:
        transPolicy:
          $ref: 'TS29554_Npcf_BDTPolicyControl.yaml#/components/schemas/TransferPolicy'
        bdtpStatus:
          $ref: '#/components/schemas/BdtPolicyStatus'

    SlicePolicyData:
      description: Contains the network slice specific policy control information.
      type: object
      properties:
        mbrUl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
```

```
        mbrDl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        remainMbrUl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        remainMbrDl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        suppFeat:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        resetIds:
          type: array
          items:
            type: string
          minItems: 1

    SlicePolicyDataPatch:
      description: Contains the modified network slice specific policy control information.
      type: object
      properties:
        remainMbrUl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        remainMbrDl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
      oneOf:
        - required: [remainMbrUl]
        - required: [remainMbrDl]

    MbsSessPolCtrlData:
      description: Represents MBS Session Policy Control Data.
      type: object
      properties:
        5qis:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/5Qi'
          minItems: 1
        maxMbsArpLevel:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ArpPriorityLevel'
        maxMbsSessionAmbr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        maxGbr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        suppFeat:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'

    MbsSessPolDataId:
      description: Represents the identifier used to access the MBS Session Policy Control Data.
      type: object
      properties:
        mbsSessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsSessionId'
        afAppId:
          type: string
      oneOf:
        - required: [mbsSessionId]
        - required: [afAppId]

# SIMPLE TYPES:

    IpIndex:
      description: >
        Represents information that identifies which IP pool or external server
        is used to allocate the IP address.
      type: integer

    OsId:
      description: Represents the Operating System of the served UE.
      type: string
      format: uuid

    ItemPath:
      description: Identifies a fragment (subset of resource data) of a given resource.
      type: string

# ENUMS:

    UsageMonLevel:
      description: Represents the usage monitoring level.
      anyOf:
```

```
      - type: string
        enum:
          - SESSION_LEVEL
          - SERVICE_LEVEL
      - type: string

    Periodicity:
      description: Represents the time period.
      anyOf:
      - type: string
        enum:
          - YEARLY
          - MONTHLY
          - WEEKLY
          - DAILY
          - HOURLY
      - type: string

    BdtPolicyStatus:
      description: Indicates the validation status of a negotiated BDT policy.
      anyOf:
      - type: string
        enum:
          - INVALID
          - VALID
      - type: string

    PolicyDataSubset:
      description: Indicates a policy data subset.
      anyOf:
        - type: string
          enum:
          - AM_POLICY_DATA
          - SM_POLICY_DATA
          - UE_POLICY_DATA
          - UM_DATA
          - OPERATOR_SPECIFIC_DATA
        - type: string
```

A.3 Nudr_DataRepository API for Application Data

For the purpose of referencing entities in the Open API file defined in this Annex, it shall be assumed that this Open API file is contained in a physical file named "TS29519_Application_Data.yaml".

```
openapi: 3.0.0

info:
  version: '-'
  title: Unified Data Repository Service API file for Application Data
  description: |
    The API version is defined in 3GPP TS 29.504
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: >
    3GPP TS 29.519 V17.16.0; 5G System; Usage of the Unified Data Repository Service for Policy
Data,
    Application Data and Structured Data for Exposure.
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.519/'

paths:
  /application-data/pfds:
    get:
      summary: Retrieve PFDs for application identifier(s)
      operationId: ReadPFDData
      tags:
        - PFD Data (Store)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
```

```
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:pfds:read
    parameters:
      - name: appId
        in: query
        description: >
          Contains the information of the application identifier(s) for the querying PFD
          Data resource. If none appId is included in the URI, it applies to all application
          identifier(s) for the querying PFD Data resource.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/ApplicationId'
          minItems: 1
      - name: supp-feat
        in: query
        description: Supported Features
        required: false
        schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    responses:
      '200':
        description: A representation of PFDs for request applications is returned.
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/PfdDataForAppExt'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /application-data/pfds/{appId}:
    get:
      summary: Retrieve the corresponding PFDs of the specified application identifier
      operationId: ReadIndividualPFDData
      tags:
        - Individual PFD Data (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:pfds:read
      parameters:
        - name: appId
          in: path
          description: >
            Indicate the application identifier for the request pfd(s). It shall apply the
            format of Data type ApplicationId.
          required: true
```

```
        schema:
          type: string
      - name: supp-feat
        in: query
        description: Supported Features
        required: false
        schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    responses:
      '200':
        description: >
          A representation of PFDs for the request application identified by the application
          identifier is returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PfdDataForAppExt'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Delete the corresponding PFDs of the specified application identifier
    operationId: DeleteIndividualPFDData
    tags:
      - Individual PFD Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:pfds:modify
    parameters:
      - name: appId
        in: path
        description: >
          Indicate the application identifier for the request pfd(s). It shall apply the
          format of Data type ApplicationId.
        required: true
        schema:
          type: string
    responses:
      '204':
        description: >
          Successful case. The Individual PFD Data resource related to the application
          identifier was deleted.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  put:
    summary: Create or update the corresponding PFDs for the specified application identifier
    operationId: CreateOrReplaceIndividualPFDData
    tags:
      - Individual PFD Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:pfds:create
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/PfdDataForAppExt'
    parameters:
      - name: appId
        in: path
        description: >
          Indicate the application identifier for the request pfd(s). It shall apply the format
          of Data type ApplicationId.
        required: true
        schema:
          type: string
    responses:
      '201':
        description: >
          The creation of an Individual PFD Data resource related to the application-identifier
          is confirmed and a representation of that resource is returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PfdDataForAppExt'
        headers:
          Location:
            description: >
              'Contains the URI of the newly created resource, according to the structure:
              {apiRoot}/nudr-dr/<apiVersion>/application-data/pfds/{appId}'
            required: true
            schema:
              type: string
      '200':
        description: >
          Successful case. The upgrade of an Individual PFD Data resource related to the
          application identifier is confirmed and a representation of that resource is returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PfdDataForAppExt'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/application-data/influenceData:
  get:
    summary: Retrieve Traffic Influence Data
    operationId: ReadInfluenceData
    tags:
      - Influence Data (Store)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:influence-data:read
    parameters:
      - name: influence-Ids
        in: query
        description: Each element identifies a service.
        required: false
        schema:
          type: array
          items:
            type: string
          minItems: 1
      - name: dnns
        in: query
        description: Each element identifies a DNN.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
          minItems: 1
      - name: snssais
        in: query
        description: Each element identifies a slice.
        required: false
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
              minItems: 1
      - name: internal-Group-Ids
        in: query
        description: Each element identifies a group of users.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
          minItems: 1
      - name: supis
        in: query
        description: Each element identifies the user.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
          minItems: 1
      - name: supp-feat
        in: query
        required: false
        description: Supported Features
        schema:
```

```
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      responses:
        '200':
          description: The Traffic Influence Data stored in the UDR are returned.
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/TrafficInfluData'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '414':
          $ref: 'TS29571_CommonData.yaml#/components/responses/414'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /application-data/influenceData/{influenceId}:
    put:
      summary: Create or update an individual Influence Data resource
      operationId: CreateOrReplaceIndividualInfluenceData
      tags:
        - Individual Influence Data (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:influence-data:create
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/TrafficInfluData'
      parameters:
        - name: influenceId
          in: path
          description: >
            The Identifier of an Individual Influence Data to be created or updated.
            It shall apply the format of Data type string.
          required: true
          schema:
            type: string
      responses:
        '201':
          description: >
            The creation of an Individual Traffic Influence Data resource is confirmed
            and a representation of that resource is returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/TrafficInfluData'
          headers:
            Location:
              description: >
                'Contains the URI of the newly created resource, according to the structure:
                {apiRoot}/nudr-dr/<apiVersion>/application-data/influenceData/{influenceId}'
              required: true
```

```
            schema:
              type: string
      '200':
        description: >
          The update of an Individual Traffic Influence Data resource is confirmed and a
          response body containing Traffic Influence Data shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/TrafficInfluData'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  patch:
    summary: Modify part of the properties of an individual Influence Data resource
    operationId: UpdateIndividualInfluenceData
    tags:
      - Individual Influence Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:influence-data:modify
    requestBody:
      required: true
      content:
        application/merge-patch+json:
          schema:
            $ref: '#/components/schemas/TrafficInfluDataPatch'
    parameters:
      - name: influenceId
        in: path
        description: >
          The Identifier of an Individual Influence Data to be updated. It shall apply
          the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '200':
        description: >
          The update of an Individual Traffic Influence Data resource is confirmed and
          a response body containing Traffic Influence Data shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/TrafficInfluData'
      '204':
        description: No content
```

```
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Delete an individual Influence Data resource
    operationId: DeleteIndividualInfluenceData
    tags:
      - Individual Influence Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:influence-data:modify
    parameters:
      - name: influenceId
        in: path
        description: >
          The Identifier of an Individual Influence Data to be deleted. It shall apply
          the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '204':
        description: The Individual Influence Data was deleted successfully.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/application-data/influenceData/subs-to-notify:
  post:
    summary: Create a new Individual Influence Data Subscription resource
    operationId: CreateIndividualInfluenceDataSubscription
    tags:
      - Influence Data Subscriptions (Collection)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
```

```
      - nudr-dr
      - nudr-dr:application-data
    - oAuth2ClientCredentials:
      - nudr-dr
      - nudr-dr:application-data
      - nudr-dr:application-data:influence-data:subscriptions:create
  requestBody:
    required: true
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/TrafficInfluSub'
  responses:
    '201':
      description: The subscription was created successfully.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/TrafficInfluSub'
      headers:
        Location:
          description: 'Contains the URI of the newly created resource'
          required: true
          schema:
            type: string
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '411':
      $ref: 'TS29571_CommonData.yaml#/components/responses/411'
    '413':
      $ref: 'TS29571_CommonData.yaml#/components/responses/413'
    '415':
      $ref: 'TS29571_CommonData.yaml#/components/responses/415'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    default:
      $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  callbacks:
    trafficInfluenceDataChangeNotification:
      '{$request.body#/notificationUri}':
        post:
          requestBody:
            required: true
            content:
              application/json:
                schema:
                  type: array
                  items:
                    oneOf:
                      - $ref: '#/components/schemas/TrafficInfluData'
                      - $ref: '#/components/schemas/TrafficInfluDataNotif'
                  minItems: 1
          responses:
            '204':
              description: No Content, Notification was successful
            '400':
              $ref: 'TS29571_CommonData.yaml#/components/responses/400'
            '403':
              $ref: 'TS29571_CommonData.yaml#/components/responses/403'
            '404':
              $ref: 'TS29571_CommonData.yaml#/components/responses/404'
            '411':
              $ref: 'TS29571_CommonData.yaml#/components/responses/411'
            '413':
              $ref: 'TS29571_CommonData.yaml#/components/responses/413'
            '415':
              $ref: 'TS29571_CommonData.yaml#/components/responses/415'
            '429':
```

```
              $ref: 'TS29571_CommonData.yaml#/components/responses/429'
            '500':
              $ref: 'TS29571_CommonData.yaml#/components/responses/500'
            '503':
              $ref: 'TS29571_CommonData.yaml#/components/responses/503'
            default:
              $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  get:
    summary: Read Influence Data Subscriptions
    operationId: ReadInfluenceDataSubscriptions
    tags:
      - Influence Data Subscriptions (Collection)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:influence-data:subscriptions:read
    parameters:
      - name: dnn
        in: query
        description: Identifies a DNN.
        required: false
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      - name: snssai
        in: query
        description: Identifies a slice.
        required: false
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      - name: internal-Group-Id
        in: query
        description: Identifies a group of users.
        required: false
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
      - name: supi
        in: query
        description: Identifies a user.
        required: false
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
    responses:
      '200':
        description: >
          The subscription information as request in the request URI query parameter(s)
          are returned.
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/TrafficInfluSub'
              minItems: 0
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
```

```
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /application-data/influenceData/subs-to-notify/{subscriptionId}:
    get:
      summary: Get an existing individual Influence Data Subscription resource
      operationId: ReadIndividualInfluenceDataSubscription
      tags:
        - Individual Influence Data Subscription (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:influence-data:subscriptions:read
      parameters:
        - name: subscriptionId
          in: path
          description: >
            String identifying a subscription to the Individual Influence Data Subscription
          required: true
          schema:
            type: string
      responses:
        '200':
          description: The subscription information is returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/TrafficInfluSub'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '414':
          $ref: 'TS29571_CommonData.yaml#/components/responses/414'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    put:
      summary: Modify an existing individual Influence Data Subscription resource
      operationId: ReplaceIndividualInfluenceDataSubscription
      tags:
        - Individual Influence Data Subscription (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:influence-data:subscriptions:modify
      requestBody:
        required: true
        content:
          application/json:
            schema:
```

```
          $ref: '#/components/schemas/TrafficInfluSub'
    parameters:
      - name: subscriptionId
        in: path
        description: >
          String identifying a subscription to the Individual Influence Data Subscription
        required: true
        schema:
          type: string
    responses:
      '200':
        description: The subscription was updated successfully.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/TrafficInfluSub'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Delete an individual Influence Data Subscription resource
    operationId: DeleteIndividualInfluenceDataSubscription
    tags:
      - Individual Influence Data Subscription (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:influence-data:subscriptions:modify
    parameters:
      - name: subscriptionId
        in: path
        description: >
          String identifying a subscription to the Individual Influence Data Subscription
        required: true
        schema:
          type: string
    responses:
      '204':
        description: The subscription was terminated successfully.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/application-data/bdtPolicyData:
  get:
    summary: Retrieve applied BDT Policy Data
    operationId: ReadBdtPolicyData
    tags:
      - BdtPolicy Data (Store)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:bdt-policy-data:read
    parameters:
      - name: bdt-policy-ids
        in: query
        description: Each element identifies a service.
        required: false
        schema:
          type: array
          items:
            type: string
          minItems: 1
      - name: internal-group-ids
        in: query
        description: Each element identifies a group of users.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
          minItems: 1
      - name: supis
        in: query
        description: Each element identifies the user.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
          minItems: 1
    responses:
      '200':
        description: The applied BDT policy Data stored in the UDR are returned.
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/BdtPolicyData'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
```

```
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /application-data/bdtPolicyData/{bdtPolicyId}:
    put:
      summary: Create an individual applied BDT Policy Data resource
      operationId: CreateIndividualAppliedBdtPolicyData
      tags:
        - Individual Applied BDT Policy Data (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:bdt-policy-data:create
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/BdtPolicyData'
      parameters:
        - name: bdtPolicyId
          in: path
          description: >
            The Identifier of an Individual Applied BDT Policy Data to be created or updated.
            It shall apply the format of Data type string.
          required: true
          schema:
            type: string
      responses:
        '201':
          description: >
            The creation of an Individual Applied BDT Policy Data resource is confirmed and a
            representation of that resource is returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/BdtPolicyData'
          headers:
            Location:
              description: >
                Contains the URI of the newly created resource, according to the structure:
                {apiRoot}/nudr-dr/<apiVersion>/application-data/bdtPolicyData/{bdtPolicyId}
              required: true
              schema:
                type: string
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '414':
          $ref: 'TS29571_CommonData.yaml#/components/responses/414'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    patch:
      summary: Modify part of the properties of an individual Applied BDT Policy Data resource
```

```
    operationId: UpdateIndividualAppliedBdtPolicyData
    tags:
      - Individual Applied BDT Policy Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:bdt-policy-data:modify
    requestBody:
      required: true
      content:
        application/merge-patch+json:
          schema:
            $ref: '#/components/schemas/BdtPolicyDataPatch'
    parameters:
      - name: bdtPolicyId
        in: path
        description: >
          The Identifier of an Individual Applied BDT Policy Data to be updated. It shall
          apply the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '200':
        description: >
          The update of an Individual Applied BDT Policy Data resource is confirmed and
          a response body containing Applied BDT Policy Data shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/BdtPolicyData'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Delete an individual Applied BDT Policy Data resource
    operationId: DeleteIndividualAppliedBdtPolicyData
    tags:
      - Individual Applied BDT Policy Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:bdt-policy-data:modify
```

```
    parameters:
      - name: bdtPolicyId
        in: path
        description: >
          The Identifier of an Individual Applied BDT Policy Data to be deleted.
          It shall apply the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '204':
        description: The Individual Applied BDT Policy Data was deleted successfully.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /application-data/iptvConfigData:
    get:
      summary: Retrieve IPTV configuration Data
      operationId: ReadIPTVCongifurationData
      tags:
        - IPTV Configuration Data (Store)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:iptv-config-data:read
      parameters:
        - name: config-ids
          in: query
          description: Each element identifies a configuration.
          required: false
          schema:
            type: array
            items:
              type: string
            minItems: 1
        - name: dnns
          in: query
          description: Each element identifies a DNN.
          required: false
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
            minItems: 1
        - name: snssais
          in: query
          description: Each element identifies a slice.
          required: false
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
                minItems: 1
        - name: supis
          in: query
```

```
        description: Each element identifies the user.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
          minItems: 1
      - name: inter-group-ids
        in: query
        description: Each element identifies a group of users.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
          minItems: 1
    responses:
      '200':
        description: The IPTV configuration data stored in the UDR are returned.
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/IptvConfigData'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /application-data/iptvConfigData/{configurationId}:
    put:
      summary: Create or update an individual IPTV configuration resource
      operationId: CreateOrReplaceIndividualIPTVConfigurationData
      tags:
        - Individual IPTV Configuration Data (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:iptv-config-data:create
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/IptvConfigData'
      parameters:
        - name: configurationId
          in: path
          description: >
            The Identifier of an Individual IPTV Configuration Data to be created or updated.
            It shall apply the format of Data type string.
          required: true
          schema:
            type: string
```

```
    responses:
      '201':
        description: >
          The creation of an Individual IPTV Configuration Data resource is confirmed and a
          representation of that resource is returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/IptvConfigData'
        headers:
          Location:
            description: 'Contains the URI of the newly created resource'
            required: true
            schema:
              type: string
      '200':
        description: The update of an Individual IPTV configuration resource.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/IptvConfigData'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  patch:
    summary: Partial update an individual IPTV configuration resource
    operationId: PartialReplaceIndividualIPTVConfigurationData
    tags:
      - Individual IPTV Configuration Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:iptv-config-data:modify
    requestBody:
      required: true
      content:
        application/merge-patch+json:
          schema:
            $ref: 'TS29522_IPTVConfiguration.yaml#/components/schemas/IptvConfigDataPatch'
    parameters:
      - name: configurationId
        in: path
        description: >
          The Identifier of an Individual IPTV Configuration Data to be updated.
          It shall apply the format of Data type string.
        required: true
        schema:
          type: string
```

```
    responses:
      '200':
        description: The update of an Individual IPTV configuration resource.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/IptvConfigData'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Delete an individual IPTV configuration resource
    operationId: DeleteIndividualIPTVConfigurationData
    tags:
      - Individual IPTV Configuration Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:iptv-config-data:modify
    parameters:
      - name: configurationId
        in: path
        description: >
          The Identifier of an Individual IPTV Configuration to be deleted. It shall
          apply the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '204':
        description: The resource was deleted successfully.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
```

```
  /application-data/serviceParamData:
    get:
      summary: Retrieve Service Parameter Data
      operationId: ReadServiceParameterData
      tags:
        - Service Parameter Data (Store)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:service-param-data:read
      parameters:
        - name: service-param-ids
          in: query
          description: Each element identifies a service.
          required: false
          schema:
            type: array
            items:
              type: string
            minItems: 1
        - name: dnns
          in: query
          description: Each element identifies a DNN.
          required: false
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
            minItems: 1
        - name: snssais
          in: query
          description: Each element identifies a slice.
          required: false
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
                minItems: 1
        - name: internal-group-ids
          in: query
          description: Each element identifies a group of users.
          required: false
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
            minItems: 1
        - name: supis
          in: query
          description: Each element identifies the user.
          required: false
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
            minItems: 1
        - name: ue-ipv4s
          in: query
          description: Each element identifies the user.
          required: false
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
            minItems: 1
        - name: ue-ipv6s
          in: query
          description: Each element identifies the user.
          required: false
```

```
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Addr'
            minItems: 1
        - name: ue-macs
          in: query
          description: Each element identifies the user.
          required: false
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
            minItems: 1
        - name: any-ue
          in: query
          description: Indicates whether the request is for any UE.
          required: false
          schema:
            type: boolean
        - name: supp-feat
          in: query
          description: Supported Features
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      responses:
        '200':
          description: The Service Parameter Data stored in the UDR are returned.
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/ServiceParameterData'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '414':
          $ref: 'TS29571_CommonData.yaml#/components/responses/414'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /application-data/serviceParamData/{serviceParamId}:
    put:
      summary: Create or update an individual Service Parameter Data resource
      operationId: CreateOrReplaceServiceParameterData
      tags:
        - Individual Service Parameter Data (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:service-param-data:create
      requestBody:
        required: true
        content:
          application/json:
            schema:
```

```
          $ref: '#/components/schemas/ServiceParameterData'
    parameters:
      - name: serviceParamId
        in: path
        description: >
          The Identifier of an Individual Service Parameter Data to be created or updated.
          It shall apply the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '201':
        description: >
          The creation of an Individual Service Parameter Data resource is confirmed
          and a representation of that resource is returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ServiceParameterData'
        headers:
          Location:
            description: >
              'Contains the URI of the newly created resource, according to the structure:
              {apiRoot}/nudr-dr/<apiVersion>/application-data/serviceParamData/{serviceParamId}'
            required: true
            schema:
              type: string
      '200':
        description: >
          The update of an Individual Service Parameter Data resource is confirmed and
          a response body containing Service Parameter Data shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ServiceParameterData'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  patch:
    summary: Modify part of the properties of an individual Service Parameter Data resource
    operationId: UpdateIndividualServiceParameterData
    tags:
      - Individual Service Parameter Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:service-param-data:modify
    requestBody:
```

```
      required: true
      content:
        application/merge-patch+json:
          schema:
            $ref: '#/components/schemas/ServiceParameterDataPatch'
    parameters:
      - name: serviceParamId
        in: path
        description: >
          The Identifier of an Individual Service Parameter Data to be updated.
          It shall apply the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '200':
        description: >
          The update of an Individual Service Parameter Data resource is confirmed
          and a response body containing Service Parameter Data shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ServiceParameterData'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Delete an individual Service Parameter Data resource
    operationId: DeleteIndividualServiceParameterData
    tags:
      - Individual Service Parameter Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:service-param-data:modify
    parameters:
      - name: serviceParamId
        in: path
        description: >
          The Identifier of an Individual Service Parameter Data to be deleted.
          It shall apply the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '204':
        description: The Individual Service Parameter Data was deleted successfully.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/application-data/am-influence-data:
  get:
    summary: Retrieve AM Influence Data
    operationId: ReadAmInfluenceData
    tags:
      - AM Influence Data (Store)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:am-influence-data:read
    parameters:
      - name: am-influence-ids
        in: query
        description: Each element identifies a service.
        required: false
        schema:
          type: array
          items:
            type: string
          minItems: 1
      - name: dnns
        in: query
        description: Each element identifies a DNN.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
          minItems: 1
      - name: snssais
        in: query
        description: Each element identifies a slice.
        required: false
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
              minItems: 1
      - name: dnn-snssai-infos
        in: query
        description: Each element identifies a combination of (DNN, S-NSSAI).
        required: false
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: 'TS29522_AMInfluence.yaml#/components/schemas/DnnSnssaiInformation'
              minItems: 1
      - name: internal-group-ids
        in: query
        description: Each element identifies a group of users.
        required: false
        schema:
          type: array
```

```
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
          minItems: 1
      - name: supis
        in: query
        description: Each element identifies the user.
        required: false
        schema:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
          minItems: 1
      - name: any-ue
        in: query
        description: Indicates whether the request is for any UE.
        required: false
        schema:
          type: boolean
      - name: supp-feat
        in: query
        required: false
        description: Supported Features
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    responses:
      '200':
        description: The AM Influence Data stored in the UDR are returned.
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/AmInfluData'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /application-data/am-influence-data/{amInfluenceId}:
    put:
      summary: Create or update an individual AM Influence Data resource
      operationId: CreateOrReplaceIndividualAmInfluenceData
      tags:
        - Individual AM Influence Data (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:am-influence-data:create
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AmInfluData'
      parameters:
```

```
      - name: amInfluenceId
        in: path
        description: >
          The Identifier of an Individual AM Influence Data to be created or updated.
          It shall apply the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '201':
        description: >
          The creation of an Individual AM Influence Data resource is confirmed and
          a representation of that resource is returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AmInfluData'
        headers:
          Location:
            description: >
              'Contains the URI of the newly created resource, according to the structure:
              {apiRoot}/nudr-dr/<apiVersion>/application-data/am-influence-data/{amInfluenceId}'
            required: true
            schema:
              type: string
      '200':
        description: >
          The update of an Individual AM Influence Data resource is confirmed and a response
          body containing AM Influence Data shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AmInfluData'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  patch:
    summary: Modify part of the properties of an individual AM Influence Data resource
    operationId: UpdateIndividualAmInfluenceData
    tags:
      - Individual AM Influence Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:am-influence-data:modify
    requestBody:
      required: true
      content:
```

```
        application/merge-patch+json:
          schema:
            $ref: '#/components/schemas/AmInfluDataPatch'
    parameters:
      - name: amInfluenceId
        in: path
        description: >
          The Identifier of an Individual AM Influence Data to be updated. It shall
          apply the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '200':
        description: >
          The update of an Individual AM Influence Data resource is confirmed and a
          response body containing AM Influence Data shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AmInfluData'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Delete an individual AM Influence Data resource
    operationId: DeleteIndividualAmInfluenceData
    tags:
      - Individual AM Influence Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:am-influence-data:modify
    parameters:
      - name: amInfluenceId
        in: path
        description: >
          The Identifier of an Individual AM Influence Data to be deleted. It shall
          apply the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '204':
        description: The Individual AM Influence Data was deleted successfully.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/application-data/subs-to-notify:
  post:
    summary: Create a subscription to receive notification of application data changes
    operationId: CreateIndividualApplicationDataSubscription
    tags:
      - ApplicationDataSubscriptions (Collection)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:subs-to-notify:create
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/ApplicationDataSubs'
    responses:
      '201':
        description: >
          Upon success, a response body containing a representation of each
          Individual subscription resource shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ApplicationDataSubs'
        headers:
          Location:
            description: 'Contains the URI of the newly created resource'
            required: true
            schema:
              type: string
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    callbacks:
      applicationDataChangeNotif:
        '{$request.body#/notificationUri}':
          post:
            requestBody:
              required: true
```

```
              content:
                application/json:
                  schema:
                    type: array
                    items:
                      $ref: '#/components/schemas/ApplicationDataChangeNotif'
                    minItems: 1
            responses:
              '204':
                description: No Content, Notification was successful
              '400':
                $ref: 'TS29571_CommonData.yaml#/components/responses/400'
              '401':
                $ref: 'TS29571_CommonData.yaml#/components/responses/401'
              '403':
                $ref: 'TS29571_CommonData.yaml#/components/responses/403'
              '404':
                $ref: 'TS29571_CommonData.yaml#/components/responses/404'
              '411':
                $ref: 'TS29571_CommonData.yaml#/components/responses/411'
              '413':
                $ref: 'TS29571_CommonData.yaml#/components/responses/413'
              '415':
                $ref: 'TS29571_CommonData.yaml#/components/responses/415'
              '429':
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
              default:
                $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    get:
      summary: Read Application Data change Subscriptions
      operationId: ReadApplicationDataChangeSubscriptions
      tags:
        - ApplicationDataSubscriptions (Collection)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:subs-to-notify:read
      parameters:
        - name: data-filter
          in: query
          description: The data filter for the query.
          required: false
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/DataFilter'
      responses:
        '200':
          description: >
            The subscription information as request in the request URI query parameter(s)
            are returned.
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/ApplicationDataSubs'
                minItems: 0
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/application-data/subs-to-notify/{subsId}:
  parameters:
   - name: subsId
     in: path
     required: true
     schema:
       type: string
  put:
    summary: Modify a subscription to receive notification of application data changes
    operationId: ReplaceIndividualApplicationDataSubscription
    tags:
      - IndividualApplicationDataSubscription (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:subs-to-notify:modify
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/ApplicationDataSubs'
    responses:
      '200':
        description: The individual subscription resource was updated successfully.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ApplicationDataSubs'
      '204':
        description: >
          The individual subscription resource was updated successfully and no
          additional content is to be sent in the response message.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Delete the individual Application Data subscription
    operationId: DeleteIndividualApplicationDataSubscription
    tags:
```

```
      - IndividualApplicationDataSubscription (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:subs-to-notify:modify
    responses:
      '204':
        description: Upon success, an empty response body shall be returned.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  get:
    summary: Get an existing individual Application Data Subscription resource
    operationId: ReadIndividualApplicationDataSubscription
    tags:
      - IndividualApplicationDataSubscription (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:subs-to-notify:read
    parameters:
      - name: subsId
        in: path
        description: >
          String identifying a subscription to the Individual Application Data Subscription
        required: true
        schema:
          type: string
    responses:
      '200':
        description: The subscription information is returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ApplicationDataSubs'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
```

```
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /application-data/eas-deploy-data:
    get:
      summary: Retrieve EAS Deployment Information Data
      operationId: ReadEasDeployData
      tags:
        - EAS Deployment Data (Store)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:application-data
          - nudr-dr:application-data:eas-deploy-data:read
      parameters:
        - name: dnn
          in: query
          description: Identifies a DNN.
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        - name: snssai
          in: query
          description: Identifies an S-NSSAI.
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        - name: internal-group-id
          in: query
          description: Identifies a group of users.
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
        - name: appId
          in: query
          description: Identifies an application.
          required: false
          schema:
            type: string
      responses:
        '200':
          description: The EAS Deployment Data stored in the UDR are returned.
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: 'TS29591_Nnef_EASDeployment.yaml#/components/schemas/EasDeployInfoData'
                minItems: 1
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '414':
          $ref: 'TS29571_CommonData.yaml#/components/responses/414'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  /application-data/eas-deploy-data/{easDeployInfoId}:
```

```
  get:
    summary: Retrieve an individual EAS Deployment Data resource
    operationId: ReadIndividualEasDeployData
    tags:
      - Individual EAS Deployment Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:eas-deploy-data:read
    parameters:
      - name: easDeployInfoId
        description: >
          String identifying an Individual EAS Deployment Information Data resource.
        in: path
        required: true
        schema:
          type: string
    responses:
      '200':
        description: >
          The EAS Deployment Data stored in the UDR for an Individual EAS Deployment
          Information Data resource is returned.
        content:
          application/json:
            schema:
              $ref: 'TS29591_Nnef_EASDeployment.yaml#/components/schemas/EasDeployInfoData'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  put:
    summary: Create or update an individual EAS Deployment Data resource
    operationId: CreateOrReplaceIndividualEasDeployData
    tags:
      - Individual EAS Deployment Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:eas-deploy-data:create
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: 'TS29591_Nnef_EASDeployment.yaml#/components/schemas/EasDeployInfoData'
    parameters:
      - name: easDeployInfoId
        in: path
        description: >
          The Identifier of an Individual EAS Deployment Data to be created or updated.
```

```
          It shall apply the format of Data type string.
        required: true
        schema:
          type: string
    responses:
      '201':
        description: >
          The creation of an Individual EAS Deployment Data resource is confirmed and a
          representation of that resource is returned.
        content:
          application/json:
            schema:
              $ref: 'TS29591_Nnef_EASDeployment.yaml#/components/schemas/EasDeployInfoData'
        headers:
          Location:
            description: >
              Contains the URI of the newly created resource, according to the structure:
              {apiRoot}/nudr-dr/<apiVersion>/application-data/eas-deploy-data/{easDeployInfoId}
            required: true
            schema:
              type: string
      '200':
        description: >
          The update of an Individual EAS Deployment Data resource is confirmed and a response
          body containing EAS Deployment Data shall be returned.
        content:
          application/json:
            schema:
              $ref: 'TS29591_Nnef_EASDeployment.yaml#/components/schemas/EasDeployInfoData'
      '204':
        description: No content
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Delete an individual EAS Deployment Data resource
    operationId: DeleteIndividualEasDeployData
    tags:
      - Individual EasDeployment Data (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:application-data
        - nudr-dr:application-data:eas-deploy-data:modify
    parameters:
      - name: easDeployInfoId
        in: path
        description: >
          The Identifier of an Individual EAS Deployment Data to be updated. It shall apply the
          format of Data type string.
        required: true
```

```
          schema:
            type: string
      responses:
        '204':
          description: The Individual Influence Data was deleted successfully.
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  schemas:

    TrafficInfluData:
      description: Represents the Traffic Influence Data.
      type: object
      properties:
        upPathChgNotifCorreId:
          type: string
          description: >
            Contains the Notification Correlation Id allocated by the NEF for the UP
            path change notification.
        appReloInd:
          type: boolean
          description: >
            Identifies whether an application can be relocated once a location of the
            application has been selected.
        afAppId:
          type: string
          description: Identifies an application.
        dnn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        ethTrafficFilters:
          type: array
          items:
            $ref: 'TS29514_Npcf_PolicyAuthorization.yaml#/components/schemas/EthFlowDescription'
          minItems: 1
          description: >
            Identifies Ethernet packet filters. Either "trafficFilters" or
            "ethTrafficFilters" shall be included if applicable.
        snssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        interGroupId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        trafficFilters:
          type: array
          items:
            $ref: 'TS29122_CommonData.yaml#/components/schemas/FlowInfo'
          minItems: 1
          description: >
            Identifies IP packet filters. Either "trafficFilters" or "ethTrafficFilters"
            shall be included if applicable.
        trafficRoutes:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/RouteToLocation'
          minItems: 1
          description: Identifies the N6 traffic routing requirement.
        traffCorreInd:
          type: boolean
        validStartTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        validEndTime:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      tempValidities:
        type: array
        items:
          $ref: 'TS29514_Npcf_PolicyAuthorization.yaml#/components/schemas/TemporalValidity'
        minItems: 1
        description: Identifies the temporal validities for the N6 traffic routing requirement.
      nwAreaInfo:
        $ref: 'TS29554_Npcf_BDTPolicyControl.yaml#/components/schemas/NetworkAreaInfo'
      upPathChgNotifUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      headers:
        description: Contains the headers provisioned by the NEF.
        type: array
        items:
          type: string
        minItems: 1
      subscribedEvents:
        type: array
        items:
          $ref: 'TS29522_TrafficInfluence.yaml#/components/schemas/SubscribedEvent'
        minItems: 1
      dnaiChgType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DnaiChangeType'
      afAckInd:
        type: boolean
      addrPreserInd:
        type: boolean
      maxAllowedUpLat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
      simConnInd:
        type: boolean
        description: >
          Indicates whether simultaneous connectivity should be temporarily
          maintained for the source and target PSA.
      simConnTerm:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
    allOf:
      - oneOf:
        - required: [afAppId]
        - required: [trafficFilters]
        - required: [ethTrafficFilters]
      - oneOf:
        - required: [supi]
        - required: [interGroupId]

  TrafficInfluDataPatch:
    description: Represents the Traffic Influence Data to be updated in the UDR.
    type: object
    properties:
      upPathChgNotifCorreId:
        type: string
        description: >
          Contains the Notification Correlation Id allocated by the NEF for the
          UP path change notification.
      appReloInd:
        type: boolean
        description: >
          Identifies whether an application can be relocated once a location of the application
          has been selected.
      ethTrafficFilters:
        type: array
        items:
          $ref: 'TS29514_Npcf_PolicyAuthorization.yaml#/components/schemas/EthFlowDescription'
        minItems: 1
        description: >
          Identifies Ethernet packet filters. Either "trafficFilters" or "ethTrafficFilters"
          shall be included if applicable.
      trafficFilters:
```

```
        type: array
        items:
          $ref: 'TS29122_CommonData.yaml#/components/schemas/FlowInfo'
        minItems: 1
        description: >
          Identifies IP packet filters. Either "trafficFilters" or "ethTrafficFilters"
          shall be included if applicable.
      trafficRoutes:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RouteToLocation'
        minItems: 1
        description: Identifies the N6 traffic routing requirement.
      traffCorreInd:
        type: boolean
      validStartTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      validEndTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      tempValidities:
        type: array
        items:
          $ref: 'TS29514_Npcf_PolicyAuthorization.yaml#/components/schemas/TemporalValidity'
        minItems: 1
        nullable: true
        description: Identifies the temporal validities for the N6 traffic routing requirement.
      nwAreaInfo:
        $ref: 'TS29554_Npcf_BDTPolicyControl.yaml#/components/schemas/NetworkAreaInfo'
      upPathChgNotifUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      headers:
        description: Contains the headers provisioned by the NEF.
        type: array
        items:
          type: string
        minItems: 1
      afAckInd:
        type: boolean
      addrPreserInd:
        type: boolean
      maxAllowedUpLat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/UintegerRm'
      simConnInd:
        type: boolean
        description: >
          Indicates whether simultaneous connectivity should be temporarily maintained
          for the source and target PSA.
      simConnTerm:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSecRm'

  TrafficInfluSub:
    description: Represents traffic influence subscription data.
    type: object
    properties:
      dnns:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        minItems: 1
        description: Each element identifies a DNN.
      snssais:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        minItems: 1
        description: Each element identifies a slice.
      internalGroupIds:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
        minItems: 1
        description: Each element identifies a group of users.
      supis:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        minItems: 1
        description: Each element identifies the user.
```

```
      notificationUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      expiry:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
    required:
      - notificationUri
    oneOf:
      - required: [dnns]
      - required: [snssais]
      - required: [internalGroupIds]
      - required: [supis]

  TrafficInfluDataNotif:
    description: Represents traffic influence data for notification.
    type: object
    properties:
      resUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      trafficInfluData:
        $ref: '#/components/schemas/TrafficInfluData'
    required:
      - resUri

  PfdDataForAppExt:
    description: Represents the PFDs and related data for the application.
    type: object
    properties:
      applicationId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ApplicationId'
      pfds:
        type: array
        items:
          $ref: 'TS29551_Nnef_PFDmanagement.yaml#/components/schemas/PfdContent'
        minItems: 1
      cachingTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      suppFeat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
      allowedDelay:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
    required:
      - applicationId
      - pfds

  BdtPolicyData:
    description: Represents applied BDT policy data.
    type: object
    properties:
      interGroupId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
      supi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
      bdtRefId:
        $ref: 'TS29122_CommonData.yaml#/components/schemas/BdtReferenceId'
      dnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      snssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      resUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
    required:
```

```
    - bdtRefId

  BdtPolicyDataPatch:
    description: >
      Represents modification instructions to be performed on the applied BDT policy data.
    type: object
    properties:
      bdtRefId:
        $ref: 'TS29122_CommonData.yaml#/components/schemas/BdtReferenceId'
    required:
     - bdtRefId

  IptvConfigData:
    description: Represents IPTV configuration data information.
    type: object
    properties:
      supi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
      interGroupId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
      dnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      snssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      afAppId:
        type: string
      multiAccCtrls:
        type: object
        additionalProperties:
          $ref: 'TS29522_IPTVConfiguration.yaml#/components/schemas/MulticastAccessControl'
        minProperties: 1
        description: >
          Identifies a list of multicast address access control information. Any string
          value can be used as a key of the map.
      suppFeat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
    required:
      - afAppId
      - multiAccCtrls
    oneOf:
      - required: [interGroupId]
      - required: [supi]

  ServiceParameterData:
    description: Represents the service parameter data.
    type: object
    properties:
      appId:
        type: string
        description: Identifies an application.
      dnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      snssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      interGroupId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
      supi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
      ueIpv4:
        $ref: 'TS29122_CommonData.yaml#/components/schemas/Ipv4Addr'
      ueIpv6:
        $ref: 'TS29122_CommonData.yaml#/components/schemas/Ipv6Addr'
      ueMac:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
      anyUeInd:
        type: boolean
      paramOverPc5:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParameterOverPc5'
      paramOverUu:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParameterOverUu'
      paramForProSeDd:
```

```
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParamForProSeDd'
      paramForProSeDc:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParamForProSeDc'
      paramForProSeU2NRelUe:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParamForProSeU2NRelUe'
      paramForProSeRemUe:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParamForProSeRemUe'
      urspGuidance:
        type: array
        items:
          $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/UrspRuleRequest'
        minItems: 1
        description: Contains the service parameter used to guide the URSP.
      deliveryEvents:
        type: array
        items:
         $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/Event'
        minItems: 1
        description: Contains the outcome of the UE Policy Delivery.
      policDelivNotifCorreId:
        type: string
        description: >
          Contains the Notification Correlation Id allocated by the NEF for the notification
          of UE Policy delivery outcome.
      policDelivNotifUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      suppFeat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      headers:
        description: Contains the headers provisioned by the NEF.
        type: array
        items:
          type: string
        minItems: 1
      resetIds:
        type: array
        items:
          type: string
        minItems: 1

  ServiceParameterDataPatch:
    description: Represents the service parameter data that can be updated.
    type: object
    properties:
      paramOverPc5:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParameterOverPc5'
      paramOverUu:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParameterOverUu'
      paramForProSeDd:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParamForProSeDd'
      paramForProSeDc:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParamForProSeDc'
      paramForProSeU2NRelUe:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParamForProSeU2NRelUe'
      paramForProSeRemUe:
        $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/ParamForProSeRemUe'
      urspInfluence:
        type: array
        items:
          $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/UrspRuleRequest'
        minItems: 1
        deprecated: true
        description: >
          Contains the service parameter used to influence the URSP. This attribute is
          deprecated by the urspGuidance attribute.
      urspGuidance:
        type: array
        items:
          $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/UrspRuleRequest'
        minItems: 1
        nullable: true
        description: >
          Contains the service parameter used to influence the URSP.
      deliveryEvents:
        type: array
        items:
```

```
          $ref: 'TS29522_ServiceParameter.yaml#/components/schemas/Event'
        minItems: 1
        description: Contains the outcome of the UE Policy Delivery.
      policDelivNotifUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      headers:
        description: Contains the headers provisioned by the NEF.
        type: array
        items:
          type: string
        minItems: 1

  AmInfluData:
    description: Represents the AM Influence Data.
    type: object
    properties:
      appIds:
        type: array
        items:
          type: string
        minItems: 1
        description: Identifies one or more applications.
      dnnSnssaiInfos:
        type: array
        items:
          $ref: 'TS29522_AMInfluence.yaml#/components/schemas/DnnSnssaiInformation'
        minItems: 1
        description: Identifies one or more DNN, S-NSSAI combinations.
      interGroupId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
      supi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
      anyUeInd:
        type: boolean
        description: Indicates whether the data is applicable for any UE.
      policyDuration:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      evSubs:
        type: array
        items:
          $ref: 'TS29522_AMInfluence.yaml#/components/schemas/AmInfluEvent'
        minItems: 1
        description: List of AM related events for which a subscription is required.
      notifUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      notifCorrId:
        type: string
        description: Notification correlation identifier.
      headers:
        type: array
        description: Contains the headers provisioned by the NEF.
        items:
          type: string
        minItems: 1
      thruReq:
        type: boolean
        description: Indicates whether high throughput is desired for the indicated UE traffic.
      covReq:
        type: array
        items:
          $ref:
'TS29534_Npcf_AMPolicyAuthorization.yaml#/components/schemas/ServiceAreaCoverageInfo'
        minItems: 1
        description: Indicates the service area coverage requirement.
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
    allOf:
      - anyOf:
        - required: [thruReq]
        - required: [covReq]
      - oneOf:
```

```
          - required: [supi]
          - required: [interGroupId]
          - required: [anyUeInd]

    AmInfluDataPatch:
      description: Represents the AM Influence Data that can be updated.
      type: object
      properties:
        appIds:
          type: array
          items:
            type: string
          minItems: 1
          description: Identifies one or more applications.
          nullable: true
        dnnSnssaiInfos:
          type: array
          items:
            $ref: 'TS29522_AMInfluence.yaml#/components/schemas/DnnSnssaiInformation'
          minItems: 1
          description: Identifies one or more DNN, S-NSSAI combinations.
          nullable: true
        evSubs:
          type: array
          items:
            $ref: 'TS29522_AMInfluence.yaml#/components/schemas/AmInfluEvent'
          minItems: 1
          description: List of AM related events for which a subscription is required.
          nullable: true
        headers:
          type: array
          description: Contains the headers provisioned by the NEF.
          items:
            type: string
          minItems: 1
        thruReq:
          type: boolean
          description: Indicates whether high throughput is desired for the indicated UE traffic.
          nullable: true
        notifUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UriRm'
        notifCorrId:
          type: string
          description: Notification correlation identifier.
          nullable: true
        covReq:
          type: array
          items:
            $ref:
'TS29534_Npcf_AMPolicyAuthorization.yaml#/components/schemas/ServiceAreaCoverageInfo'
          minItems: 1
          description: Indicates the service area coverage requirement.
          nullable: true
        policyDuration:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSecRm'

    ApplicationDataSubs:
      description: Identifies a subscription to application data change notification.
      type: object
      properties:
        notificationUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        dataFilters:
          type: array
          items:
            $ref: '#/components/schemas/DataFilter'
          minItems: 1
        expiry:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        immRep:
          type: boolean
          description: Immediate reporting indication.
        amInfluEntries:
          type: array
          items:
            $ref: '#/components/schemas/AmInfluData'
          minItems: 1
          description: The AM Influence Data entries stored in the UDR that match a subscription.
```

```
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
    required:
      - notificationUri

  ApplicationDataChangeNotif:
    description: Contains changed application data for which notification was requested.
    type: object
    properties:
      iptvConfigData:
        $ref: '#/components/schemas/IptvConfigData'
      pfdData:
        $ref: 'TS29551_Nnef_PFDmanagement.yaml#/components/schemas/PfdChangeNotification'
      bdtPolicyData:
        $ref: '#/components/schemas/BdtPolicyData'
      resUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      serParamData:
        $ref: '#/components/schemas/ServiceParameterData'
      amInfluData:
        $ref: '#/components/schemas/AmInfluData'
    required:
      - resUri

  DataFilter:
    description: Identifies a data filter.
    type: object
    properties:
      dataInd:
        $ref: '#/components/schemas/DataInd'
      dnns:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        minItems: 1
      snssais:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        minItems: 1
      internalGroupIds:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
        minItems: 1
      supis:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        minItems: 1
      appIds:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ApplicationId'
        minItems: 1
      ueIpv4s:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
        minItems: 1
      ueIpv6s:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Addr'
        minItems: 1
      ueMacs:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
        minItems: 1
      anyUeInd:
        type: boolean
        description: Indicates the request is for any UE.
```

```
        dnnSnssaiInfos:
          description: Indicates the request is for any DNN and S-NSSAI combination present in the
array.
          type: array
          items:
            $ref: 'TS29522_AMInfluence.yaml#/components/schemas/DnnSnssaiInformation'
          minItems: 1
      required:
        - dataInd

    DataInd:
      anyOf:
      - type: string
        enum:
          - PFD
          - IPTV
          - BDT
          - SVC_PARAM
          - AM
      - type: string
        description: >
          This string provides forward-compatibility with future
          extensions to the enumeration but is not used to encode
          content defined in the present version of this API.
      description: |
        Possible values are
        - PFD
        - IPTV
        - BDT
        - SVC_PARAM
        - AM
```

A.4 Nudr_DataRepository API for Exposure Data

For the purpose of referencing entities defined in the Open API file defined in this Annex, it shall be assumed that this Open API file is contained in a physical file termed "TS29519_Exposure_Data.yaml".

```
openapi: 3.0.0
info:
  version: '-'
  title: Unified Data Repository Service API file for structured data for exposure
  description: |
    The API version is defined in 3GPP TS 29.504
    © 2022, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.
externalDocs:
  description: >
    3GPP TS 29.519 V17.7.0; 5G System; Usage of the Unified Data Repository Service for Policy Data,
    Application Data and Structured Data for Exposure.
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.519/'

paths:
  /exposure-data/{ueId}/access-and-mobility-data:
    put:
      summary: Creates and updates the access and mobility exposure data for a UE
      operationId: CreateOrReplaceAccessAndMobilityData
      tags:
        - AccessAndMobilityData
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
          - nudr-dr:exposure-data:access-and-mobility-data:create
      parameters:
        - name: ueId
          in: path
          description: UE id
          required: true
```

```
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AccessAndMobilityData'
      responses:
        '201':
          description: >
            Successful case. The resource has been successfully created and a response
            body containing a representation of the access and mobility data shall be returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AccessAndMobilityData'
          headers:
            Location:
              description: 'Contains the URI of the newly created resource'
              required: true
              schema:
                type: string
        '200':
          description: >
            Successful case. The resource has been successfully updated and a response
            body containing a representation of the access and mobility data shall be returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AccessAndMobilityData'
        '204':
          description: Successful case. The resource has been successfully updated and no additional
content is to be sent in the response message.
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    get:
      summary: Retrieves the access and mobility exposure data for a UE
      operationId: QueryAccessAndMobilityData
      tags:
        - AccessAndMobilityData
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
          - nudr-dr:exposure-data:access-and-mobility-data:read
      parameters:
        - name: ueId
          in: path
          description: UE id
          required: true
          schema:
```

```
          $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
      - name: supp-feat
        in: query
        description: Supported Features
        required: false
        schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    responses:
      '200':
        description: The response body contains the access and mobility data
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AccessAndMobilityData'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Deletes the access and mobility exposure data for a UE
    operationId: DeleteAccessAndMobilityData
    tags:
      - AccessAndMobilityData
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:exposure-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:exposure-data
        - nudr-dr:exposure-data:access-and-mobility-data:modify
    parameters:
      - name: ueId
        in: path
        description: UE id
        required: true
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
    responses:
      '204':
        description: Upon success, an empty response body shall be returned
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  patch:
    summary: Updates the access and mobility exposure data for a UE
```

```
      operationId: UpdateAccessAndMobilityData
      tags:
        - AccessAndMobilityData
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
          - nudr-dr:exposure-data:access-and-mobility-data:modify
      parameters:
        - name: ueId
          in: path
          description: UE id
          required: true
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
      requestBody:
        required: true
        content:
          application/merge-patch+json:
            schema:
              $ref: '#/components/schemas/AccessAndMobilityData'
      responses:
        '204':
          description: >
            Successful case. The resource has been successfully updated and no additional
            content is to be sent in the response message.
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  /exposure-data/{ueId}/session-management-data/{pduSessionId}:
    put:
      summary: Creates and updates the session management data for a UE and for an individual PDU
session
      operationId: CreateOrReplaceSessionManagementData
      tags:
        - PduSessionManagementData
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
          - nudr-dr:exposure-data:session-management-data:create
      parameters:
        - name: ueId
          in: path
          description: UE id
          required: true
          schema:
```

```
          $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
      - name: pduSessionId
        in: path
        description: PDU session id
        required: true
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/PduSessionManagementData'
    responses:
      '201':
        description: >
          Successful case. The resource has been successfully created and a response
          body containing a representation of the session management data shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PduSessionManagementData'
        headers:
          Location:
            description: 'Contains the URI of the newly created resource'
            required: true
            schema:
              type: string
      '200':
        description: >
          Successful case. The resource has been successfully updated and a response body
          containing a representation of the session management data shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PduSessionManagementData'
      '204':
        description: >
          Successful case. The resource has been successfully updated and no additional
          content is to be sent in the response message.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  get:
    summary: Retrieves the session management data for a UE and for an individual PDU session
    operationId: QuerySessionManagementData
    tags:
      - PduSessionManagementData
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:exposure-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:exposure-data
        - nudr-dr:exposure-data:session-management-data:read
```

```
    parameters:
      - name: ueId
        in: path
        description: UE id
        required: true
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
      - name: pduSessionId
        in: path
        description: PDU session id
        required: true
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
      - name: ipv4-addr
        in: query
        description: IPv4 Address of the UE
        required: false
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
      - name: ipv6-prefix
        in: query
        description: IPv6 Address Prefix of the UE
        required: false
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Prefix'
      - name: dnn
        in: query
        description: DNN of the UE
        required: false
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      - name: fields
        in: query
        description: attributes to be retrieved
        required: false
        schema:
          type: array
          items:
            type: string
          minItems: 1
      - name: supp-feat
        in: query
        description: Supported Features
        required: false
        schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    responses:
      '200':
        description: The response body contains the session management data
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PduSessionManagementData'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '414':
        $ref: 'TS29571_CommonData.yaml#/components/responses/414'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Deletes the session management data for a UE and for an individual PDU session
    operationId: DeleteSessionManagementData
    tags:
      - PduSessionManagementData
```

```
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:exposure-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:exposure-data
        - nudr-dr:exposure-data:session-management-data:modify
    parameters:
      - name: ueId
        in: path
        description: UE id
        required: true
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
      - name: pduSessionId
        in: path
        description: PDU session id
        required: true
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
    responses:
      '204':
        description: Upon success, an empty response body shall be returned
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  /exposure-data/subs-to-notify:
    post:
      summary: Create a subscription to receive notification of exposure data changes
      operationId: CreateIndividualExposureDataSubscription
      tags:
        - ExposureDataSubscriptions (Collection)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
          - nudr-dr:exposure-data:subs-to-notify:create
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ExposureDataSubscription'
      responses:
        '201':
          description: Created
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ExposureDataSubscription'
          headers:
            Location:
              description: 'Contains the URI of the newly created resource'
              required: true
              schema:
```

```
            type: string
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    callbacks:
      exposureDataChangeNotification:
        '{$request.body#/notificationUri}':
          post:
            requestBody:
              required: true
              content:
                application/json:
                  schema:
                    type: array
                    items:
                      $ref: '#/components/schemas/ExposureDataChangeNotification'
                    minItems: 1
            responses:
              '204':
                description: No Content, Notification was successful
              '400':
                $ref: 'TS29571_CommonData.yaml#/components/responses/400'
              '401':
                $ref: 'TS29571_CommonData.yaml#/components/responses/401'
              '403':
                $ref: 'TS29571_CommonData.yaml#/components/responses/403'
              '404':
                $ref: 'TS29571_CommonData.yaml#/components/responses/404'
              '411':
                $ref: 'TS29571_CommonData.yaml#/components/responses/411'
              '413':
                $ref: 'TS29571_CommonData.yaml#/components/responses/413'
              '415':
                $ref: 'TS29571_CommonData.yaml#/components/responses/415'
              '429':
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
              default:
                $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  /exposure-data/subs-to-notify/{subId}:
    put:
      summary: updates a subscription to receive notifications of exposure data changes
      operationId: ReplaceIndividualExposureDataSubscription
      tags:
        - IndividualExposureDataSubscription (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:exposure-data
          - nudr-dr:exposure-data:subs-to-notify:modify
```

```
    parameters:
      - name: subId
        in: path
        description: Subscription id
        required: true
        schema:
          type: string
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/ExposureDataSubscription'
    responses:
      '200':
        description: >
          Resource was successfully modified and a body with the modified subscription
          to notifications about exposure data is returned..
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ExposureDataSubscription'
      '204':
        description: The individual subscription resource was successfully modified.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  delete:
    summary: Deletes the individual Exposure Data subscription
    operationId: DeleteIndividualExposureDataSubscription
    tags:
      - IndividualExposureDataSubscription (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:exposure-data
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:exposure-data
        - nudr-dr:exposure-data:subs-to-notify:modify
    parameters:
      - name: subId
        in: path
        description: Subscription id
        required: true
        schema:
          type: string
    responses:
      '204':
        description: Resource was successfully deleted.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
```

```
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
components:
  schemas:
    AccessAndMobilityData:
      description: Represents Access and Mobility data for a UE.
      type: object
      properties:
        location:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
        locationTs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        timeZone:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/TimeZone'
        timeZoneTs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        accessType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
        regStates:
          type: array
          items:
            $ref: 'TS29518_Namf_EventExposure.yaml#/components/schemas/RmInfo'
        regStatesTs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        connStates:
          type: array
          items:
            $ref: 'TS29518_Namf_EventExposure.yaml#/components/schemas/CmInfo'
        connStatesTs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        reachabilityStatus:
          $ref: 'TS29518_Namf_EventExposure.yaml#/components/schemas/UeReachability'
        reachabilityStatusTs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        smsOverNasStatus:
          $ref: 'TS29518_Namf_Communication.yaml#/components/schemas/SmsSupport'
        smsOverNasStatusTs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        roamingStatus:
          type: boolean
          description: >
            True  The serving PLMN of the UE is different from the HPLMN of the UE; False
            The serving PLMN of the UE is the HPLMN of the UE.
        roamingStatusTs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        currentPlmn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnId'
        currentPlmnTs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        ratType:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
        ratTypesTs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        suppFeat:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        resetIds:
          type: array
          items:
            type: string
          minItems: 1
    PduSessionManagementData:
      description: Represents Session management data for a UE and a PDU session.
      type: object
      properties:
        pduSessionStatus:
          $ref: '#/components/schemas/PduSessionStatus'
        pduSessionStatusTs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
```

```
      dnai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnai'
      dnaiTs:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      n6TrafficRoutingInfo:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RouteToLocation'
      n6TrafficRoutingInfoTs:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      ipv4Addr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
      ipv6Prefix:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Prefix'
        minItems: 1
        description: UE IPv6 prefix.
      ipv6Addrs:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Addr'
        minItems: 1
      pduSessType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionType'
      ipAddrTs:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      dnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      pduSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
      suppFeat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
  ExposureDataSubscription:
    description: Represents a Subscription to Notifications about Exposure Data.
    type: object
    properties:
      notificationUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      monitoredResourceUris:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        minItems: 1
      expiry:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      resetIds:
        type: array
        items:
          type: string
        minItems: 1
    required:
      - notificationUri
      - monitoredResourceUris
  ExposureDataChangeNotification:
    description: Represents changed exposure data for one UE for which Notification was requested.
    type: object
    properties:
      ueId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/VarUeId'
      accessAndMobilityData:
        $ref: '#/components/schemas/AccessAndMobilityData'
      pduSessionManagementData:
        type: array
        items:
          $ref: '#/components/schemas/PduSessionManagementData'
        minItems: 1
      delResources:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
```

```
      minItems: 1
  PduSessionStatus:
    anyOf:
    - type: string
      enum:
        - "ACTIVE"
        - "RELEASED"
    - type: string
      description: >
        This string provides forward-compatibility with future
        extensions to the enumeration but is not used to encode
        content defined in the present version of this API.
    description: >
      Possible values are
      - "ACTIVE"
      - "RELEASED"
```

Annex B (informative): Change history

Change history							
Date	**Meeting #**	**TSG Doc.**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New**
2018-01	CT3#94					TS skeleton of Usage of the Unified Data Repository service for Policy Data and Structured Data for exposure	0.0.0
2018-01	CT3#94					Inclusion of C3-180352, C3-180326, C3-180327	0.1.0
2018-03	CT3#95					Inclusion of C3-181145, C3-181288, C3-181289, C3-181313	0.2.0
2018-04	CT3#96					Inclusion of C3-182402, C3-182476, C3-182456, C3-182404, C3-182457, C3-182458	0.3.0
2018-05	CT3#97					Inclusion of C3-183872, C3-183705, C3-183704, C3-183873, C3-183732, C3-183706 C3-183874, C3-183709, C3-183876, C3-183263, C3-183877, C3-183712, C3-183715 and editorial changes by Rapporteur	0.4.0
2018-06	CT#80	CP-181033				TS sent to plenary for approval	1.0.0
2018-06	CT#80	CP-181033				TS approved by plenary	15.0.0
2018-09	CT#81	CP-182015	0006		F	Scope and Overview	15.1.0
2018-09	CT#81	CP-182015	0007		F	Introduction subclauses 5.1, 6.1 and 7.1	15.1.0
2018-09	CT#81	CP-182015	0008		F	Resource overview subclauses 5.2.1, 6.2.1 and 7.2.1	15.1.0
2018-09	CT#81	CP-182015	0009	3	F	Nudr reused data types	15.1.0
2018-09	CT#81	CP-182015	0010	6	B	AM policy data resource definition	15.1.0
2018-09	CT#81	CP-182015	0011	8	B	SM policy data resource definition	15.1.0
2018-09	CT#81	CP-182015	0012	3	B	Sponsored data connectivity profiles	15.1.0
2018-09	CT#81	CP-182015	0014	5	B	Background Data Transfer data	15.1.0
2018-09	CT#81	CP-182015	0015	4	B	Policy data change subcription	15.1.0
2018-09	CT#81	CP-182015	0016	5	B	Data change notification	15.1.0
2018-09	CT#81	CP-182015	0017	2	F	Error handling subcaluses	15.1.0
2018-09	CT#81	CP-182015	0018	1	F	Feature negotiation	15.1.0
2018-09	CT#81	CP-182015	0019	1	F	Resource PFD Datas and Data types definition	15.1.0
2018-09	CT#81	CP-182015	0020	1	F	OpenAPI schema for Policy Data	15.1.0
2018-09	CT#81	CP-182015	0021	3	F	Stage 2 Allignments for Structured Data for Exposure	15.1.0
2018-09	CT#81	CP-182015	0022	4	B	Subscription for Structured Data for Exposure	15.1.0
2018-09	CT#81	CP-182015	0023	3	F	Get query parameters for structured data for exposure	15.1.0
2018-09	CT#81	CP-182015	0024	5	F	HTTP error responses for structured data for exposure	15.1.0
2018-09	CT#81	CP-182015	0025	1	F	Description of Structured data types	15.1.0
2018-09	CT#81	CP-182032	0026	3	B	Resource Structure for policy data	15.1.0
2018-09	CT#81	CP-182015	0027		F	n6 Traffic Routing Information encoding	15.1.0
2018-09	CT#81	CP-182015	0028	1	F	Removal of ExternalDocs section in OpenApi file for Exposure Data	15.1.0
2018-09	CT#81	CP-182015	0029	1	F	VarUeId used in TS 29.519	15.1.0
2018-09	CT#81	CP-182015	0034	1	F	OpenAPI schema for PolicyData	15.1.0
2018-09	CT#81	CP-182015	0035	3	F	External information stored in UDR for Application Data	15.1.0
2018-09	CT#81	CP-182015	0036		F	Clean up the Editor Note in applicatoin data	15.1.0
2018-09	CT#81	CP-182015	0037	4	F	Correction the data type of application data	15.1.0
2018-09	CT#81	CP-182015	0038	3	B	Introduce the operator policy data	15.1.0
2018-09	CT#81	CP-182015	0039	2	B	OpenAPI of Application data	15.1.0
2018-09	CT#81	CP-182015	0040	1	F	Update of resource structure of application data 29.519	15.1.0
2018-09	CT#81	CP-182015	0041	1	F	Ethernet packet filter for traffic influence data of applicatoin data 29.519	15.1.0
2018-12	CP#82	CP-183205	0042	1	F	DNAI reused data	15.2.0
2018-12	CP#82	CP-183205	0043	1	F	ExternalDocs field	15.2.0
2018-12	CP#82	CP-183205	0044		F	PduSessionStatus	15.2.0
2018-12	CP#82	CP-183205	0045	1	F	Query parameters correction on application data	15.2.0
2018-12	CP#82	CP-183205	0046	1	F	PUT operation for Individual Influence Data Subscription	15.2.0
2018-12	CP#82	CP-183205	0047	3	F	Background Data Transfer Data	15.2.0
2018-12	CP#82	CP-183205	0048	1	F	Cardinality for array and map	15.2.0
2018-12	CP#82	CP-183205	0049	1	F	Resource URI for ExposureData	15.2.0
2018-12	CP#82	CP-183205	0050		F	UeId definition	15.2.0
2018-12	CP#82	CP-183205	0052	1	F	Error status codes for HTTP responses	15.2.0
2018-12	CP#82	CP-183205	0053		F	Service name	15.2.0
2018-12	CP#82	CP-183205	0054	1	F	Reference to the yaml file of Npcf_BDTPolicyControl API	15.2.0
2018-12	CP#82	CP-183205	0055		F	Missed GET method on UEPolicySetResource	15.2.0

2018-12	CP#82	CP-183205	0055	2	F	Missed GET method on UEPolicySetResource	15.2.0
2018-12	CP#82	CP-183205	0056	4	F	Corrections on AM Policy and UE Policy data	15.2.0
2018-12	CP#82	CP-183205	0057	6	F	Corrections on SM Policy Data	15.2.0
2018-12	CP#82	CP-183205	0058	1	F	Correct TrafficRoute data type	15.2.0
2018-12	CP#82	CP-183205	0060	4	F	Correct subscription for application data	15.2.0
2018-12	CP#82	CP-183205	0062	3	F	Correction to the AF Influence on traffic steering control	15.2.0
2018-12	CP#82	CP-183205	0063	3	F	PLMN ID as key for UE data sets	15.2.0
2018-12	CP#82	CP-183205	0066	1	F	Supported features	15.2.0
2018-12	CP#82	CP-183205	0067	1	F	Add AF application ID for traffic influence	15.2.0
2018-12	CP#82	CP-183205	0068		F	Correct referenced file in openAPI for application data	15.2.0
2018-12	CP#82	CP-183205	0069		F	Correct referenced file in openAPI for exposure data	15.2.0
2018-12	CP#82	CP-183205	0070	1	F	Use correct naming convention	15.2.0
2018-12	CP#82	CP-183205	0071	1	F	Correct subscription for Application Data	15.2.0
2018-12	CP#82	CP-183205	0072	1	F	Status code update for Application Data	15.2.0
2018-12	CP#82	CP-183205	0073	1	F	Correction on UsageThreshold	15.2.0
2018-12	CP#82	CP-183205	0075	2	F	Status code update for Exposure Data	15.2.0
2018-12	CP#82	CP-183205	0076	2	F	Status code update for Policy Data	15.2.0
2018-12	CP#82	CP-183205	0077	2	F	Incorrect response code of PUT method for individual PFD Data creation	15.2.0
2018-12	CP#82	CP-183205	0078	1	F	corrections on individual Influence Data	15.2.0
2018-12	CP#82	CP-183205	0079	2	F	Adding timestamp for allowed usage storage	15.2.0
2018-12	CP#82	CP-183205	0080		F	Missing 201 response body for /policy-data/subs-to-notify	15.2.0
2018-12	CP#82	CP-183205	0081	1	F	Location header	15.2.0
2018-12	CP#82	CP-183205	0083	4	F	Corrections on Usage Monitoring information	15.2.0
2018-12	CP#82	CP-183205	0087		F	Correction on PduSessionManagementData	15.2.0
2019-03	CP#83	CP-190115	0090	1	F	Correction on ChargingInformation data type reference in the OpenAPI file	15.3.0
2019-03	CP#83	CP-190115	0091	1	F	Formatting of structured data types in query parameters	15.3.0
2019-03	CP#83	CP-190115	0092	1	F	Correcting resource definitions	15.3.0
2019-03	CP#83	CP-190115	0093		F	References to data types	15.3.0
2019-03	CP#83	CP-190115	0095	2	F	Correction on Policy Data change notification	15.3.0
2019-03	CP#83	CP-190115	0096	1	F	Correction on BDT data	15.3.0
2019-03	CP#83	CP-190115	0098	1	F	Support of allowed Route Selection data	15.3.0
2019-03	CP#83	CP-190115	0099	1	F	Support of PEI, OSid and ANDSP support indication in UE Policy Set	15.3.0
2019-03	CP#83	CP-190115	0100		F	Support of PRA information	15.3.0
2019-06	CP#84	CP-191077	0101		F	Correction of OpenAPI file and main body misalignments for Notification messages	15.4.0
2019-06	CP#84	CP-191077	0102	2	F	Corrections in PolicyDataSubscription and PolicyDataChangeNotification types	15.4.0
2019-06	CP#84	CP-191077	0106	1	F	Precedence of OpenAPI file	15.4.0
2019-06	CP#84	CP-191077	0113		F	Correcting the name of URI to receive notifications	15.4.0
2019-06	CP#84	CP-191077	0103	4	F	"201 Created" response without body and other HTTP response errors	15.4.0
2019-06	CP#84	CP-191077	0116	1	F	Expiry time in subscription data types	15.4.0
2019-06	CP#84	CP-191184	0117	3	F	Notification of resource deletion	15.4.0
2019-06	CP#84	CP-191208	0118	3	F	Copyright Note in YAML file	15.4.0
2019-06	CP#84	CP-191077	0119	1	F	Missed Traffic Influence information	15.4.0
2019-06	CP#84	CP-191183	0120	4	F	Notification of resource deletion for Application and Exposure data sets	15.4.0
2019-06	CP#84	CP-191089	0105	2	B	PFD extension for domain name protocol	16.0.0
2019-06	CP#84	CP-191071	0109	3	B	AF acknowledgement to be expected	16.0.0
2019-06	CP#84	CP-191071	0110	2	B	UE IP address preservation indication	16.0.0
2019-06	CP#84	CP-191100	0114	1	B	NEF stored exposure data	16.0.0
2019-09	CT#85	CP-192169	0125	2	F	UDR Application Errors	16.1.0
2019-09	CT#85	CP-192145	0127	2	A	Missed any UE traffic influence data	16.1.0
2019-09	CT#85	CP-192169	0128	1	B	Multiple temporal validity conditions for AF traffic influence	16.1.0
2019-09	CT#85	CP-192244	0130	2	A	Remove PFD change notification	16.1.0
2019-09	CT#85	CP-192156	0131	1	B	Support a set of MAC addresses in traffic filter	16.1.0
2019-09	CT#85	CP-192196	0138	4	B	Application Data update to support xBDT	16.1.0
2019-09	CT#85	CP-192152	0140	3	B	Subscription and Notification of IPTV configuration data change	16.1.0
2019-09	CT#85	CP-192152	0141	2	B	Resource structure of Nudr API for Application Data	16.1.0
2019-09	CT#85	CP-192152	0142	2	B	Storage of IPTV configuration data	16.1.0
2019-09	CT#85	CP-192156	0143	2	F	Subscription/unsubscription for PFDs update notifications and related callback correction	16.1.0
2019-10	CT#85	-	-	-	-	Addition of missing yaml files	16.1.1

2019-12	CT#86	CP-193223	0144	2	B	Retrieval of BDT policy data for a set of BDT reference identifiers	16.2.0
2019-12	CT#86	CP-193181	0145	1	F	Open issue for AddrPreservation feature	16.2.0
2019-12	CT#86	CP-193223	0149	3	B	Support of BDT reference Id within Session Management data	16.2.0
2019-12	CT#86	CP-193191	0151	3	B	Partial update of IPTVConfiguration API	16.2.0
2019-12	CT#86	CP-193210	0152	2	B	Patch Report to Nudr_DataRepository API for Policy data	16.2.0
2019-12	CT#86	CP-193223	0153		F	Reference correction to BdtReferenceId	16.2.0
2019-12	CT#86	CP-193212	0155	1	F	Update of TS version_Rel-16	16.2.0
2019-12	CT#86	CP-193197	0156	2	B	MCS Priority Indication	16.2.0
2019-12	CT#86	CP-193187	0158	-	A	Definition of BdtData in OpenAPI	16.2.0
2019-12	CT#86	CP-193197	0159	-	F	Usage of BdtReferenceId data type	16.2.0
2019-12	CT#86	CP-193223	0161	1	B	include S-NSSAI and DNN in Application Data for xBDT	16.2.0
2019-12	CT#86	CP-193223	0162		F	remove 204 response on PUT request for AppliedBDTPolicyData	16.2.0
2019-12	CT#86	CP-193210	0164		B	Feature Negotiation for OperatorSpecificData resource	16.2.0
2019-12	CT#86	CP-193210	0165		B	Feature Negotiation for Influence Data resource	16.2.0
2019-12	CT#86	CP-193126	0166		B	URI correction on resource Individual Applied BDT Policy Data	16.2.0
2020-03	CT#87e	CP-200141	0167	2	B	Application data change notification	16.3.0
2020-03	CT#87e	CP-200203	0168	1	B	Complete the IPTV configuration	16.3.0
2020-03	CT#87e	CP-200263	0169	2	B	Nudr_DataRepository API update for Service Parameter provisioning	16.3.0
2020-03	CT#87e	CP-200218	0170	1	B	Indication of traffic correlation	16.3.0
2020-03	CT#87e	CP-200136	0171	2	B	Include influenceId into TrafficInfluData for Notification	16.3.0
2020-03	CT#87e	CP-200219	0172	2	B	Correct the mapping of ASPid for xBDT	16.3.0
2020-03	CT#87e	CP-200207	0173	1	B	Subscription and notification to data changes related to a subset of resource data, Policy Data set	16.3.0
2020-03	CT#87e	CP-200203	0174		F	Correct content type in PATCHing IPTV application data	16.3.0
2020-03	CT#87e	CP-200198	0175		B	Additional PDU session status data	16.3.0
2020-03	CT#87e	CP-200219	0176		F	Correct content type in PATCHing BDT application data	16.3.0
2020-03	CT#87e	CP-200219	0177		F	Removal of the BDT policy from the "bdtRefIds" object	16.3.0
2020-06	CT#88e	CP-201220	0179	2	A	internalGroupId in Influence Data	16.4.0
2020-06	CT#88e	CP-201238	0180	2	F	Definition of ServiceParameterData in openAPI	16.4.0
2020-06	CT#88e	CP-201246	0181	3	F	Data Types for Application Data	16.4.0
2020-06	CT#88e	CP-201228	0183	1	F	Correction to IPTV Configuration	16.4.0
2020-06	CT#88e	CP-201253	0184	1	F	Correction to BDT Policy	16.4.0
2020-06	CT#88e	CP-201220	0185	1	A	Correction to notifications of Operator Specific Data changes	16.4.0
2020-06	CT#88e	CP-201220	0188		A	Correct content type in PATCHing traffic influence application data	16.4.0
2020-06	CT#88e	CP-201233	0189		F	Correct DataFilter presence condition	16.4.0
2020-06	CT#88e	CP-201233	0190		F	Correct resourceId in required field	16.4.0
2020-06	CT#88e	CP-201233	0191	1	B	Support local traffic routing in session subscription	16.4.0
2020-06	CT#88e	CP-201244	0192	1	F	Storage of YAML files in ETSI Forge	16.4.0
2020-06	CT#88e	CP-201276	0194	1	F	Traffic descriptor for xBDT	16.4.0
2020-06	CT#88e	CP-201256	0195	1	F	URI of the Nudr_DataRepository service for Policy, Application and Exposure data	16.4.0
2020-06	CT#88e	CP-201313	0197		F	Data type of internalGroupId attribute	16.4.0
2020-06	CT#88e	CP-201228	0199		F	Data model correction for IPTV	16.4.0
2020-06	CT#88e	CP-201228	0200		F	Remove feature for IPTV data configuration	16.4.0
2020-06	CT#88e	CP-201244	0201		F	Optionality of ProblemDetails	16.4.0
2020-06	CT#88e	CP-201233	0202	2	F	Corrections on supported features definition	16.4.0
2020-06	CT#88e	CP-201244	0203		F	required field in OpenAPI file	16.4.0
2020-06	CT#88e	CP-201244	0204	1	F	Supported headers, Resource Data type	16.4.0
2020-06	CT#88e	CP-201255	0206		F	Update in TS version in externalDocs field	16.4.0
2020-09	CT#89e	CP-202059	0207		F	Include resouceURI in IptvConfigData for change notification association	16.5.0
2020-09	CT#89e	CP-202069	0208		F	Include resouceURI in ServiceParameterData for change notification association	16.5.0
2020-09	CT#89e	CP-202209	0209		F	Include resouceURI in TrafficInfluData for change notification association	16.5.0
2020-09	CT#89e	CP-202082	0210		F	Include resouceURI in BdtPolicyData for change notification association	16.5.0
2020-09	CT#89e	CP-202073	0211		F	Removal of sibling elements	17.0.0
2020-09	CT#89e	CP-202073	0212		B	Resource Level Authorization for Policy Data, Application Data, and Exposure Data	17.0.0
2020-09	CT#89e	CP-202085	0214		F	Update of TS version in externalDocs field	17.0.0
2020-12	CT#90e	CP-203139	0216		A	TS 29.519 Essential Corrections and alignments	17.1.0

2020-12	CT#90e	CP-203110	0218	1	A	Correction to support NEF binding indication	17.1.0
2020-12	CT#90e	CP-203129	0220	1	A	Correction to the BDT policy re-negotiation	17.1.0
2020-12	CT#90e	CP-203139	0222		A	Storage of YAML files in 3GPP Forge	17.1.0
2020-12	CT#90e	CP-203147	0223	2	B	Data retrieval of multiple policy data sets	17.1.0
2020-12	CT#90e	CP-203154	0225		A	MA Support indication within UE Context policy control subscription information	17.1.0
2020-12	CT#90e	CP-203139	0227	1	A	Callback URI correction	17.1.0
2020-12	CT#90e	CP-203153	0229		F	Update of TS version in externalDocs field	17.1.0
2021-03	CT#91e	CP-210226	0230	1	B	Allow the retrieval of a set of policy data subsets for a UE	17.2.0
2021-03	CT#91e	CP-210218	0231		F	Adding "description" field for map data types	17.2.0
2021-03	CT#91e	CP-210218	0232		F	OpenAPI reference	17.2.0
2021-03	CT#91e	CP-210226	0233		F	Encoding of Snssai as key of a map	17.2.0
2021-03	CT#91e	CP-210219	0235		F	Adding some missing description fields to data type definitions in OpenAPI specification files	17.2.0
2021-03	CT#91e	CP-210198	0238	1	A	UsageMonData: definition of "scopes" attribute	17.2.0
2021-03	CT#91e	CP-210198	0242	1	A	PUT method for Exposure Data modification	17.2.0
2021-03	CT#91e	CP-210206	0244	1	A	extend bdtpStatus to include VALID	17.2.0
2021-03	CT#91e	CP-210240	0248		F	Update of TS version in externalDocs field	17.2.0
2021-06	CT#92e	CP-211242	0250	1	F	Correction to Charging Information	17.3.0
2021-06	CT#92e	CP-211211	0252	3	B	Correction to OSD handling	17.3.0
2021-06	CT#92e	CP-211121	0253	4	B	5G ProSe related updates to Nudr	17.3.0
2021-06	CT#92e	CP-211218	0254	2	B	Support of User Plane Latency requirement	17.3.0
2021-06	CT#92e	CP-211218	0257	1	B	UDR application data update to support AF influence on URSP	17.3.0
2021-06	CT#92e	CP-211265	0259		F	Update of TS version in externalDocs field	17.3.0
2021-09	CT#93e	CP-212242	0260	1	B	AM Influence UDR models for DCAMP	17.4.0
2021-09	CT#93e	CP-212206	0261	1	B	PDU session policy data extension for UE-Slice-MBR	17.4.0
2021-09	CT#93e	CP-212206	0263	1	B	Policy Data extension for Slice-MBR	17.4.0
2021-09	CT#93e	CP-212224	0264	1	F	Corrections to IPTV configuration data	17.4.0
2021-09	CT#93e	CP-212229	0266	1	F	Correction of cardinality for xBDT	17.4.0
2021-09	CT#93e	CP-212190	0269		F	Correction of data types of resetPeriod and resetTime in openAPI file	17.4.0
2021-09	CT#93e	CP-212188	0271		B	Removal of some 5G ProSe related Ens	17.4.0
2021-09	CT#93e	CP-2112223	0272		F	Update of externalDocs field	17.4.0
2021-12	CT#94e	CP-213230	0275	1	B	Correction on Network Slice Policy Control Subscription information	17.5.0
2021-12	CT#94e	CP-213230	0276		B	Correction on PolicyDataChangeNotification	17.5.0
2021-12	CT#94e	CP-213222	0277	1	B	Nudr impacts for the notification on the outcome of UE Policies delivery due to service specific parameter provisioning	17.5.0
2021-12	CT#94e	CP-213194	0278		F	AM Influence Data: alignment with naming convention	17.5.0
2021-12	CT#94e	CP-213230	0279	1	F	Presence conditions for attributes in SlicePolicyData	17.5.0
2021-12	CT#94e	CP-213234	0280	1	B	Time Synch data in UDR	17.5.0
2021-12	CT#94e	CP-213239	0281	1	F	Clarification about the data type used for UE policy section information	17.5.0
2021-12	CT#94e	CP-213194	0282	1	B	AM Influence data model updates	17.5.0
2021-12	CT#94e	CP-213213	0283	1	B	Updates to the 5G ProSe service parameters	17.5.0
2021-12	CT#94e	CP-213223	0286	1	B	Clarification of AF preference for the user plane latency	17.5.0
2021-12	CT#94e	CP-213168	0288	2	F	Feature negotiation at PFD data retrieval	17.5.0
2021-12	CT#94e	CP-213170	0290	2	F	Reference PfdDataForApp data type	17.5.0
2021-12	CT#94e	CP-213223	0291	1	B	AF Request for Simultaneous Connectivity over Source and Target PSA at Edge Relocation	17.5.0
2021-12	CT#94e	CP-213239	0293	1	F	Naming Convention	17.5.0
2021-12	CT#94e	CP-213246	0294		F	Update of externalDocs field	17.5.0
2022-03	CT#95e	CP-220198	0296	2	B	Resolution of DCAMP open issues such as applicable filters and immediate reporting	17.6.0
2022-03	CT#95e	CP-220201	0297		B	Support of Operator-Specific Data with no keys	17.6.0
2022-03	CT#95e	CP-220176	0298	2	A	Alignment of "Application Errors" clauses with SBI TS template	17.6.0
2022-03	CT#95e	CP-220185	0299	1	B	Updates to support AF Guidance on URSP for Application Data	17.6.0
2022-03	CT#95e	CP-220327	0300	2	B	resetId assignment to UDR consumers (PCF, NEF)	17.6.0
2022-03	CT#95e	CP-220195	0301	1	D	Clause Correction	17.6.0
2022-03	CT#95e	CP-220186	0302	1	B	API definition for EAS Deployment Information Data	17.6.0
2022-03	CT#95e	CP-220330	0303	2	B	OpenAPI to support EAS Deployment Information Data	17.6.0
2022-03	CT#95e	CP-220183	0304	1	B	Time Synch data updates	17.6.0
2022-03	CT#95e	CP-220176	0306		A	Correction of query parameter content type	17.6.0
2022-03	CT#95e	CP-220215	0308	2	F	Introduction of ServiceParameterDataPatch	17.6.0
2022-03	CT#95e	CP-220185	0309		F	Removal of Editor's Notes for URSP encoding and DeliveryOutcome feature	17.6.0

2022-03	CT#95e	CP-220186	0310	1	B	Support of binding principles for UE Policy Delivery outcome	17.6.0
2022-03	CT#95e	CP-220198	0312	1	F	Updates of AM Influence Data resource	17.6.0
2022-03	CT#95e	CP-220198	0313	1	F	Query for AM Influence data for Any UE	17.6.0
2022-03	CT#95e	CP-220197	0314	1	B	Support of service area coverage notifications	17.6.0
2022-03	CT#95e	CP-220198	0315	1	F	Solving Editor's Note on query parameters	17.6.0
2022-03	CT#95e	CP-220201	0316	1	F	Formatting of description fields	17.6.0
2022-03	CT#95e	CP-220334	0318	2	F	Query for Service Parameter Data for Any UE	17.6.0
2022-03	CT#95e	CP-220196	0320	1	F	Subscription to notifications for Any UE	17.6.0
2022-03	CT#95e	CP-220195	0321	1	F	Updates of Traffic Influence resource	17.6.0
2022-03	CT#95e	CP-220195	0322	2	F	Correction to default values	17.6.0
2022-03	CT#95e	CP-220194	0324		F	Update of info and externalDocs fields	17.6.0
2022-06	CT#96	CP-221157	0326		F	Removal of allowEmptyValue parameter	17.7.0
2022-06	CT#96	CP-221153	0327	2	F	Support of allowedDelay for PFD data	17.7.0
2022-06	CT#96	CP-221126	0328		F	Correction of supported features for Edge Computing functionality	17.7.0
2022-06	CT#96	CP-221159	0329	2	F	Handling of query parameters for AM Influence Data resource	17.7.0
2022-06	CT#96	CP-221157	0330	1	B	Policy Data Subscription enhancements	17.7.0
2022-06	CT#96	CP-221157	0331	1	F	Handling of combinations of attributes of DataFilter data type	17.7.0
2022-06	CT#96	CP-221158	0332	2	F	Handling of query parameters in Applied BDT Data	17.7.0
2022-06	CT#96	CP-221158	0333	2	F	Handling of query parameters in IPTV Configuration resource	17.7.0
2022-06	CT#96	CP-221157	0334	1	F	Handling of query parameters for PFD Data	17.7.0
2022-06	CT#96	CP-221158	0335	2	F	Handling of query parameters in Traffic Influence Data	17.7.0
2022-06	CT#96	CP-221158	0336	2	F	Handling of query parameters in Service Parameter Data resource	17.7.0
2022-06	CT#96	CP-221155	0337	1	B	Nudr_DataRepository Application data Access Scope update for API, resource, operation	17.7.0
2022-06	CT#96	CP-221155	0338	1	B	Nudr_DataRepository Exposure Data Access Scope update for API, resource, operation	17.7.0
2022-06	CT#96	CP-221155	0339	1	B	Nudr_DataRepository Policy data Access Scope update for API, resource, operation	17.7.0
2022-06	CT#96	CP-221126	0340	1	F	Update on EAS Deployment Information Data	17.7.0
2022-06	CT#96	CP-221159	0341	1	F	Clarification about subscription to cha	17.7.0
2022-06	CT#96	CP-221154	0342	1	F	Removing the apiVersion placeholder from the Resource URI variables tables	17.7.0
2022-06	CT#96	CP-221154	0343		F	Nudr_DataRepository Policy data description field update	17.7.0
2022-06	CT#96	CP-221158	0344	1	F	Dependencies between UM resource(s) and SM resource	17.7.0
2022-06	CT#96	CP-221151	0345		F	Update of info and externalDocs fields	17.7.0
2022-09	CT#97e	CP-222123	0346	2	F	Correction in the handling of individual AM Influence Data resource	17.8.0
2022-09	CT#97e	CP-222123	0347	2	F	Correction in the handling of individual Applied BDT Policy Data resource	17.8.0
2022-09	CT#97e	CP-222123	0348	2	F	Correction in the handling of individual Influence Data resource	17.8.0
2022-09	CT#97e	CP-222123	0349	2	F	Correction in the handling of individual IPTV configuration resource	17.8.0
2022-09	CT#97e	CP-222123	0350	2	F	Correction in the handling of individual Service Parameter Data resource	17.8.0
2022-09	CT#97e	CP-222123	0351		F	OpenAPI file for policy data: tabs removal and "description" field adding	17.8.0
2022-09	CT#97e	CP-222123	0352	1	F	Application data API: definitions of 403 and 404 responses	17.8.0
2022-09	CT#97e	CP-222099	0353		F	Support of GET for Individual EAS Deployment Information Data resource	17.8.0
2022-09	CT#97e	CP-222099	0354		F	Incomplete implementation of CR #0308	17.8.0
2022-09	CT#97e	CP-222093	0357		A	Correction to UE Dedicated PRAs	17.8.0
2022-09	CT#97e	CP-222099	0358		F	Corrections to Group Id	17.8.0
2022-09	CT#97e	CP-222095	0359	1	B	Supporting MBS policy control	17.8.0
2022-09	CT#97e	CP-222121	0360		F	Update of info and externalDocs fields	17.8.0
2022-12	CT#98e	CP-223196	0366	1	F	Correction related to applicability of traffic correlation indicator	17.9.0
2022-12	CT#98e	CP-223166	0367		F	Correction to MbsSessPolCtrlData data type	17.9.0
2023-03	CT#99	CP-230165	0374	1	F	Partial subscriptions corrections	17.10.0
2023-03	CT#99	CP-230163	0382		F	Correction to allowedDelay for PFD data	17.10.0
2023-03	CT#99	CP-230164	0385		A	internal-Group-Ids attribute correction	17.10.0
2023-03	CT#99	CP-230129	0396	1	A	Correction of a misalignment in an attribute's name	17.10.0
2023-09	CT#101	CP-232114	0415	1	F	Correction in ServiceParameterDataPatch data type	17.11.0
2023-09	CT#101	CP-232106	0433	-	F	Correction on headers attribute	17.11.0

2023-09	CT#101	CP-232118	0441		F	Update of externalDocs fields	17.11.0
2023-12	CT#102	CP-233254	0447	1	A	Corrections on attribute names of SM Policy Data	17.12.0
2023-12	CT#102	CP-233272	0476		F	Correction of anyUeInd attribute	17.12.0
2023-12	CT#102	CP-233260	0477		F	Update of externalDocs fields	17.12.0
2024-06	CT#104	CP-241134	0495	1	F	Data type misalignment for group identifier	17.13.0
2024-06	CT#104	CP-241109	0526		F	Update of externalDocs fields	17.13.0
2024-09	CT#105	CP-242122	0539	1	F	OAuth2 scope correction	17.14.0
2024-09	CT#105	CP-242139	0541	1	F	Non-workable OAuth2 scope	17.14.0
2024-09	CT#105	CP-242122	0548		F	Update of externalDocs fields	17.14.0
2025-06	CT#108	CP-251097	0605		F	Wrong attribute name	17.15.0
2026-03	CT#111	CP-260048	0633		A	Policy and Exposure Data Corrections	17.16.0
2026-03	CT#111	CP-260053	0636		F	AM Influence Data Corrections	17.16.0
2026-03	CT#111	CP-260055	0639		F	Traffic Influence Data Corrections	17.16.0
2026-03	CT#111	CP-260079	0643		F	Update of externalDocs fields	17.16.0
2026-09	CT#113	CP-262129	0655	1	F	MCS Priority Indication description	17.17.0

History

Version	Date	Status
V17.6.0	May 2022	Publication
V17.7.0	June 2022	Publication
V17.8.0	September 2022	Publication
V17.9.0	January 2023	Publication
V17.10.0	April 2023	Publication
V17.11.0	September 2023	Publication
V17.12.0	January 2024	Publication
V17.13.0	July 2024	Publication
V17.14.0	September 2024	Publication
V17.15.0	June 2025	Publication
V17.16.0	March 2026	Publication
V17.17.0	October 2026	Publication